

From Tweets to the Streets: Twitter and Extremist Protests in the United States*

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Abstract

How does social media affect the composition of political protests in the United States? Using early adoption of Twitter as a plausibly exogenous source of variation in county-level Twitter penetration, and comprehensive data on protest events, this paper finds that Twitter penetration increases the frequency of protests overall, but also radicalizes them. Twitter disproportionately fuels protests with participation of “extreme” groups—groups that are particularly militant, radical, or hateful—both on the political left and right. Survey data suggest that coordination is not the only mechanism: County-level Twitter penetration also intensifies opinions, plausibly increasing individuals’ willingness to partake in protests involving extreme groups.

Keywords: Twitter, Collective action, Protests, Social media, Information Technology

JEL codes: P11, P16, D70, L86, L82.

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1 Introduction

Since the late 2000s, the world has witnessed increasing waves of protest movements such as the Arab Spring and the Black Lives Matter Movement (Ortiz et al., 2013; Cantoni et al., 2024). These developments have coincided with the rapid expansion of Information and Communication Technologies (ICTs) in general and social media in particular, creating the widespread perception of their role in facilitating protest coordination (Economist, 2019; Ovide, 2020). More recently in Western democracies, particularly the United States, where far-right demonstration activity has been surging (ACLED, 2022), reports document not only rising protest activity but increased hostility and aggression during protest events (ACLED, 2020c,a,b). Journalists and politicians have suggested that social media may also play an important role in these trends (Brewster, 2021; Hadero, 2023). While increased violence during demonstrations is worrying, understanding the influences of social media on the composition of political protests is important as protests can influence attitudes, voting behaviour and policy (Madestam et al., 2013; Skoy, 2021; Klein Teeselink & Melios, 2021; Caesmann et al., 2021; Casanueva, 2023).

The literature has explored the mechanisms through which ICTs can facilitate protests and confirmed them empirically (e.g., Little, 2016; Enikolopov, Makarin, & Petrova, 2020). However, the best available evidence is limited to either rigorous studies focusing on protests related to one particular social movement (e.g., Enikolopov, Makarin, & Petrova, 2020) or studies taking a more global approach, treating the nature of different movements as a black box (e.g., Manacorda & Tesei, 2020). Does social media disproportionately fuel protest mobilization among political fringes? Does social media influence aggression and violence during protest events? This paper contributes to the understanding of these issues by studying the effect of Twitter (now X) penetration on various protest movements, depending on the characteristics of the movements and of the protest events themselves in the United States. It finds that Twitter penetration causes a disproportionate increase in protest events that involve “extreme” groups—groups that are particularly militant, radical, or hateful—both on the left and on the right.

Like traditional media, social media can provide citizens with politically relevant information, increasing their willingness to protest for a given cause. In addition, social media can facilitate coordination through horizontal interaction between users, alleviating collective-action problems (Olson, 1965). These coordination effects can stem from reduced costs of acquiring and exchanging logistical and tactical information on protest events (Little, 2016; Enikolopov, Makarin, & Petrova, 2020), from updating beliefs about how many others are willing to protest (Edmond, 2013; Barbera et al., 2020; González, 2020; Passarelli & Tabellini, 2017) or by altering how one can project their social image via protest participation (Enikolopov, Makarin, Petrova, & Polishchuk, 2020; Cantoni et al., 2019).

Social media may disproportionately fuel extremist mobilization through various channels. Prominent social commentators and scientists have long warned that through fierce competition for users’ attention, social media platforms create informational environments that trigger fear, anger and stoke public outrage (CBS, 2022; Rose-Stockwell, 2023). These warnings contend that algorithmic content recommendations designed to maximize engagement may disproportionately suggest content which triggers emotional responses, or more radical versions of whatever content the user was already consuming (Tufekci, 2018; Rose-Stockwell, 2017). Social psychologists have also argued that the way social media turns much communication into public performance, evaluated by the public in forms of likes and other metrics, may tilt incentives towards expressing more anger, aggression and radical positions than one would in other forms of communication (Haidt & Rose-Stockwell, 2019).¹ This constant push towards more radical content and exposure to aggressive online communication may drive partisan hostility, persuade individuals towards taking more radical positions and thus disproportionately fuel extreme protest movements.²

Even if no radicalization process occurs, social media may be a more powerful coordination tool among those already radicalized. First, as extreme voices diffuse more widely (Barberá et al., 2015; Brady et al., 2017), social media could be especially conducive to connecting extremists to like-minded individuals, thus facilitating tactical coordination among extremists disproportionately. Second, the “outrage machine” may alter people’s beliefs about the popularity of extreme views. If individuals’ decisions to participate in protests are strategic complements, as often is theorized in the literature (Edmond, 2013; Passarelli & Tabellini, 2017; Barbera et al., 2020), this may increase extremists’ willingness to act upon their opinions in the form of public demonstration.³

¹Empirical investigations have shown that exposure to hate speech online is prevalent and Twitter users who post about politics tend to have more extreme ideological preferences than the general population (Oksanen et al., 2014; Barberá et al., 2015). Evidence suggests that tweets using more moral or emotional language—plausibly correlated with the intensity or extremity of opinion—diffuse more intensely on Twitter (Brady et al., 2017). Finally, Beknazar-Yuzbashev et al. (2023) show in a randomized experiment that hiding “toxic” text content from users on Facebook and Twitter reduces content consumption, implying that the toxic content indeed boosts engagement.

²It is worth noting that the empirical literature on the consequences of social media in terms of political polarization and partisan hostility remains inconclusive. For example, Boxell et al. (2017) find that recent growth in political polarization in the United States is driven by demographic groups that are the least likely to use the Internet and social media and Barberá (2015) finds that in Germany, Spain and the United States, most social media users have ideologically diverse networks online and that exposure to this diversity increases political moderation. On the other hand, Melnikov (2023) finds that access to 3G internet contributed to increased political polarization between Democratic and Republican voters, Lelkes et al. (2017) find that broadband internet availability increases partisan hostility and segregation in the consumption of partisan media, and Allcott et al. (2020) find that deactivation of Facebook accounts reduced political polarization of the account owner. See Zhuravskaya et al. (2020) and Aridor et al. (2024) for a review of the literature.

³The idea of strategic complementarity in protest participation assumes that the cost of participation is lower when the protest is larger (e.g., the likelihood of being arrested may be lower in a larger crowd) or that the likelihood of “success” (e.g., the extent to which protesters can influence policy or the attitudes of the general population) increases with size. Evidence of strategic complementarity in protest

Following [Müller & Schwarz \(2023\)](#) and [Fujiwara et al. \(2023\)](#), this paper relies on a plausibly exogenous shock to Twitter’s early adoption for identification: The South by Southwest festival (SXSW) in Austin, Texas in March 2007. Promotion of Twitter at the festival sparked its popularity among attendees and, through network effects, in their home counties. The promotion was so successful that county-level festival attendance, proxied by the number of current Twitter users who created their Twitter account in March 2007 and follow the official account of SXSW, is predictive of contemporary county-level Twitter penetration. Further, counties with SXSW Twitter followers who created their accounts in 2006 and follow SXSW, and counties with SXSW Twitter followers who created their accounts in March 2007 are statistically similar in terms of observable characteristics, while only the latter Twitter users are predictive of contemporary Twitter penetration. This paper uses the variation in Twitter penetration predicted by SXSW followers who created their accounts in March 2007, while controlling for interest in the festival proxied by the SXSW followers who created their accounts in 2006, in an instrumental variable framework.

Using the Armed Conflict Location and Event data (ACLED), aggregated at the county-level, I classify protest events based on whether “extreme” groups are involved. These groups are defined by their willingness to use violence, militant tactics, or radically alter current political institutions to achieve their goals, or by being outright hate groups. I also classify protest events depending on the topic, political leaning, and aggressive behaviour exhibited during the event. Extremity is defined by the methods a group is willing to use to achieve its goals, not by the goals themselves—distinguishing this dimension from the classification of protest topics

In accordance with the literature (e.g., [Enikolopov, Makarin, & Petrova, 2020](#)), I find that Twitter penetration has substantially facilitated protest activity, but has also disproportionately fueled protest events where “extreme” groups are involved, both for extreme left and extreme right groups. A standard deviation increase in the log number of Twitter users (residualized by population) implies a 169% increase in the frequency of protests where extreme groups are involved and a 55.2% increase in the frequency of other protests between January 2020 and 12th of November 2021. This translates into roughly one additional extreme and six additional non-extreme protests compared to the means. In counties with at least one protest event during the sample period, the same increase in Twitter penetration implies a 2 percentage point increase in the share of protests where extreme groups are involved, from an average share of 2.4%.

participation is found in some settings ([Manacorda & Tesei, 2020](#)) but not always. [Cantoni et al. \(2019\)](#) find experimental evidence of strategic substitutability in the context of Hong Kong’s ongoing anti-authoritarian movement. They provide suggestive evidence of three sources of strategic substitutability: Public good characteristics of protest participation, greater perceived likelihood of government crackdown in larger protests, and that subjects perceive participating in smaller protests as a stronger signal of their ideological “type.”

Twitter penetration also implies an increase in the relative frequency of protest events where protesters exhibit aggressive or violent behaviour. The effect of Twitter penetration on protest frequency does not vary substantially by the protest topic or by the political leaning of the protest events, implying that the disproportionate effects for extreme groups are not purely topic driven.

A potential threat to identification is that pre-Twitter protest activity correlates both with Twitter penetration and with the instrument, SXSW followers who created their accounts in March 2007. I argue that these correlations are not important drivers of the results through a series of identification checks: First, all main results are remarkably stable across alternative specifications without or with extensive controls for pre-Twitter protest activity, implying that they are not an important confounding factor. Second, I construct a matched sample where counties with SXSW March 2007 followers are compared to counties similar in terms of observables, but with no such followers. In this sample, all main results are replicated and the instrument only correlates with protest activity from 2012 onward, consistent with Twitter becoming a significant engine for protest mobilization as the platform’s popularity grew. Therefore, the result that Twitter penetration disproportionately fuels extreme protests is unlikely to be driven by unobservables related to pre-Twitter protest activity.

Using individual responses from the Cooperative Election Study, this paper finds that county-level Twitter penetration intensifies opinions on race and gender issues without increasing general political knowledge. These effects are only detectable from 2019 onward, three years after Twitter introduced an algorithmically curated timeline (Pierce, 2016). While the literature has highlighted how social media fuels protest activity by reducing the cost of coordination (e.g., Enikolopov, Makarin, & Petrova, 2020; Fergusson & Molina, 2021), this individual-level evidence suggests that Twitter may not only serve as a more powerful coordination tool in the hands of extreme groups, but that Twitter penetration could increase the appeal of such groups. However, since the analysis examines regional-level exposure and individual-level outcomes, it reflects the effects of Twitter use in the respondent’s county, not their individual use. Therefore, these results should be interpreted only as suggestive of this mechanism.

This paper contributes to the literature on the effects of media on social and political outcomes (Enikolopov et al., 2011; Gentzkow et al., 2011; Gentzkow, 2006; DellaVigna & Gentzkow, 2010; Campante et al., 2022), and especially the role of the Internet and social media (e.g., Falck et al., 2014; Guriev et al., 2021; Petrova et al., 2021; Bessone et al., 2022; Boken et al., 2023; Manacorda et al., 2023; Enikolopov et al., 2018, 2024; Jiménez Durán et al., 2024; Hatte et al., 2025), recently surveyed by Zhuravskaya et al. (2020) and Aridor et al. (2024). By examining specifically protests with the involvement of extreme groups on either end of the political spectrum, it contributes to understanding on the links between the internet and social media, and political polarization (Gentzkow & Shapiro,

2011; Halberstam & Knight, 2016; Boxell et al., 2017; Barberá, 2015; Yanagizawa-Drott et al., 2020; Melnikov, 2023; Lelkes et al., 2017). Further, as the extreme right groups classified in this paper are hate groups associated with white supremacy ideologies, it contributes to the literature on media's, in particular social media's, role in propagating violence and hateful sentiments (DellaVigna et al., 2014; Yanagizawa-Drott, 2014; Adena et al., 2015; Bursztyn et al., 2019; Müller & Schwarz, 2021, 2023; Jiménez Durán et al., 2024)

This paper is directly related to the growing literature on the ability of information and communication technologies and social media to facilitate protest activity. Amorim et al. (2022) find that broadband Internet availability increased the probability of an Occupy Movement protest event. Enikolopov, Makarin, & Petrova (2020) show that VKontakte (VK), the most popular social media application in Russia, facilitated protests during a protest wave in 2011 triggered by electoral fraud benefiting the incumbent party, United Russia. Casanueva et al. (2024) show that late adoption of Twitter during the coronavirus pandemic mobilized Black Lives Matter protesters in the United States, in counties that had not protested in support of the movement before. However, these papers focus on a single protest movement and thus cannot explore differential effects depending on the characteristics of movements. Closely related to this study, Boyer et al. (2024) show that street protests during the Yellow Vest movement in France sparked online communities around the movement, and that these online communities grew increasingly hostile and negative in their communication over time. This study thus complements their work by showing how such online radicalization can translate into off-line action. Other studies take a more aggregate approach. Mobile phone coverage has been shown to increase protest activity across Africa (Manacorda & Tesei, 2020) and globally (Christensen & Garfias, 2018). Qin et al. (2024) show that communication on Sina Weibo increased the geographical spread of protests and increased the probability of very large protest waves in China. Fergusson & Molina (2021) find that Facebook's release in a new language increases protest activity in countries where the language is commonly spoken.

The main contribution to the literature is the classification of protest events by the involvement of extreme groups, protest topic, and political leaning, and the direct analysis of how social media penetration affects protests across these dimensions. Further, compared to Christensen & Garfias (2018) and Fergusson & Molina (2021), who find that, globally, ICTs can limit violent conflict and government repression during protest events,⁴ this paper finds that Twitter penetration increases the occurrence of violent behaviour during protests in the United States.

⁴Consistent with the conjecture that ICTs can increase visibility of and thus deter violent action (Durante & Zhuravskaya, 2018).

2 Data and Background

2.1 Protests in the United States

The United States provide a good setting to test whether social media influences the composition of political protest. Protest activity in recent years has been high (ACLED, 2020c) and U.S. citizens engage in protests around a diverse set of issues. In 2020, three issues were particularly common causes of protest: a) protests associated with the Black Lives Matter (BLM) movement; b) protests related to the 2020 presidential elections; and c) protests related to the coronavirus pandemic (ACLED, 2020a). The ACLED data records numerous other causes for protests in the United States, ranging from laborers' rights and compensation to environmental and immigration issues.

The sample period was further characterized by increasing signs of political instability and aggression during protest events. Comparing 2020 to 2019, the start of ACLED data collection in the United States, ACLED reports that counter-protests became more frequent, government authorities increasingly engaged with and exhibited force against protesters, and armed non-state actors were increasingly visible in protest events, either participating in protests, engaging in "peacekeeping," or intimidating protesters (ACLED, 2020b).

2.1.1 Protest data and main variables

Data on protests come from the Armed Conflict Location and Event Data (ACLED) project (Raleigh et al., 2010). ACLED is an event-level dataset, where events are manually compiled from regional, national, and international media outlets deemed reliable, and reports from international institutions and NGOs. Protest events, the focus of this paper, are defined as all physical congregations of three or more people when they are directed against a political entity, government institution, policy, group or individual, tradition or event, businesses, or other private institutions. For each event, the data includes information on the associated groups and organizations, as well as a short qualitative description of the event,⁵ both of which I use to classify protest events. Unfortunately, the data does not include information on the number of participants in each protest. This paper uses data from January 2020 to November 12, 2021, the full time span of available data at the time of collection.

The main focus is to identify protest events where "extreme" groups are involved. I define a political group as "extreme" if one of the following conditions holds: a) The group is aggressive and willing to use violence or militant tactics to achieve its goals.

⁵An example of an event description is the following: "On 12 November 2021, a group of protesters gathered outside of Ascension Via Christi hospital in Wichita (Kansas) to protest against coronavirus vaccine mandates for health workers. Unvaccinated employees joined the protest after finishing their final shift before the mandate took effect that day."

For example, the Black Lives Matter movement is defined as mainstream, while “Detroit Will Breathe,” a political group also protesting against police brutality towards black lives but explicitly self-defining as militant, is classified as extreme. b) The group is “revolutionary,” in the sense that it aims to radically alter current political systems or institutions to achieve its goals (e.g., alter electoral institutions to facilitate environmental protection, alter property rights to protect economically vulnerable groups). c) The group subscribes to nationalist ideologies or is a hate group. The full list of political groups and organizations defined as extreme is found in Table A1 in the Appendix. Thus, a group is defined as “extreme” depending on the methods they are willing to use to achieve their goals, not depending on the goals themselves, which distinguishes the dimension of “extremity” from the classification of topics.

Next, events including a set of keywords in their description and events with associated actors corresponding to a particular topic are coded as such. This paper codes five topics covering the majority of protest events during the time period: protests related to a) abortion rights; b) public health measures related to the coronavirus pandemic; c) the post-presidential election unrest in 2020, i.e. the “Stop The Steal” movement and the “Count All Votes” counter-movement; d) the Black Lives Matter (BLM) movement and counter-movements; and e) a broader racial issues topic category. Note that the categories described are not mutually exclusive. One protest event can be included in e.g., both the “Covid related” category if it includes the covid-related keywords, and in the “BLM related” category if organizations and movements usually protesting against police brutality against black people are involved. Further, a protest event can be defined as extreme and e.g. “racial issues related”, if the extreme group concerns racial issues.

For the extreme protests and all topics except for the “Covid related” category, I classify events as broadly left- or right leaning. For example, pro-choice abortion-related protests would be defined as left-leaning, while pro-life protests would be defined as right-leaning. A full list of keywords, phrases, and associated actors used to code each of the topics is provided in Tables A2 to A6 in the Appendix.

Finally, each event has information on the level of violence or interference. This paper examines heterogeneous effects for a) peaceful protests, b) protests where protesters themselves are peaceful, but they are targeted with violence or there is an attempt to suppress the protest, c) demonstration events where protesters engage in disruptive or violent behavior, or d) demonstration events where violent protesters interact violently with counter-rioters, civilians, or armed groups without the *use* of lethal weapons.

Figure 1 shows the spatial distribution of the main outcome variables. The upper map shows the number of overall protest events (per capita) between January 2020 and November 2021. The lower map shows the share of protests where extreme groups are involved in counties where at least one such protest occurred. In total, 56% of counties had protest events over the period of analysis, and 11% had protests with extreme group

involvement. Figure A1 in the Appendix shows additional maps of the raw frequency of protests overall and extreme protests, as well as extreme protests per capita.

[Figure 1 around here]

Table 1 shows summary statistics for the protest classification. There are 34,185 protest events in the data over the sample period. Of these, 4.8% have extreme group involvement, 3% left-leaning extreme groups, 1.5% right-leaning extreme groups, and 0.2% involve both extreme left and extreme right groups, i.e., where there is a counter protest. Panel B shows that 52.1% of all protests are classified into at least one of the five topic categories explained above, and 47.9% remain unclassified. The largest share of protests fall into BLM-related or, more generally, race-related protests, 36.7% and 39.5% respectively. Extremist groups are involved in about 3-6% of protests in each topic category, except for presidential election unrest protests, where 23% have extreme group involvement. Panel C shows that about half of all protests could be classified along the left-right axis. Around 41% are classified as left-leaning, while 6% are right-leaning. 7% of left-leaning protests, and 20% of right-leaning protests, have extreme group involvement. Finally, Panel D shows that the vast majority (93.6%) of protests are peaceful, but extremist groups are more likely to be involved with protest events which are not fully peaceful.

[Table 1 around here]

Figures A2 to A5 in the Appendix show time series of the weekly number of protests in each of the protest categories from Table 1. In Figure A2 we see that non-extreme protest activity surged in the summer of 2020, which is the peak of the BLM protest wave, while protests where extreme groups are involved occur more evenly over time. Figure A3 shows that while BLM related protests dwarf any other protest topic, each topic is characterized by waves of protest activity, so that in a short period of time there are many protests in a given topic category. Protests involving extreme groups occur quite consistently over the time period as opposed to the waves of protests in each topic, further illustrating that extreme groups are partaking in protests related to any of these major topics of protest. Figure A4 shows that mainstream left protests, unsurprisingly, peak around the same time as the BLM movement, while mainstream right protests pick up steam in summer 2020 until the end of 2020. Extreme right groups protest the most at the same time period as mainstream right protests, while extreme left groups' protest involvement increases steadily from summer 2020 until summer 2021. Figure A5 shows that both protests where peaceful protesters are met with intervention or violence, or those where protesters themselves are violent, peak in summer 2020.

2.2 Twitter and the South by Southwest festival

Survey data from 2021 indicate that a large majority of Americans self-report having ever used a social media site, and that the share of American has remained relatively stable over half a decade. These data suggest that the most popular platforms are YouTube, Facebook, and Instagram, used by 81%, 69% and 40% of those who respond to survey questions. 23% of U.S. adults used Twitter (now X). As with most other social media platforms, Twitter was most popular among young, educated, and urban adults ([Pew Research Center, 2021](#)).

Although Twitter was founded in March 2006, it remained largely unknown until the South by Southwest festival (SXSW hereafter) in March 2007, where early adoption took off. At the festival, Twitter installed hallway screens and created an event-specific feature allowing users to easily create an account and broadcast posts. Those who created accounts through this feature were automatically made to follow “Twitter ambassadors”, festival attendees already on the platform ([Quora, 2011](#)). [Fujiwara et al. \(2023\)](#) and [Müller & Schwarz \(2023\)](#) thoroughly document the immediate growth in Twitter penetration after the festival in 2007, showing that this growth occurred primarily in the counties of festival-goers. Activity went from an average of 5,000 tweets per day in 2007 to 300,000 in 2008, gradually reaching 500 million tweets per day in 2019.

The SXSW festival itself is an ambitious project combining film, interactive media and music festivals with professional conferences focused on creative industries and technology, and takes place annually in mid-March in Austin, Texas (see the official homepage: <https://www.sxsw.com/about/>). The festival is wildly popular; in 2016, it attracted around 300,000 official attendees ([Theis, 2016](#)). By 2007, it was already drawing speakers, performers, and attendees from across the United States and beyond. Broadly speaking, the festival’s content combines a wide range of Western pop-culture entertainment with professional themes of technological innovation.⁶

2.2.1 Twitter and South by Southwest festival attendance data

Twitter penetration at the county level is proxied with a sample of 475 million geo-coded tweets collected by [Kinder-Kurlanda et al. \(2017\)](#). The tweets are from the continental United States, and were collected from June 1st to the end of November, in 2014 and 2015 through the Twitter Streaming API, which returns a 1% sample of daily tweets as long as it is called. Only tweets that can be geo-located are included in the sample. Twitter allows users to “geo-tag” their tweets, adding the latitude/longitude information

⁶For example, in 2007 the festival included performances from musicians across the United States and international acts such as Peter, Bjorn & John from Sweden, Amy Winehouse from the UK, and YB from South Korea. Judd Apatow’s popular comedy film “Knocked Up” premiered at the festival, and it included panel sessions on topics such as “Virtual Worlds and Virtual Humans” with more than 450 speakers from the media and information technology sector. The program for 2006 had similar themes, scope and size: <https://www.sxsw.com/about/history/>.

from which the tweet was posted. To get a measure of users per county, the user profiles for each tweet in the database are collected and the users are then assigned to the county from which they tweet most frequently.

As geo-tagging tweets is optional, the observed sample is self-selected. In fact, only about 1% of users chose to geo-locate their tweets. Further, during the data-collection period, Twitter introduced the option of tagging tweets with the name of the location, as opposed to the latitude/longitude pair. The data collection did not catch tweets sharing this location information, but this new option substantially reduced the popularity of “geo-tagging” tweets (Kinder-Kurlanda et al., 2017). However, Fujiwara et al. (2023) show that measures using these tweets closely resemble other measures of Twitter penetration, such as estimates of the number of users over time from Statista or survey estimates of county-level number of Twitter users, alleviating concerns about self-selection biases.

[Figure 2 around here]

South by Southwest festival attendance is proxied using data collected through the Twitter API. Information on each user following the official Twitter account of the SXSW festival (@SXSW) is webscraped and assigned to counties based on the location users report in their profile. The two crucial variables constructed with this data are: a) The county-level number of SXSW followers who created their Twitter account in March 2007, 2007 SXSW followers hereafter, and b) the number of SXSW followers that created their Twitter account in 2006 (the founding year of Twitter), 2006 SXSW followers hereafter. The number of SXSW followers who created accounts in March 2007 is taken as a proxy for county-level attendance at the festival, the excluded instrument, while those creating accounts in 2006 are used to control for general interest in the festival. A total of 855 SXSW followers who created their accounts in March 2007 could be located within the United States, into a total of 155 counties. A total of 209 followers created their accounts in 2006 were located, into a total of 67 counties. Table A7 shows the distribution of SXSW March 2007 followers, Figure 2 shows the spatial distribution of Twitter users per capita and of the SXSW followers across continental United States, while Figure A6 in the Appendix shows a map of the number of Twitter users per county.

2.3 Additional data sources

As ACLED data on protest events only dates back to 2020, this paper uses information on “pre-Twitter” protest activity from the GDELT project. GDELT collects information on political events, including protests, through algorithmic coding of digitized newspapers, news agencies and web-based news aggregators such as Google News. Compared to ACLED, GDELT data includes less detailed information and thus cannot be used to categorize protest events by extremity or violence levels. Therefore, this paper uses

GDELT data from 2000 onward to document overall protest activity in pre-Twitter and early-Twitter years. The automated data collection and coding of events in GDELT are known to lead to duplication of events (Wang et al., 2016). If Twitter penetration influences the intensity of reporting on a given event (Weidmann, 2016; Hatte et al., 2022; Cagé et al., 2024), it may exacerbate the issue and lead to more duplication of events in counties with high Twitter penetration. Therefore, this paper considers various de-duplication strategies for the GDELT data. As a main measure, this paper treats all protest events occurring in the same coordinates and on the same day as one protest. All major results are robust to other de-duplication strategies.

Data to measure political attitudes come from the Cooperative Election Study (CES), waves from 2006 to 2022 (Schaffner et al., 2023). Respondents were asked about their attitudes towards various statements on the police, racial, gender/sexuality or environmental issues. Intensity of opinion is measured using dummy variables equal to one when individuals select the most intense response option (e.g., “strongly agree” rather than “somewhat agree”). This paper utilizes all questions from the survey where intensity of opinion can be inferred, but the specific questions vary considerably between waves. Table A8 lists the questions and the years in which they are available. The focus is on the survey wave from 2020, as the main outcome variable is measured in 2020-2021 and the CES wave from 2020 includes many questions on attitudes towards the police, and race and gender issues. The 2020 wave included 61,000 respondents and was sampled to represent the general adult population from each state.

Various demographic, geographical, socio-economic, crime, media consumption, and voting result variables are included as controls. Most of these variables are from 2009 and 2010, which can be considered a “pre-treatment” period as Twitter did not gain immense popularity until later.⁷ Demographic information comes from the 2010 U.S. Census. Geographical variables such as the distance between Austin, Texas (the location of the SXSW festival) and the centroid of each county are calculated using the U.S. Census Tigerline 2020 shapefile. Estimates of poverty rates and median household income are from the 2009 U.S. Census Bureau’s SAIPE program. Estimates of educational attainment in 2010 come from the American Community Survey, and employment statistics in 2009 from the Bureau of Labor Statistics. Crime controls come from the FBI’s Uniform Crime Reporting (UCR) program, and media consumption controls from SimplyAnalytics, both computed by Müller & Schwarz (2023). Data on county-level presidential election results in 2000 (George W. Bush), 2004 (George W. Bush), and 2008 (John McCain) are from the MIT Election Lab. Table A9 in the Appendix describes the full set of controls.

⁷In 2010, worldwide Twitter users were around 30-50 million, about 10-15% of the 300 million users estimated in 2015. Since 2015, the growth of Twitter users has remained relatively modest, with between 300 and 350 million users until 2019 (Statista, 2019).

3 Empirical Strategy

If Twitter were an important facilitator of protests, we would expect to see greater protest activity in areas where Twitter penetration is higher. The relationship of interest is:

$$Y_c = \beta_0 + \beta_1 \text{Log}(\text{TwitterUsers}_c) + \beta_2 \text{Log}(1 + \text{SXSW}_c^{2006}) + \mathbf{X}'_c \boldsymbol{\beta} + \epsilon_c \quad (1)$$

Where Y_c represents a protest variable in county c (e.g., the share of protests involving extreme groups), TwitterUsers_c represents a measure of Twitter penetration, here the natural logarithm of unique Twitter users in the [Kinder-Kurlanda et al. \(2017\)](#) data set, in county c , $\mathbf{X}'_c$ is a vector of covariates including pre-Twitter protest activity, β 's are coefficients to be estimated and ϵ_c is the error term. When looking at the frequency of protests, we would expect β_1 to be positive, representing the positive effect of Twitter penetration on protest activity.

Twitter penetration may be endogenous to protest activity for various reasons, giving alternative interpretations to β_1 . First, unobserved county characteristics might drive both Twitter adoption and protest activity. As an example, as Twitter allows users to follow politicians and express their political opinions, differences in inherent interest in politics might drive county-level differences in Twitter adoption and in protest activity. Second, as social media such as Twitter facilitate protest event coordination, one might suspect that high protest activity drives individuals to join Twitter, to ease coordination for protests they would have attended even without the existence of Twitter.

To circumvent these concerns, identification relies on the plausibly exogenous shock to early Twitter adoption in the United States connected to the 2007 SXSW festival, inspired by [Fujiwara et al. \(2023\)](#) and [Müller & Schwarz \(2023\)](#). As discussed above, the 2007 conferences provided a tipping point in the rise of Twitter's popularity. Consistent with the literature on path dependence in technology adoption ([Arrow, 2000](#); [Liebowitz & Margolis, 1995](#)) and following [Fujiwara et al. \(2023\)](#) and [Müller & Schwarz \(2023\)](#), a persistent effect of festival attendance on the spatial distribution of Twitter penetration is found. This setting lends itself well to an instrumental variable strategy where county-level Twitter penetration is instrumented using county-level attendance at SXSW in 2007 and subsequent adoption, measured by SXSW Twitter account followers who created their account around the time of the festival.

A crucial identification concern is that counties with festival-goers or general interest in the festival might be systematically different from other counties, in ways that correlate with protest activity. Information on the exact timing of Twitter adoption by SXSW followers helps alleviate these concerns. Consistent with prior evidence, SXSW followers that created their accounts in March 2007, presumably after attending the festival, are robustly and persistently predictive of the spatial distribution of Twitter penetration,

while the county-level number of SXSW followers who created their accounts in 2006 are not predictive of Twitter penetration. Further, counties with SXSW March 2007 followers are statistically similar to counties with SXSW 2006 followers as shown in Table A10. This allows to disentangle the effect of general interest in the SXSW festival and early adoption of Twitter from the effect of current Twitter penetration. By including the number of SXSW followers who created their account prior to the festival (in 2006), interest in SXSW and early interest in Twitter is effectively controlled for. Moreover, this provides a placebo test by testing whether inherent interest in the festival has an effect on protest activity. It turns out that throughout the analysis, SXSW followers who created their accounts in 2006 rarely correlate with protest outcomes—in the few cases where they correlate to protest outcomes, the correlation is of the opposite sign compared to 2007 followers, i.e., negative.

All main specifications control for pre-Twitter protest activity, ensuring that results are not driven by time-invariant unobserved determinants of protest activity. Unsurprisingly, protest activity between 2000 and 2007 correlates with protest activity in 2020–2021. However, protest activity between 2000 and 2007 does not correlate with the share of protests where extreme groups are involved, the main outcome of this study, and all main coefficients of interest remain remarkably stable across specifications with or without controls for pre-Twitter protest activity. This implies that the relationship between the instrument and the outcomes of interest is unrelated to time-invariant determinants of protest activity.

Formally, I estimate the following equations:

$$\begin{aligned} \text{Log}(\text{TwitterUsers}_c) = & \alpha_0 + \alpha_1 \text{Log}(1 + \text{SXSW}_c^{\text{March2007}}) + \alpha_2 \text{Log}(1 + \text{SXSW}_c^{2006}) \\ & + \mathbf{X}'_c \boldsymbol{\alpha} + v_c \end{aligned} \quad (2)$$

$$\begin{aligned} Y_c = & \gamma_0 + \gamma_1 \text{Log}(1 + \text{SXSW}_c^{\text{March2007}}) + \gamma_2 \text{Log}(1 + \text{SXSW}_c^{2006}) \\ & + \mathbf{X}'_c \boldsymbol{\gamma} + \mu_c \end{aligned} \quad (3)$$

Where equation (2) represents the first stage and equation (3) the reduced-form estimation.⁸ $\text{SXSW}^{\text{March2007}}$ represents the number of 2007 SXSW followers and SXSW^{2006} represents the number of 2006 SXSW followers in county c . α 's and γ 's are coefficients to be estimated, and u_c and μ_c are error terms. Other notation remains the same as for equation (1). The accompanying second stage equation can be written as:

$$Y_c = \lambda_0 + \lambda_1 \text{Log}(\hat{\text{TwitterUsers}}_c) + \lambda_2 \text{Log}(1 + \text{SXSW}_c^{2006}) + \mathbf{X}'_c \boldsymbol{\lambda} + \phi_c \quad (4)$$

⁸In the Appendix, all main analyses are repeated using the inverse hyperbolic sine of SXSW followers instead of the logarithm of one plus SXSW followers in the first and second stage. Results are robust to this alternative specification.

Where $\hat{Log}(TwitterUsers_c)$ refers to the predicted value of the county-level measure of Twitter penetration from equation (2), λ 's are coefficients to be estimated, ϕ is the error term, and other notation is the same as before. Here, λ_1 represents the Local Average Treatment Effect (LATE), the effect of increased Twitter penetration for counties where Twitter adoption would be strongly affected by being the home of an SXSW festival-goer in 2007. Intuitively, early adoption of Twitter among festival attendees is likely to increase Twitter penetration in attendees' home counties if the SXSW attendees have large offline social networks in their home counties and their networks are receptive to early adoption of a new social media platform. This is likely the case in counties with a non-negligible population of "tech savvy" individuals interested in adopting a new social media platform, similar to SXSW enthusiasts in terms of socio-demographics or cultural interest. However, as discussed above, the festival offers a wide variety of popular culture attractions and should attract the interest of diverse groups of people. The LATE should thus not be excessively particular, although one should not overgeneralize the results.

The main focus of this paper is the effect of Twitter penetration on a given protest category, relative to other protests. Therefore, the natural outcome of interest is the share of protests that fall in a given category (e.g., the share of protests with extreme group involvement) and the sample of interest is counties where at least one protest occurred over the period. To rule out the possibility that the share of one protest category increases due to a reduction in protests of another category, this paper also reports effects of Twitter penetration on the number of protests in each category for the full sample of continental United States counties.

As the relationship between Twitter penetration and the frequency of protest events is likely proportional, and a substantial proportion of counties have zero events, this paper uses "log-like" transformations for the outcome variables. "Log-like" transformations of outcome variables such as the inverse hyperbolic sine or $\log(1 + Y)$ have come under scrutiny recently, in particular as treatment effect estimates may be sensitive to arbitrary choices regarding the unit of measurement (Aihounton & Henningsen, 2021; Mullahy & Norton, 2022; Chen & Roth, 2023). The fundamental problem is that percentage changes from zero are not defined, and the unit of measurement implicitly determines the weight on the extensive margin responses relative to the intensive margin. Following the recommendations of Chen & Roth (2023), this paper makes this weighting explicit. In the main text, results using a log transformation of the outcome variable are reported, weighting an increase from 0 to 1 as a 150% increase (as a change from 1 to 2 is a 100% increase, a change from 2 to 3 a 50% increase, and so on). I also report additional results in the Appendix with alternative functional forms and weighting schemes. Note that the core argument of this paper concerns the effects of Twitter penetration on one category of protest relative to others. Therefore, the effect size for one protest category is not of utmost importance, as long as the effect sizes can be compared across protest categories.

For individual-level survey data, the empirical strategy follows the county-level analysis closely. I estimate the following equations:

$$\begin{aligned} \text{Log}(\text{TwitterUsers}_{ic}) &= \theta_0 + \theta_1 \text{Log}(1 + \text{SXSW}_c^{\text{March2007}}) + \theta_2 \text{Log}(1 + \text{SXSW}_c^{2006}) \\ &\quad + \mathbf{X}'_{ic} \boldsymbol{\theta} + \eta_{ic} \end{aligned} \quad (5)$$

$$\begin{aligned} \text{Attitude}_{ic} &= \pi_0 + \pi_1 \text{Log}(\hat{\text{TwitterUsers}}_c) + \pi_2 \text{Log}(1 + \text{SXSW}_c^{2006}) \\ &\quad + \mathbf{X}'_{ic} \boldsymbol{\pi} + \psi_{ic} \end{aligned} \quad (6)$$

Where equation (5) represents the first stage and equation (6) the second stage. $\text{Log}(\text{TwitterUsers}_{ic})$ is the natural logarithm of unique Twitter users in individual i 's county c , $\text{SXSW}_c^{\text{March2007}}$ and SXSW_c^{2006} are the number of SXSW followers in March 2007 and 2006 respectively, in county c , and $\mathbf{X}'_{ic}$ is a vector of county-level and individual-level covariates. $\text{Log}(\hat{\text{TwitterUsers}}_c)$ is the predicted value of $\text{Log}(\text{TwitterUsers}_{ic})$ from the first stage and Attitude_{ic} is the response to survey questions for individual i residing in county c . θ 's and π 's are coefficients to be estimated and η_{ic} and ψ_{ic} are error terms.

The individual-level analyses study the effect of county-level Twitter penetration on individual-level attitudes. Thus, they cannot tell us directly how individuals' usage of Twitter influences their attitudes, but rather whether Twitter popularity in one's local environment influences attitudes.

4 Main results

4.1 The first stage

Table 2 shows regression results for the first-stage relationship between the county-level number of new SXSW followers in March 2007 and Twitter penetration, measured as the natural logarithm of the number of Twitter users from the sample of tweets from 2014-2015. It shows that, as discussed above, the SXSW 2007 followers correlate quite robustly with the logarithm of Twitter users, while the SXSW 2006 followers do not. Quantitatively, a standard deviation increase in the natural logarithm of SXSW March 2007 followers plus one (0.319) implies a 12% increase in the number of Twitter users, given the estimate of 0.381 in column (8).

[Table 2 around here]

The robust F-statistic is well above the traditional rule-of-thumb threshold of 10 and takes values around 35-40 in the most stringent specifications. As the model is just-identified (i.e., does not include more excluded instruments than instrumented variables of interest), the robust F-statistic (sometimes also called the Kleibergen Paap) corresponds

to the effective F-statistic developed by [Olea & Pflueger \(2013\)](#). The F-statistics are thus above the threshold for a potential bias of 10% and a significance level of 5%, which is 23. Further, [Andrews et al. \(2019\)](#) show that for higher-than-10 F-statistics the weak-instrument problem is unlikely to affect the validity of conventional t-statistics in the case of clustered standard errors. This alleviates concerns that the results presented while controlling properly for county characteristics suffer from a weak first stage. Table [A11](#) in the Appendix reports the first stage using the inverse hyperbolic sine of SXSW 2007 and 2006 followers instead of the logarithm of one plus SXSW followers, where results remain robust: SXSW 2007 followers correlate robustly with the logarithm of Twitter users while the SXSW 2006 followers do not, and F-statistics are around 40 in the most stringent specifications.

[Figure 3 around here]

Figure 3 visualizes the first stage of the main specification via binned scatter plots. Panel (a) presents a binned scatter plot of the relationship between the natural logarithm of Twitter users and the number of SXSW March 2007 followers and panel (b) plots the same relationship between the natural logarithm of Twitter users and the SXSW 2006 followers. The variables on the x-axis and the y-axis are residualized by partialing out all covariates used in column (8) of Table 2, as well as the SXSW follower variable not on the x-axis. The data are binned into 20 equal-interval bins on the x-axis and the dots represent average values of both variables within each bin. The red line represents an OLS regression line using the unbinned data. We see a clear positive relationship between the natural logarithm of (one plus) SXSW March 2007 followers and Twitter penetration in panel (a), while no such relationship exists between the 2006 SXSW followers and the natural logarithm of Twitter users in panel (b).

4.2 Extreme protests

Table 3 presents estimations from the Two-Stage Least Squares (2SLS) model. Panel A shows the effect on the number of protests with extreme group involvement, and Panel B shows the effect on the number of protests without the involvement of extreme groups. The table presents results using the log number of protest events as the outcome variable, treating a change from 0 to 1 as a 150% increase. Table [A12](#) in the Appendix shows results from a 2SLS model where the outcome is the overall number of log protests irrespective of extreme group involvement.

In both panels of Table 3, across all specifications, the estimates indicate a positive effect of Twitter penetration on the number of protests, statistically significant at the 1% level. The coefficient estimate of the effect of SXSW 2006 followers is small, negative, and statistically indistinguishable from zero. After adding basic controls for population,

demographics and geography, coefficients remain mostly stable. Finally, we see that pre-Twitter GDELT protests correlate with non-extreme ACLED protests in 2020-2021, but not with extreme protests.

In each specification, the estimated effect of Twitter penetration is substantially larger for extreme protests than for non-extreme protests. To compare the coefficients explicitly, the bottom of the table reports p-values from a test of equality of the reduced form effect (the full reduced form results are reported in Table A13 in the Appendix) on extreme protests and non-extreme protests. These tests are performed by simultaneously estimating the reduced form equation for both protest types using seemingly unrelated regressions (SUR). In all specifications, the reduced form effect for extreme protests is statistically significantly larger than for non-extreme protests at least at the 5% level. Table A14 in the Appendix reports the same reduced form effects using the inverse hyperbolic sine of SXSX 2007 as the excluded instrument, and results remain unchanged.

[Table 3 around here]

The estimated effects are large. In the most stringent specification, a percentage increase in Twitter penetration implies a 0.8% increase in non-extreme protests and a 1.8% increase in extreme protests. To put these magnitudes into context: After residualizing the number of Twitter users using population deciles (as population is highly predictive of the number of Twitter users), the standard deviation is 0.55 log points, implying a 169% increase in extremist protests and a 55.2% increase in non-extremist protests.⁹ From the means of 0.524 extreme protests and 10.48 non-extreme protests per county, a standard deviation increase in residualized Twitter penetration thus implies roughly 0.9 additional protests where extreme groups are involved, and 5.8 additional non-extreme protests.

Further, Appendix Table A15 shows second stage results using an indicator for having at least one protest event as the outcome variable, and alternative weighting of the extensive margin. Twitter penetration increases the probability that an extreme protest occurs, but not the probability of a non-extreme protest. Using the same quantification as before: A standard deviation increase in residualized Twitter penetration (0.55 log points) implies a 12.7 percentage point increase in the probability of an extreme protest, while 11% of counties experience such a protest. In columns (2)-(4) the outcome is the log number of protests, where the extensive margin is valued as a 0%, 100%, or 200% increase, respectively. Irrespective of the weighting scheme, the effect on extreme protests is larger than the effect on non-extreme protests, and a simultaneous estimation of the reduced form effects yields a statistically significant difference between the two.

To further explore the heterogeneous effect depending on extreme group involvement, and to alleviate concerns that the heterogeneity is driven by extensive margin weighting

⁹ $e^{\beta \cdot 0.55} - 1$

choices, Table 4 reports on the same 2SLS model, estimated on a sample restricted to counties with at least one protest event, and the outcome variable is the share of protests involving extreme groups. Panel A reports on the second stage results, Panel B shows the reduced-form, and Panel C shows the OLS correlation between Twitter users and the share of extreme protests.

[Table 4 around here]

In Panel A, the estimated effect of Twitter users on the share of protests where extreme groups are involved is positive and significant at the 1% level. The first stage in this sample (reported in Table A16 in the Appendix) is strong, like in the full sample of counties. Magnitudes are large; a standard deviation increase in Twitter penetration (residualized by population deciles, 0.55 log points), implies a 2 percentage point increase in the share of protests where extreme groups are involved, from an average share of 2.2%. Similarly, in Panel B, we see that the reduced form effect of SXSW 2007 followers on the share of protests with extreme group involvement is positive, significant at the 1% level, and stable across specifications. In Panel C, we see that Twitter users, uninstrumented for, do not correlate with the share of extreme protests, nor do the SXSW 2006 followers nor pre-Twitter protest activity.

Table A17 in the Appendix replicates the second stage results on the share of protests with extreme group involvement, using the inverse hyperbolic sine of SXSW followers as the excluded instrument, where results remain stable. Tables A18 and A19 in the Appendix show that the results are remarkably stable across alternative specifications without or with extensive controls for pre-Twitter protest activity, and using alternative de-duplication strategies for GDELT pre-Twitter protests.

Figure 4 visualizes the reduced form corresponding to the regression in column (8) of Panel B in Table 4, using binned scatter plots. The y-axis shows the share of protests involving extreme groups and in panel (a), the x-axis represents the natural logarithm of (one plus) SXSW 2007 followers while in panel (b) the x-axis represents the natural logarithm of (one plus) SXSW 2006 followers. The variables on the x-axis and the y-axis are residualized by partialing out all covariates used in column (8) of Table 4, and the SXSW follower variable not represented in the x-axis of each panel. The data are binned into 15 equal-interval bins on the x-axis and the dots represent average values of both variables within each bin. The red line represents an OLS regression line using the unbinned data. Again, we see a positive relationship between the natural logarithm of (one plus) SXSW March 2007 followers and the share of extreme protests in panel (a). In Panel (b) there is clearly no positive relationship between SXSW 2006 followers and the share of extreme protests. In fact, the relationship is negative.

[Figure 4 around here]

Taken together, these results can be taken as evidence that Twitter may facilitate protest activity. These correlations are unlikely to be driven by selection of SXSW enthusiasts and attendees into areas with high protest activity as the number of SXSW followers who created their accounts in 2006 is controlled for, and they also do not correlate with protest activity. However, these analyses also show that Twitter penetration has, quite robustly, a disproportionate effect on protests where extreme groups are involved, relative to other protests.

As pointed out by Müller & Schwarz (2023), some uncertainty remains on the extent to which the causal effects estimated can be exclusively attributed to Twitter, as opposed to other social media platforms. While the SXSW festival in 2007 likely only influenced Twitter adoption initially, it is possible that spillovers in the adoption of other social media platforms existed in the medium-run. Therefore, the estimated effect might not be a pure “Twitter effect.” Such spillovers should however not change the core argument of the paper, i.e., that these estimates and the ones presented below refer to the causal effect of social media diffusion on protest frequency and the composition of political protests.

4.3 Identification checks: Matched sample

A potential threat to identification is the fact that pre-Twitter protest activity correlates both with Twitter penetration and with the instrument (see column (8) in Table 2 and Table A20). This section argues that these correlations are not important drivers of the main results.

First, note that Boyer et al. (2024) find that offline protests spark online communities on social media around the topic of the protest. A plausible interpretation of the correlation between Twitter penetration and pre-Twitter protest activity is that indeed protest activity during the advent of Twitter causally led participants to become users on the platform. This endogeneity would be dealt with by the instrumental variable strategy, as we only use variation in Twitter penetration stemming from festival attendance. Second, all the main results are stable across alternative specifications without or with extensive controls for pre-Twitter protest activity.

To further alleviate this concern, this section reports on a matching exercise. A propensity score for having SXSW March 2007 followers (“treatment”) is computed via a probit model estimated on the sample of counties with at least one protest, using pre-Twitter protest activity, SXSW 2006 followers, state fixed effects, and the full set of controls. Each county with an SXSW March 2007 follower is then matched to the county with the closest propensity score, which has no such followers, to construct a “matched sample.” Only match-pairs where both counties are within common support

(from 0.002-0.93, see Figure A7 in the Appendix for the distribution of propensity scores) are included in the sample. The matched sample contains 120 treatment counties and 67 unique control counties, as a single county can appear as a control for more than one treatment county. There are no statistically detectable differences in observables between treated and control counties in the matched sample, as reported on in Table A21 in the Appendix.

Table 5 presents 2SLS estimations from the matched sample, weighting control observations appropriately, using the log number of (one plus) SXSW March 2007 followers as the excluded instrument, as before. Columns (1) and (2) show first stages with or without an extensive set of controls, and columns (3)-(8) show second stages. The first stage is strong, the coefficient to the instrument is precisely estimated, of similar magnitude as in the original sample, with a cluster robust F-statistic of around 20. The point estimate for SXSW 2006 followers is larger than in the full sample, but this correlation is statistically insignificant and not strong: The cluster-robust F-statistic for removing the SXSW 2006 variable is 1.37. In columns (3)-(6), we see that the main results are replicated in this subsample, and point estimates are very similar. A percentage increase in Twitter penetration implies 0.8% increase in overall protest activity, a similar magnitude to in the main sample (reported in Table A12 in the Appendix). A standard deviation increase in Twitter penetration (residualized by population deciles, 0.55 log points) implies a significant increase in the share of protests involving extreme groups by 1.5 percentage points, from an average share of 4%. Finally, columns (7)-(8) perform the placebo test and confirm that the instrument does not correlate with pre-Twitter protest activity in the matched sample.

[Table 5 around here]

Figure 5 shows the dynamic effect of Twitter penetration on GDELT protests in the matched sample. For each year, the plot presents a cross-sectional regression with the same right-hand-side variables as column (8) of Table 5, with the yearly log number of GDELT protests as the outcome. Events at the same coordinates and date are treated as one. In panel (a), we see that the effect of Twitter penetration is indistinguishable from zero from 2007-2011, but from 2012 onward we see a positive effect on protest frequency. Panel (b) confirms that SXSW 2006 followers remain uncorrelated with protest activity in the matched sample. Figures A8-A10 show the same plots using alternative de-duplication strategies. Figure A8 performs no de-duplication. In Figure A9 all protest events of the same type occurring in the same coordinates and on the same date are considered as one event, and in Figure A10 all protest events occurring in the same county on the same date are counted as one event. The results are robust to these different de-duplication strategies: Twitter penetration has no detectable effect on pre-Twitter protest frequency,

but generally starts to have a statistically significant and positive effect on protest activity from 2012 onward. On the contrary, the 2006 SXSW followers do not correlate with protest activity at any point during the sample period. This suggests that the results are not driven by Twitter penetration influencing the reporting of protest events in a way that correlates with GDELT’s failure to de-duplicate events.

[Figure 5 around here]

To sum up, this section shows that in a matched sample where pre-Twitter protest activity does not systematically correlate with Twitter penetration or the SXSW instrument in concerning ways, Twitter penetration still implies an increase in protest activity overall and an increase in the share of protests where extreme groups are involved. Therefore, the main results of this paper are unlikely to be driven by unobservables related to pre-Twitter protest activity.

5 Alternative compositional changes

5.1 Topic composition

Twitter’s content may be skewed towards particular political topics and disproportionately facilitate protests concerning those topics. This could drive the extremity results if extreme groups are more likely to attend protests concerning those topics.¹⁰ This section examines this possibility.

Table 6 estimates the effect of Twitter penetration on the frequency of protests that fall under the topic categories explained in Section 2.1. Panel A shows the 2SLS estimates with the full set of controls, and the log number of protests in a given topic category as the outcome, where an increase from 0 to 1 is valued as a 150% increase. Panel A is replicated in Table A22 in the Appendix using alternative weighting of the extensive margin, where results remain comparable. Panel B restricts the sample to counties where at least one protest event occurred during the sample period, and runs the same specification on the share of protests that fit the indicated topic category.

Panel A shows that Twitter penetration has a positive effect on protests concerning all topics. The effects are precisely estimated for all topic categories and do not exhibit much heterogeneity. Point estimates are always considerably lower than the effect for extreme protests (an elasticity of 1.8), and simultaneous estimation of the reduced form effect on protests in each topic and the effect on extreme protests yields a statistically

¹⁰Suppose that Twitter penetration would disproportionately fuel protest activity related to the BLM movement, for or against. If extreme groups are more likely to participate in (pro or anti) BLM protests, Twitter penetration would imply an increase in the share of protests involving extreme groups.

significantly lower point estimate for each of the topic categories.

[Table 6 around here]

Panel B estimates the effect of Twitter penetration on the share of all protests falling into a given topic category, for counties where at least one protest occurs over the sample period. First, we cannot reject a null effect for the share of presidential election, abortion rights, BLM, or racial issues related protests at any reasonable significance threshold. That is, we cannot rule out that protests in these categories increase proportionately to protests overall. Twitter penetration implies a decrease in the share of protests that are coronavirus related. To give a sense of magnitude, a standard deviation increase in residualized Twitter penetration (0.55 log points) leads to a decrease in the share of coronavirus-related protests by about 2.8 percentage points, from an average of 9.6%.

These results indicate that while Twitter penetration disproportionately fuels protests where extreme groups are involved, effects do not vary substantially for various salient protest topics during the sample period. Therefore, it is unlikely that the large effect for extreme protests is purely topic driven.

5.2 Political leaning

Table 7 studies whether the effect of Twitter on protests depends on the political leaning of the protest event. That is, depending on whether the topic of protest, or the groups involved in the protest align with the political left or the right in the United States. Panel A shows effects on the log number of protests (with an increase from 0 to 1 valued as a 150% increase) in a given category. Panel B shows regressions on a sub-sample of counties where at least one protest occurred and the outcome variable is the share of all protests in the county which fall into a given protest category. In column (1) we look at “mainstream left leaning” protests: These are all left leaning protests that fall in any topic category from Table 6, in addition to protests associated with the Democratic party and a small set of additional left-leaning protest movements, where no extreme group was involved. Column (2), in contrast, shows effects on protests where an extreme left group was involved. Columns (3) and (4) show the same comparison for “Mainstream right leaning” protests versus protests where extreme right groups are involved. A full description of protest categorization is found in Tables A1 to A6. Table A23 in the Appendix reports estimates of the effects on the number and share of protests with extreme left or extreme right group involvement when Twitter users are instrumented for using the inverse hyperbolic sine of SXSW followers, where results remain comparable. As before, the same specifications as in Panel A are replicated using alternative transformations of the outcome variable and alternative weighting of the extensive margin in Table A24 in the Appendix. Those results are consistent with what is presented here.

In Panel A, we see that the point estimates for mainstream left or right protests are quite similar, and less than one standard error away from the average effect on all protests (an elasticity of 0.84 found in column (8) of Table A12). The effect sizes for extreme left and extreme right protests are larger, a percentage increase in Twitter penetration implies a 1.9% increase in extreme left protests and a 1.3% increase in extreme right protests. We can reject equality of the reduced form effects for extreme versus mainstream left from a simultaneous estimation at the 1% significance level, and reject equality of the reduced form effect for extreme v.s. mainstream right at the 10% significance level.

[Table 7 around here]

Panel B studies counties where at least one protest occurs. Increased Twitter penetration implies an increase in the share of protests involving extreme left groups, and also the share of protest involving extreme right groups, statistically significant at least at the 5% level. Again, to give a sense of magnitude, a standard deviation increase in the log number of Twitter users (residualized by population) implies around a 1.4 percentage point increase in the share of protests involving extreme left groups are involved and 0.9 points for extreme right groups, from averages of 0.9% and 1.3%, respectively.

For mainstream left and mainstream right leaning protests, we cannot rule out null effects on shares, that is, that the effect of Twitter penetration on these protest categories is proportional to the average effect. Further, Table A25 takes separately the topic categories except for Coronavirus related protests,¹¹ splits them up by political leaning and compares effects on e.g., left leaning presidential election related protests (the “Count every vote” movement) versus right leaning presidential election related protests (the “Stop the Steal” movement). No heterogeneity by political leaning is found within any of the salient topics studied.

While the conceptual definition of extreme groups remains the same irrespective of political leaning, the following pattern emerges: On the left, extreme groups can be classified into two broad categories. First, there are political groups who protest the same societal conditions as the broad Black Lives Matter movement (i.e., police brutality and discrimination against the black population), but are more aggressive and willing to use militant tactics than the mainstream movement, or envision revolutionary remedies to these problems. Second, there are various revolutionary political groups which are willing to use militant tactics or oppose current political systems or property rights institutions, in order to achieve their goals. These groups include, for example, communist and socialist parties, and radical environmentalist groups. Table A26 shows that the effects found for

¹¹The way Coronavirus-related protests are identified via keywords does not allow to distinguish protests calling for *stricter* public health measures versus those calling for more lax public health measures, sadly. Thus, it is not possible to compare left-leaning versus right-leaning coronavirus related protests

extreme left groups hold both for the “radical racial justice” groups, and for the “other extreme left” groups. On the right, extreme groups are unified by white supremacy ideologies. One can thus conceptualize the effects on these groups as social media’s effects on individuals’ willingness to publicly display their hateful ideologies. Prior research has found that social media can act as a propagating mechanism for hateful sentiments and hate crime (Bursztyn et al., 2019; Müller & Schwarz, 2021). In particular in the United States, this effect may be driven by hateful messages from influential opinion makers to the general public (Müller & Schwarz, 2023), thus influencing norms about what is socially acceptable (Bursztyn et al., 2020). As social image concerns likely influence protest participation decisions (Enikolopov, Makarin, Petrova, & Polishchuk, 2020; Cantoni et al., 2019), this potential shift of social norms may be a driving factor for the disproportionate effects found for white supremacy groups.

Taken together, these results confirm and strengthen the main finding: Twitter penetration facilitates protest mobilization, it does so irrespective of the topic of protest and the effect does not depend on political leaning. However, Twitter penetration disproportionately fuels protests where extreme groups are involved, both extreme groups to the left and to the right.

5.3 Violent v.s. peaceful protests

Table 8 analyzes whether Twitter plays a role in reported increases of hostility, violence, and repression during protest events. It examines whether there are heterogeneous effects depending on the nature of violent or disruptive behavior.

Panel A shows effects on the log number of protests (with an increase from 0 to 1 valued as a 150% increase) in a given category. Panel B shows regressions on a subsample of counties where at least one protest occurred, and the outcome variable is the share of all protests in the county that fall into a given protest category. Column (1) studies entirely peaceful protests, column (2) protests where outside groups or individuals interact non-peacefully with otherwise peaceful protesters, column (3) studies events where the protesters themselves are violent or exhibit disruptive behavior, and column (4) studies events where violent protesters interact violently with violent outside group, i.e. there are riots. Table A27 in the Appendix repeats the analysis in Panel A with alternative weighting of the extensive margin response, where results remain comparable.

Panel A shows that Twitter penetration increases both peaceful and non-peaceful protests significantly. Compared to the effect on peaceful protests, the effects for events where peaceful protesters are met with intervention or violence, and where protesters themselves are violent, are larger. Comparing the effect sizes explicitly by simultaneously estimating the reduced form effects reveals that both the effects for events where peaceful protesters are met with intervention and violence, and for events with violent protesters

are statistically significantly larger than the effect for peaceful protests. However, the effect for riots is much smaller than the effect for peaceful protests, and statistically significantly so.

[Table 8 around here]

In Panel B, we also see some indication that Twitter penetration increases the relative frequency of non-peaceful or aggressive behavior during protest events. Point estimates are rather imprecisely estimated, but imply that Twitter penetration increases the share of protests where protesters themselves are violent, significantly distinguishable from a null effect at the 10% significance level.

In sum, the evidence in Table 8 suggests that Twitter penetration increases the relative frequency of violence and aggression during protest events. There is strong evidence for an effect on the relative frequency of protest events where protesters themselves are violent, and indications that increased Twitter penetration implies that peaceful protesters are more frequently met with intervention or violence from outside groups.

6 Mechanisms

As discussed in the introduction, social media may fuel extremists' protests disproportionately through two mechanisms: It may be a source of radicalization or alternatively, social media may serve as a more powerful coordination tool for extremists. To gauge the likelihood of the former mechanism, this section studies the effect of county-level Twitter penetration on opinions and attitudes of individuals in those counties.

Table 9 studies the effect of county-level Twitter penetration on strong opinions concerning statements on race or gender issues among respondents in the 2020 Cooperative Election Study. Column (1) shows the first stage in these individual-level regressions, and columns (2)-(7) show second stages. The outcome variables are constructed from dummy variables equal to one if respondents hold a strong opinion on a particular statement. In column (2) the outcome variable is the sum of strong opinions on 5 statements concerning race issues, in column (3) the outcome is the sum of strong opinions on 7 statements concerning race and gender issues, and in column (4) the outcome is the sum of strong opinions on 10 statements concerning race, gender and partisanship. In columns (5)-(7) the outcome variables are the first principal component constructed from the same 5, 7, or 10 dummies. We see that for all these outcome variables, Twitter penetration implies an increase in holding strong opinions, significantly different from 0 at the 5% or 10% level.

[Table 9 around here]

Table A28 in the Appendix studies the impact of Twitter penetration on holding strong opinions on each individual statement on race or gender issues, where results are consistent. Table A29 verifies that the effect on strong opinions is not one directional, i.e., it does not represent a shift to the left or right. For each statement, two dummy variables are created: One for strongly liberal and one for strongly conservative opinions. We see statistically significant effects both in the conservative and liberal direction. Even though point estimates are often imprecisely estimated, they are positive for six or seven out of seven statements whether we look at strongly liberal or strongly conservative opinions. Overall, these results provide support for the idea that Twitter penetration in one’s local environment increases one’s intensity of opinion.

Figure 6 studies the dynamic effect on strong opinions. For each year from 2006-2022, the plot presents a cross-sectional regression, as in Table 9, where the outcome is the sum of statements on which respondents hold strong opinions. The questions asking respondents to report the intensity of their opinion changes considerably during the time period, but they broadly capture attitudes towards race issues, gender or sexuality issues, environmental issues, and ideology or partisanship. The full list of questions used is reported in Table A8 in the Appendix.

[Figure 6 around here]

Panel a) of Figure 6 shows that Twitter penetration had no effect on strong opinions in the “pre-treatment” period, and did not start to intensify opinions until 2019. In 2019, 2020 and 2022 we have consistently positive coefficients and similar effect sizes. The point estimate for 2021 is very close to zero, likely as in 2021 there were no questions related to race or environmental issues and only one question on gender issues (about the legality of abortions), while in 2019, 2020, and 2022 there are many questions on both race and gender/sexuality issues. In panel b), we see that the SXSW 2006 followers are not systematically correlated with strong opinions. Figure A11 in the Appendix repeats this exercise with the first principal component of strong opinions as the outcome variable; results are consistent with what is reported here.

Intense opinions can be a prerequisite to radicalization, but social media also facilitates information flows. Twitter’s effect on strong opinions could be interpreted as a sign of higher information. To explore the plausibility of this explanation for intensified opinion, Table A30 in the Appendix looks at Twitter’s effect on political knowledge in the 2020 wave of the Cooperative Election Study. The table reports null effects on knowing the party of one’s governor, senators, congressperson or the majority party in the house of representatives or the senate. Thus, the evidence does not indicate that the effect on strong opinions is driven by increased knowledge about politics.

Taken together, this analysis presents evidence consistent with the conjecture that

Twitter penetration in one’s local environment intensifies individuals’ opinions without increasing their general political knowledge. Intense opinions may increase individuals’ willingness to partake in protests associated with extremist groups. Thus, this individual-level analysis suggests that Twitter may not only be a more powerful coordination tool in the hands of extremists, but that it may also *radicalize* the population to some extent.

7 Conclusion

This paper investigates whether social media can affect the composition of political protests. Following Müller & Schwarz (2023); Fujiwara et al. (2023) and Casanueva et al. (2024), it leverages a plausibly exogenous shock in early adoption of Twitter stemming from county-level attendance at the South by Southwest festival in 2007 as an instrument for contemporary county-level Twitter penetration.

Consistent with prior research (e.g., Enikolopov, Makarin, & Petrova, 2020; Manacorda & Tesei, 2020), it finds that Twitter penetration can substantially facilitate protests, but disproportionately fuels protests involving extreme groups—groups that are particularly militant, radical, or hateful. This holds both for extreme left and extreme right groups. A standard deviation increase in Twitter penetration (residualized by population) implies a 169% increase in protests where extremist groups are involved and a 55.2% increase in protests without the involvement of extremist groups between January 2020 and November 2021. This amounts to roughly one additional extremist protest and 6 additional non-extremist protests, compared to the means. In a sample of counties where at least one protest occurred, the same increase in Twitter penetration increases the share of protests with extreme group involvement by 2 percentage points, from an average share of 2.2%.

In addition, this paper finds that the effect of Twitter penetration does not differ systematically between various topics of protest or by the political leaning of protests. However, Twitter penetration also increases the relative frequency of violent or aggressive behavior by protesters during protest events.

While social media’s ability to facilitate the coordination of collective action has been highlighted in the literature (e.g., Enikolopov, Makarin, & Petrova, 2020; Fergusson & Molina, 2021), this paper finds evidence from survey data suggesting that Twitter may not only be a more powerful coordination tool for political fringe groups, but that county-level Twitter penetration may increase the appeal of extreme groups. County-level Twitter penetration intensifies opinions on statements related to race or gender issues among county residents.

These results speak to current debates on social media as an “outrage machine.” They show that social media can increase the visibility of political fringes, not only online but also off-line through protests.

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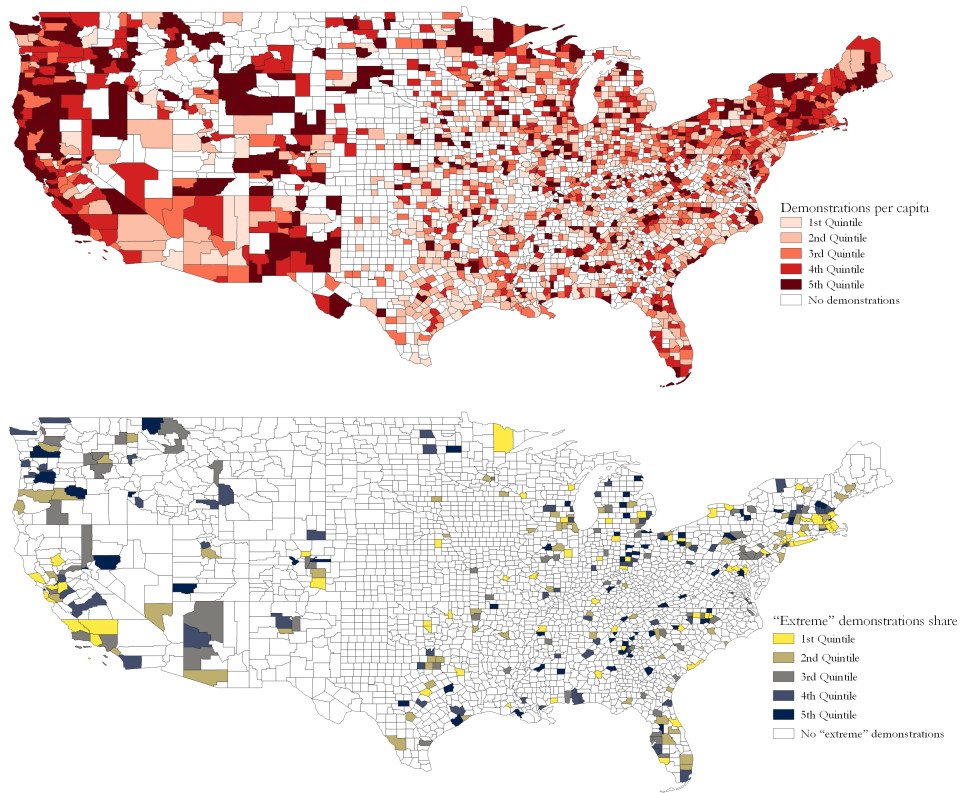


Figure 1: Spatial distribution of protest events

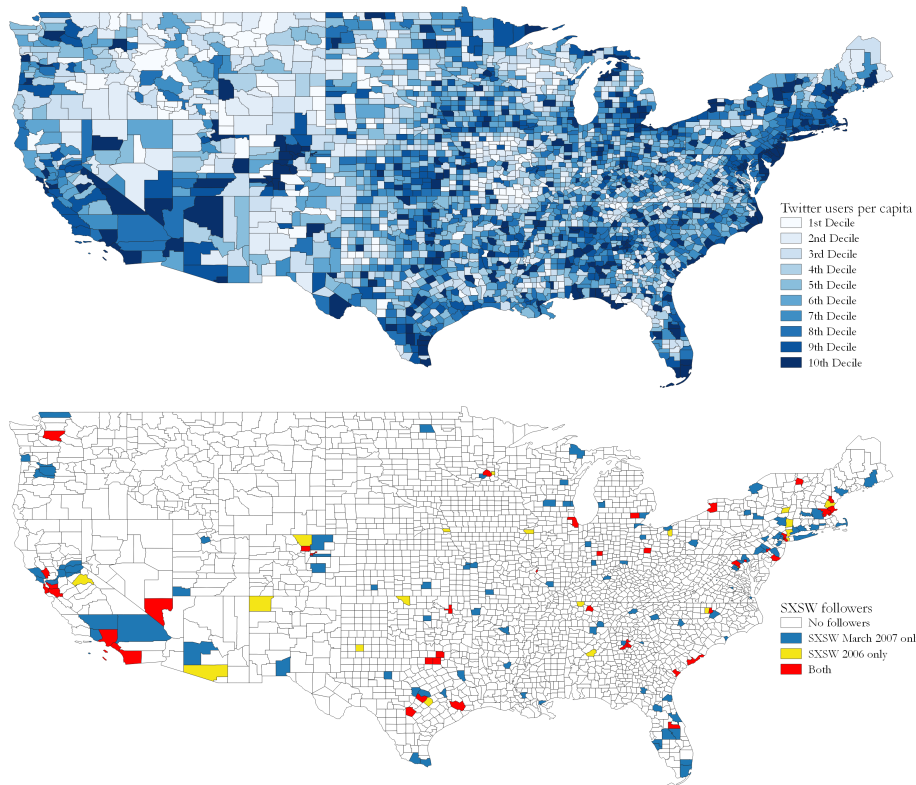


Figure 2: Spatial distribution of Twitter users per capita and SXSW followers

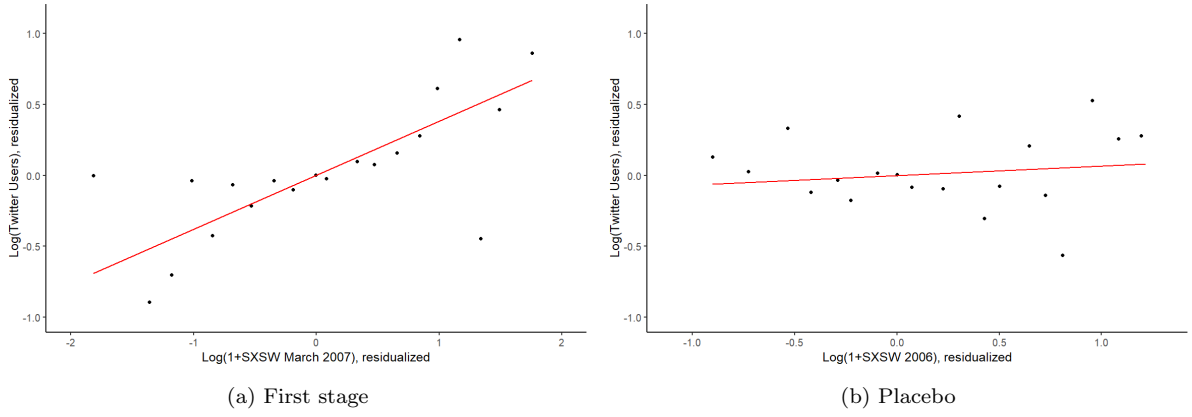


Figure 3: Binned scatter plots, visualizing the first stage

Note: This figure presents binned scatter plots of the relationship between county-level Twitter users in 2014-2015 and the number of SXSW followers in the full sample ($N=3,104$). In panel (a) the x-axis represents the logarithm of (one plus) the number of SXSW followers who joined Twitter in March 2007 and in panel (b) it represents the logarithm of (one plus) the number of SXSW followers who joined Twitter in 2006. Variables are residualized by partialing out state fixed effects, population deciles, demographic, geographic, socio-economic, media, crime, pre-2010 election, and pre-Twitter protest activity controls (see description of controls in Table A9 in the Appendix), as well as the SXSW follower variable not represented in the x-axis of each panel. The data is binned into 20 equal-interval bins on the x-axis and dots represent average values of both variables within each bin. The red line represents an OLS regression line using the unbinned data.

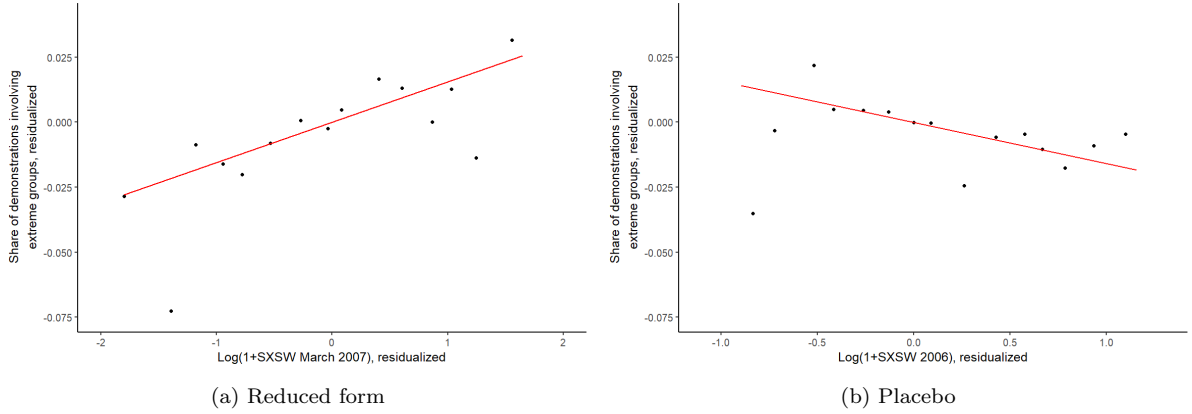


Figure 4: Binned scatter plots, visualizing the reduced form effect on the share of protests involving extreme groups

Note: This figure presents binned scatter plots of the relationship between county-level share of protests where extreme groups are involved and the number of SXSW followers in the sample of counties where at least one protest occurred ($N=1,776$). In panel (a) the x-axis represents the logarithm of (one plus) the number of SXSW followers who joined Twitter in March 2007 and in panel (b) it represents the logarithm of (one plus) the number of SXSW followers who joined Twitter in 2006. Variables are residualized by partialing out state fixed effects, population deciles, demographic, geographic, socio-economic, media, crime, pre-2010 election, and pre-Twitter protest activity controls (see description of controls in Table A9 in the Appendix), as well as the SXSW follower variable not represented in the x-axis of each panel. The data is binned into 15 equal-interval bins on the x-axis and dots represent average values of both variables within each bin. The red line represents an OLS regression line using the unbinned data.

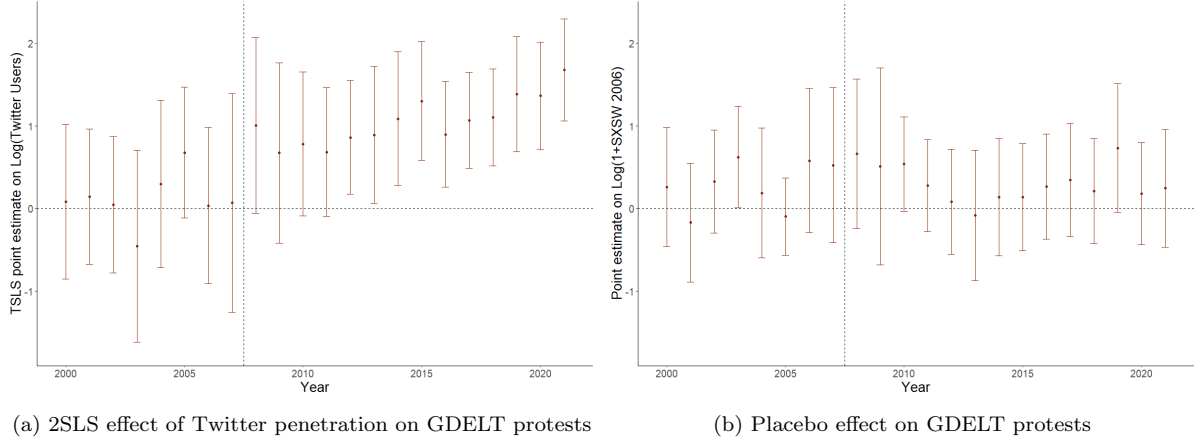


Figure 5: Twitter and GDELT protest over time – 2SLS

Note: This figure presents point estimates and 95% confidence intervals for λ_1 and λ_2 from the 2SLS model represented in equation (4), estimated on the matched sample ($N=187$) where each control observation is weighted by the number of treated observations it is matched to. The outcome variables are the log number of yearly protests in the GDELT database from 2000-2021. An increase from 0 to 1 is weighted as a 100% increase. Panel (a) plots the point estimate and standard error for λ_1 , measuring the effect of Twitter penetration (measured in 2014-2015) on protest frequency. Panel (b) plots the point estimate and standard error for λ_2 from the same regressions, measuring the correlation between the log number of (one plus) SXSW followers who created their account in 2006, and the outcome variable. All regressions control for state fixed effects, population deciles, demographic, geographic, socio-economic, media, crime, and pre-2010 election controls (see description of controls in Table A9 in the Appendix). Red dots represent point estimates and whiskers represent 95% confidence intervals based on robust standard errors clustered by state. For ease of interpretation, a horizontal dashed line is placed at 0 and a vertical dashed line is placed between the years 2007 and 2008.

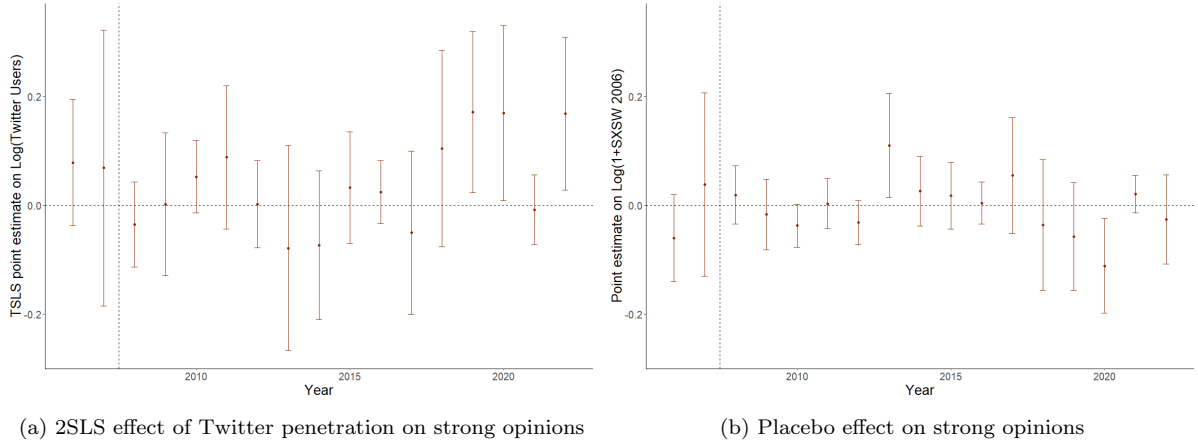


Figure 6: Twitter and strong opinions over time – 2SLS

Note: This figure presents point estimates and 90% confidence intervals for π_1 and π_2 from the 2SLS model represented in equation (6), estimated on a sample of all respondents from the Cooperative election study, with non-missing answers, from the full sample of counties ($N=5,873-63,584$, depending on year). The outcome variables are the sum of strong opinions to questions related to race issues, gender/sexuality/environmental issues, or partisanship. Panel (a) plots the point estimate and standard error for π_1 , measuring the effect of Twitter penetration in one's local environment (measured in 2014-2015) on holding strong opinions. Panel (b) plots the point estimate and standard error for π_2 from the same regressions, measuring the correlation between the log number of (one plus) SXSW followers who created their account in 2006, and the outcome variable. All regressions control for state fixed effects and the full set of county-characteristics (see description of controls in Table A9 in the Appendix), and the following individual-level controls: Age, gender, race, ethnicity, religion, education, marital status, employment status, family income, home ownership, union membership, military service, and number of children. Red dots represent point estimates and whiskers represent 90% confidence intervals based on robust standard errors clustered by county. For ease of interpretation, a horizontal dashed line is placed at 0 and a vertical dashed line is placed between the years 2007 and 2008.

Table 1: Protest classification – Summary statistics

	(1)	(2)	(3)	(4)	(5)	(6)
	Frequency	Share of all protests	Share with extreme group involvement	Share with extreme left (only) group involvement	Share with extreme right (only) group involvement	Share with extreme left and right group involvement
Panel A: All protests						
Total	34,185	1	0.048	0.030	0.015	0.002
Panel B: Topics (not mutually exclusive)						
Covid Related	3,411	0.100	0.047	0.005	0.040	0.002
Abortion related	602	0.018	0.033	0.018	0.008	0.007
Presidential election related	632	0.018	0.228	0.033	0.157	0.038
BLM related	12,530	0.367	0.038	0.023	0.010	0.005
Race related	13,506	0.395	0.060	0.025	0.029	0.005
Unclassified topic	16,381	0.479	0.043	0.041	0.003	0.000
Panel C: Political alignment (mutually exclusive)						
Left (only)	14,142	0.414	0.071	0.071	0	0
Right (only)	2,074	0.061	0.202	0	0.202	0
Left and right	640	0.019	0.319	0.049	0.164	0.106
Unclassified political alignment	17,329	0.507	0	0	0	0
Panel D: Violence (mutually exclusive)						
Peaceful	32,024	0.936	0.045	0.030	0.014	0.001
Peaceful protesters met with intervention or violence	1,145	0.033	0.071	0.042	0.027	0.002
Violent protesters	936	0.027	0.113	0.036	0.040	0.037
Violent protesters interact with violent outside group	80	0.002	0.075	0.025	0.050	0.000

Note: This table shows the frequency of protests in each classification category along three dimensions: Topics, political alignment, and nature of violence. Column (2) shows the share of all protests which fall into a given category. Columns (3)-(6) show the share of protests in a given category where extreme groups are involved

Table 2: First stage analysis

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent variable:	Log(Twitter Users)							
Log(1+SXSW March2007)	0.617*** (0.062)	0.465*** (0.055)	0.444*** (0.055)	0.430*** (0.055)	0.425*** (0.056)	0.418*** (0.055)	0.412*** (0.055)	0.381*** (0.055)
Log(1+SXSW 2006)	0.212*** (0.073)	0.099 (0.074)	0.073 (0.076)	0.105 (0.076)	0.099 (0.074)	0.087 (0.071)	0.082 (0.071)	0.067 (0.071)
Log(1+Protests 2000-2007)								0.055*** (0.015)
R ²	0.925	0.939	0.939	0.939	0.945	0.945	0.947	0.947
Cluster robust F-statistic	84.46	53.33	49.18	39.48	38.11	38.60	37.50	34.99
p-value: March 2007 = 2006	0.010**	0.014**	0.011**	0.032**	0.032**	0.026**	0.027**	0.030**
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Population deciles	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Demographic controls		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Geographic controls			Yes	Yes	Yes	Yes	Yes	Yes
Socio-economic controls				Yes	Yes	Yes	Yes	Yes
Media controls					Yes	Yes	Yes	Yes
Crime controls						Yes	Yes	Yes
Pre 2010 election controls							Yes	Yes
Observations	3,108	3,108	3,108	3,108	3,107	3,107	3,106	3,106

Note: This table presents county-level regressions where the natural logarithm of our Twitter users from 2014-2015 is the dependent variable. SXSW March2007 is the number of Twitter users who joined Twitter in March 2007 and follow South by Southwest (@SXSW). SXSW 2006 is the number of Twitter users who joined in 2006 and follow SXSW. Regressions include the indicated control variables (see Table A9 in the Appendix for their descriptions). Robust standard errors in parentheses are clustered by state. *p<0.1; **p<0.05; ***p<0.01

Table 3: 2SLS: The effect of Twitter penetration on protests by extremity

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A								
Dependent variable:	Log(Number of extreme protests)							
Log(Twitter Users)	1.422*** (0.193)	1.679*** (0.311)	1.742*** (0.312)	1.780*** (0.297)	1.788*** (0.303)	1.821*** (0.314)	1.848*** (0.319)	1.828*** (0.341)
Log(1+SXSW 2006)	-0.094 (0.209)	-0.032 (0.187)	-0.009 (0.188)	-0.056 (0.175)	-0.054 (0.173)	-0.048 (0.174)	-0.042 (0.173)	-0.045 (0.171)
Log(1+Protests 2000-2007)								0.014 (0.035)
Mean number of protests	0.524	0.524	0.524	0.524	0.524	0.524	0.524	0.524
Panel B								
Dependent variable:	Log(Number of non-extreme protests)							
Log(Twitter Users)	1.001*** (0.122)	0.872*** (0.115)	0.971*** (0.132)	0.921*** (0.140)	0.898*** (0.139)	0.895*** (0.138)	0.879*** (0.142)	0.804*** (0.159)
Log(1+SXSW 2006)	-0.109 (0.107)	-0.103 (0.091)	-0.066 (0.080)	-0.029 (0.083)	-0.036 (0.082)	-0.039 (0.081)	-0.038 (0.080)	-0.047 (0.083)
Log(1+Protests 2000-2007)								0.055* (0.030)
Mean number of protests	10.48	10.48	10.48	10.48	10.48	10.48	10.48	10.48
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Population deciles	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Demographic controls		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Geographic controls			Yes	Yes	Yes	Yes	Yes	Yes
Socio-economic controls				Yes	Yes	Yes	Yes	Yes
Media controls					Yes	Yes	Yes	Yes
Crime controls						Yes	Yes	Yes
Pre 2010 election controls							Yes	Yes
Observations	3,108	3,108	3,108	3,108	3,107	3,107	3,106	3,106
Cluster robust F-statistic	84.46	53.33	49.18	39.48	38.11	38.60	37.34	34.99
p-value: $\beta_{Extreme}^{RF} = \beta_{Non-extreme}^{RF}$	0.027**	0.000***	0.001***	0.000***	0.000***	0.000***	0.000***	0.000***

Note: This table presents county-level second stage regressions for a 2SLS model of the effect of Twitter penetration on protest frequency from January 2020 to November 2021, depending on extreme group involvement. An extensive margin increase in protests from 0 to 1 is valued as a 150% increase. Log(Twitter Users) is the natural logarithm of our Twitter users from 2014-2015. They are instrumented for via the log number of (one plus) Twitter users who joined Twitter in March 2007 and follow South by Southwest (@SXSW). SXSW 2006 is the number of Twitter users who joined in 2006 and follow SXSW. Extreme refers to demonstrations involving a political group defined as extreme. The groups classified as extreme are listed in Table A1. Regressions include the indicated control variables (see Table A9 in the Appendix for their descriptions). Robust standard errors in parentheses are clustered by state. *p<0.1; **p<0.05; ***p<0.01

Table 4: Effect of Twitter penetration on the share of extreme protests

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent variable:	Share of extreme protests							
Panel A: 2SLS Second Stage								
Log(Twitter Users)	0.028*** (0.007)	0.031*** (0.009)	0.030*** (0.011)	0.033*** (0.011)	0.033*** (0.011)	0.035*** (0.011)	0.037*** (0.011)	0.038*** (0.013)
Log(1+SXSW2006)	-0.013 (0.009)	-0.018** (0.009)	-0.018** (0.009)	-0.019** (0.009)	-0.020** (0.009)	-0.020** (0.009)	-0.019** (0.009)	-0.019** (0.009)
Log(1+Protests 2000-2007)								-0.001 (0.003)
Cluster robust F-statistic	88.40	57.60	60.64	50.13	50.04	50.16	51.11	47.88
Panel B: Reduced-form								
Log(1+SXSW March2007)	0.018*** (0.005)	0.015*** (0.004)	0.014*** (0.005)	0.015*** (0.005)	0.015*** (0.005)	0.015*** (0.005)	0.016*** (0.005)	0.016*** (0.005)
Log(1+SXSW 2006)	-0.008 (0.007)	-0.015* (0.008)	-0.016* (0.009)	-0.016* (0.008)	-0.016** (0.008)	-0.016** (0.008)	-0.015* (0.008)	-0.016* (0.008)
Log(1+Protests 2000-2007)								0.002 (0.002)
R ²	0.064	0.079	0.080	0.085	0.089	0.090	0.094	0.095
Panel C: OLS								
Log(Twitter Users)	0.005* (0.003)	0.005 (0.004)	0.006 (0.004)	0.003 (0.004)	0.004 (0.005)	0.006 (0.005)	0.006 (0.005)	0.005 (0.005)
Log(1+SXSW 2006)	0.008* (0.005)	-0.003 (0.006)	-0.006 (0.007)	-0.003 (0.006)	-0.004 (0.006)	-0.004 (0.006)	-0.002 (0.006)	-0.004 (0.006)
Log(1+Protests 2000-2007)								0.002 (0.002)
R ²	0.063	0.078	0.080	0.084	0.088	0.089	0.093	0.093
Observations	1,758	1,758	1,758	1,758	1,757	1,757	1,756	1,756
Mean dependent variable	0.022	0.022	0.022	0.022	0.022	0.022	0.022	0.022
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Population deciles	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Demographic controls		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Geographic controls			Yes	Yes	Yes	Yes	Yes	Yes
Socio-economic controls				Yes	Yes	Yes	Yes	Yes
Media controls					Yes	Yes	Yes	Yes
Crime controls						Yes	Yes	Yes
Pre 2010 election controls							Yes	Yes

Note: This table presents county-level regressions for an OLS and a 2SLS model of the effect of Twitter penetration on the share of protests with extreme group involvement between January 2020 to November 2021. The groups classified as extreme are listed in Table A1. The sample only includes counties where at least one protest occurred. Log(Twitter Users) is the natural logarithm of our Twitter users from 2014-2015. SXSW March2007 is the number of Twitter users who joined Twitter in March 2007 and follow South by Southwest (@SXSW). SXSW 2006 is the number of Twitter users who joined in 2006 and follow SXSW. Regressions include the indicated control variables (see Table A9 in the Appendix for their descriptions). Robust standard errors in parentheses are clustered by state. *p<0.1; **p<0.05; ***p<0.01

Table 5: 2SLS on matched sample

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Specification:	First Stage		Second Stage					
Dependent variable:	log(Twitter Users)		log(Protests 2020-2021)		Extreme protests share		log(Protests 2000-2007)	
log(1+SXSW March 2007)	0.538*** (0.069)	0.368*** (0.095)						
log(Twitter Users)			0.875*** (0.154)	0.809*** (0.115)	0.026* (0.014)	0.028** (0.012)	0.509 (0.406)	-0.171 (0.681)
log(1+SXSW 2006)	0.587*** (0.178)	0.220 (0.199)	-0.089 (0.200)	-0.001 (0.157)	-0.023** (0.011)	-0.024* (0.014)	0.445 (0.452)	0.595 (0.527)
Log(1+Protests 2000-2007)		0.107** (0.051)		-0.038 (0.032)		0.002 (0.003)		
R ²	0.721	0.906						
Cluster Robust F-statistic log(1+SXSW March 2007)	30.18	18.92	30.18	18.92	30.18	18.92	30.18	18.92
Cluster Robust F-statistic log(1+SXSW 2006)	9.83	1.37	9.83	1.37	9.83	1.37	9.83	1.37
Weighted mean outcome variable	8.78	8.78	3.78	3.78	0.04	0.04	1.42	1.42
Observations	187	187	187	187	187	187	187	187
Treatment observations	120	120	120	120	120	120	120	120
Unique control observations	67	67	67	67	67	67	67	67
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Population deciles	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Demographic controls		Yes		Yes		Yes		Yes
Geographic controls		Yes		Yes		Yes		Yes
Socio-economic controls		Yes		Yes		Yes		Yes
Media controls		Yes		Yes		Yes		Yes
Crime controls		Yes		Yes		Yes		Yes
Pre 2010 election controls		Yes		Yes		Yes		Yes

Note: This table presents county-level 2SLS regressions using a matched sample. Control observations are weighted by the number of treatment observations they are matched to. An extensive margin increase in protests from 0 to 1 is valued as a 150% increase. Log(Twitter Users) is the natural logarithm of our Twitter users from 2014-2015. Extreme refers to protests involving a political group defined as extreme. The groups classified as extreme are listed in Table A1 in the Appendix. Data on protests from 2000-2007 come from GDELT. See Table A9 in the Appendix for control variable descriptions. Robust standard errors clustered by state in parentheses. *p<0.1; **p<0.05; ***p<0.01

Table 6: 2SLS protests by topic

	(1)	(2)	(3)	(4)	(5)
Topic of protest:	Coronavirus	Presidential elections	Abortion rights	BLM related	Race related
Panel A					
Dependent variable:	Log(Number of protests)				
Log(Twitter Users)	0.906*** (0.246)	1.127*** (0.320)	1.211*** (0.215)	0.932*** (0.192)	0.939*** (0.193)
Log(1+SXSW 2006)	-0.102 (0.167)	0.036 (0.142)	-0.087 (0.154)	-0.081 (0.098)	-0.085 (0.094)
Log(1+Protests 2000-2007)	0.077** (0.035)	0.064* (0.035)	-0.017 (0.030)	0.004 (0.028)	0.009 (0.029)
Mean number of protests	1.10	0.20	0.19	4.03	4.35
Cluster robust F-statistic	34.99	34.99	34.99	34.99	34.99
Observations	3,106	3,106	3,106	3,106	3,106
p-value: $\beta^{RF} = \beta_{Extreme}^{RF}$	0.000***	0.000***	0.001***	0.000***	0.000***
Panel B					
Dependent variable:	Share among all protests				
Log(Twitter Users)	-0.050** (0.021)	-0.0002 (0.012)	-0.001 (0.009)	-0.009 (0.043)	-0.005 (0.043)
Log(1+SXSW 2006)	0.007 (0.014)	-0.007 (0.005)	0.003 (0.006)	0.019 (0.028)	0.017 (0.028)
Log(1+Protests 2000-2007)	0.011 (0.007)	0.004** (0.002)	-0.002* (0.001)	-0.023** (0.011)	-0.023** (0.011)
Mean dependent variable	0.096	0.013	0.017	0.490	0.506
Cluster robust F-statistic	47.88	47.88	47.88	47.88	47.88
Observations	1,756	1,756	1,756	1,756	1,756

Note: This table presents county-level 2SLS regressions. Panel A presents 2SLS results where the log number of protests from January 2020 to November 2021, fitting the given column category is the dependent variable, weighting a change from 0 to 1 as a 150% increase. Panel B presents 2SLS results on a subsample of counties where at least one protest event occurred, where the dependent variable is the share of protests that fit the column category. Log(Twitter Users) is the natural logarithm of our Twitter users from 2014-2015. They are instrumented for via the log number of (one plus) Twitter users who joined Twitter in March 2007 and follow South by Southwest (@SXSW). SXSW 2006 is the number of Twitter users who joined Twitter in 2006 and follow South by Southwest (@SXSW). Coronavirus refers to protests related to public health measures due to the Covid pandemic. Presidential elections refers to protests related to the 2020 post-presidential election unrest (including those calling for results to be respected). Abortion rights refers to protests related to reproductive rights. BLM related refers to protests concerning police brutality against minorities, and counter movements. Race related refers to protests concerning racial tensions. A detailed description of topic categorization is presented in Tables A2 to A6 in the Appendix. All regressions control for population via decile dummies, and the full set of controls (as in column (8) in Table 2), see Table A9 in the Appendix for their descriptions. Robust standard errors in parentheses are clustered by state.

*p<0.1; **p<0.05; ***p<0.01

Table 7: 2SLS protests by political leaning and extremity

	(1)	(2)	(3)	(4)
Political leaning:	Left		Right	
Extremity:	Mainstream	Extreme	Mainstream	Extreme
Panel A				
Dependent variable:	Log(Number of protests)			
Log(Twitter Users)	1.007*** (0.185)	1.886*** (0.360)	0.876*** (0.237)	1.271*** (0.298)
Log(1+SXSW 2006)	-0.073 (0.088)	-0.064 (0.191)	-0.095 (0.136)	0.068 (0.161)
Log(1+Protests 2000-2007)	-0.008 (0.032)	-0.007 (0.040)	0.064* (0.035)	0.012 (0.029)
Mean number protests	4.40	0.36	0.68	0.19
Cluster robust F-statistic	34.99	34.99	34.99	34.99
Observations	3,106	3,106	3,106	3,106
p-value: $\beta_{Extreme}^{RF} = \beta_{Mainstream}^{RF}$		0.000***		0.059*
Panel B				
Dependent variable:	Share among all protests			
Log(Twitter Users)	-0.016 (0.041)	0.025*** (0.009)	0.001 (0.020)	0.016** (0.008)
Log(1+SXSW 2006)	0.032 (0.025)	-0.009 (0.007)	0.00000 (0.009)	-0.010** (0.004)
Log(1+Protests 2000-2007)	-0.026** (0.011)	0.00003 (0.002)	0.001 (0.005)	-0.001 (0.002)
Cluster robust F-statistic	47.88	47.88	47.88	47.88
Observations	1,756	1,756	1,756	1,756
Mean dependent variable	0.495	0.009	0.078	0.013

Note: This table presents county-level 2SLS regressions. Panel A presents results where the log number of protests from January 2020 to November 2021, fitting the given column category is the dependent variable, weighting a change from 0 to 1 as a 150% increase. Panel B presents results on a subsample of counties where at least one protest event occurred, where the dependent variable is the share of protests that fit the column-category. Log(Twitter Users) is the natural logarithm of our Twitter users from 2014-2015. They are instrumented for via the log number of (one plus) Twitter users who joined Twitter in March 2007 and follow South by Southwest (@SXSW). SXSW 2006 is the number of Twitter users who joined Twitter in 2006 and follow South by Southwest (@SXSW). Mainstream left refers to protests which are left-leaning and no extreme groups were involved, mainstream right refers to right-leaning protests where no extreme groups were involved, Extreme left and right refer to protests where extreme left or extreme right groups are involved, respectively. A detailed description of protest categorization can be found in Tables A2 to A6 in the Appendix. All regressions control for population via decile dummies, and the full set of controls (as in column (8) in Table 2), see Table A9 in the Appendix for their descriptions. Robust standard errors in parentheses are clustered by state. *p<0.1; **p<0.05; ***p<0.01

Table 8: 2SLS protests by aggression

	(1)	(2)	(3)	(4)
Type of protest:	Peaceful	Peaceful Protesters met with intervention or violence	Violent Protesters	Violent protesters interact with violent outside group
Panel A				
Dependent variable:	Log(Number of protests)			
Log(Twitter Users)	0.821*** (0.165)	1.421*** (0.268)	1.482*** (0.289)	0.357** (0.164)
Log(1+SXSW 2006)	-0.054 (0.087)	0.064 (0.173)	0.017 (0.207)	0.254* (0.131)
Log(1+ Protests 2000-2007)	0.044 (0.028)	0.028 (0.037)	-0.012 (0.030)	0.012 (0.013)
Mean number of protests	10.31	0.37	0.30	0.03
Cluster robust F-statistic	34.99	34.99	34.99	34.99
Observations	3,106	3,106	3,106	3,106
p-value: $\beta^{RF} = \beta_{Peaceful}^{RF}$		0.018**	0.009***	0.008***
Panel B				
Dependent variable:	Share among all protests			
Log(Twitter Users)	-0.026 (0.020)	0.007 (0.013)	0.022* (0.012)	-0.002 (0.003)
Log(1+SXSW 2006)	0.010 (0.014)	-0.004 (0.008)	-0.006 (0.008)	0.001 (0.002)
Log(1+Protests 2000-2007)	-0.007 (0.005)	0.009** (0.004)	-0.003* (0.002)	0.001 (0.001)
Cluster robust F-statistic	47.88	47.88	47.88	47.88
Observations	1,756	1,756	1,756	1,756
Mean dependent variable	0.966	0.022	0.009	0.003

Note: This table presents county-level 2SLS regressions. Panel A presents results where the log number of protests from January 2020 to November 2021, fitting the given column category is the dependent variable, weighting a change from 0 to 1 as a 150% increase. Panel B presents results on a subsample of counties where at least one protest event occurred, where the dependent variable is the share of protests that fit the column-category. Log(Twitter Users) is the natural logarithm of our Twitter users from 2014-2015. They are instrumented for via the log number of (one plus) Twitter users who joined Twitter in March 2007 and follow South by Southwest (@SXSW). SXSW 2006 is the number of Twitter users who joined Twitter in 2006 and follow South by Southwest (@SXSW). All regressions control for population via decile dummies, and the full set of controls (as in column (8) in Table 2), see Table A9 in the Appendix for their descriptions. Robust standard errors in parentheses are clustered by state. *p<0.1; **p<0.05; ***p<0.01

Table 9: Effects on strong opinions, Cooperative Election Study 2020

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Specification:	First stage	Second Stage					
Dependent variable:	Log(Twitter users)	Sum of strong opinions			1st principal component		
Issue bundle:		Race	Race+Gender	Race+Gender +Partisanship	Race	Race+Gender	Race+Gender +Partisanship
Log(1+SXSW March 2007)	0.432*** (0.004)						
Log(Twitter Users)		0.119** (0.049)	0.161** (0.069)	0.169* (0.098)	0.130** (0.061)	0.143** (0.068)	0.133* (0.076)
Log(1+SXSW 2006)	0.002 (0.006)	-0.080*** (0.031)	-0.095** (0.038)	-0.112** (0.053)	-0.086** (0.034)	-0.086** (0.035)	-0.084** (0.040)
Log(1+Protests 2000-2007)	0.115*** (0.002)	0.002 (0.013)	-0.0004 (0.019)	0.001 (0.024)	-0.002 (0.015)	-0.003 (0.017)	-0.002 (0.018)
R ²	0.955						
Cluster robust F-statistic	2347.33						
p-value: March 2007 = 2006	0.000***						
Mean dependent variable		2.59	3.39	4.88	0	0	0
Number of statements		5	7	10	5	7	10
Observations	49,718	49,718	49,379	49,718	49,379	49,718	49,379

Note: This table presents individual-level regressions. Log(Twitter Users) is the natural logarithm of unique Twitter users from a large sample of Tweets from 2014-2015. SXSW March2007 and SXSW 2006 are the number of Twitter users who follow South by Southwest (@SXSW) and joined Twitter in March 2007 and in 2006 respectively. Column (1) presents first stage regressions, while columns (2)-(6) present second stages. All regressions include the following individual-level controls: Age, gender, education, race, ethnicity, employment status, home-ownership, family income, number of children and internet connection, and the full set of county-characteristics (as in column (8) in Table 2, see Table A9 in the Appendix for their descriptions). The statements on race issues are the following: "Do the police make you feel...?" where "Mostly safe" is considered the strongly conservative option and "Mostly unsafe" is the strongly liberal option. "White people in the U.S. have certain advantages because of the color of their skin," where "Strongly disagree" is considered the strongly conservative opinion and "Strongly agree" is the strongly liberal option. "Racial problems in the U.S. are rare, isolated situations," "Irish, Italians, Jewish and many other minorities overcame prejudice and worked their way up. Blacks should do the same without any special favors," "Generations of slavery and discrimination have created conditions that make it difficult for blacks to work their way out of the lower class," where "Strongly agree" is the strongly liberal option and "Strongly disagree" is the strongly conservative option. The statements on gender issues are the following: Women are too easily offended, Women seek to gain power by getting control over men, where "Strongly agree" is the strongly conservative option and "Strongly disagree" is the strongly liberal option. Partisanship statements are the following: "Generally speaking, do you think of yourself as...?" where "Strong Democrat" is the strongly liberal option and "Strong Republican" is the strongly conservative option. "Do you approve of the way President Trump is doing their job" where "Strongly approve" is the strongly conservative option and "Strongly disapprove" is the strongly liberal option. "How would you be your own ideology" where "Very liberal" is the strongly liberal option and "Very conservative" is the strongly conservative option. Observations are weighted to represent adult Americans and robust standard errors are clustered by state. *p<0.1; **p<0.05; ***p<0.01

A Appendix

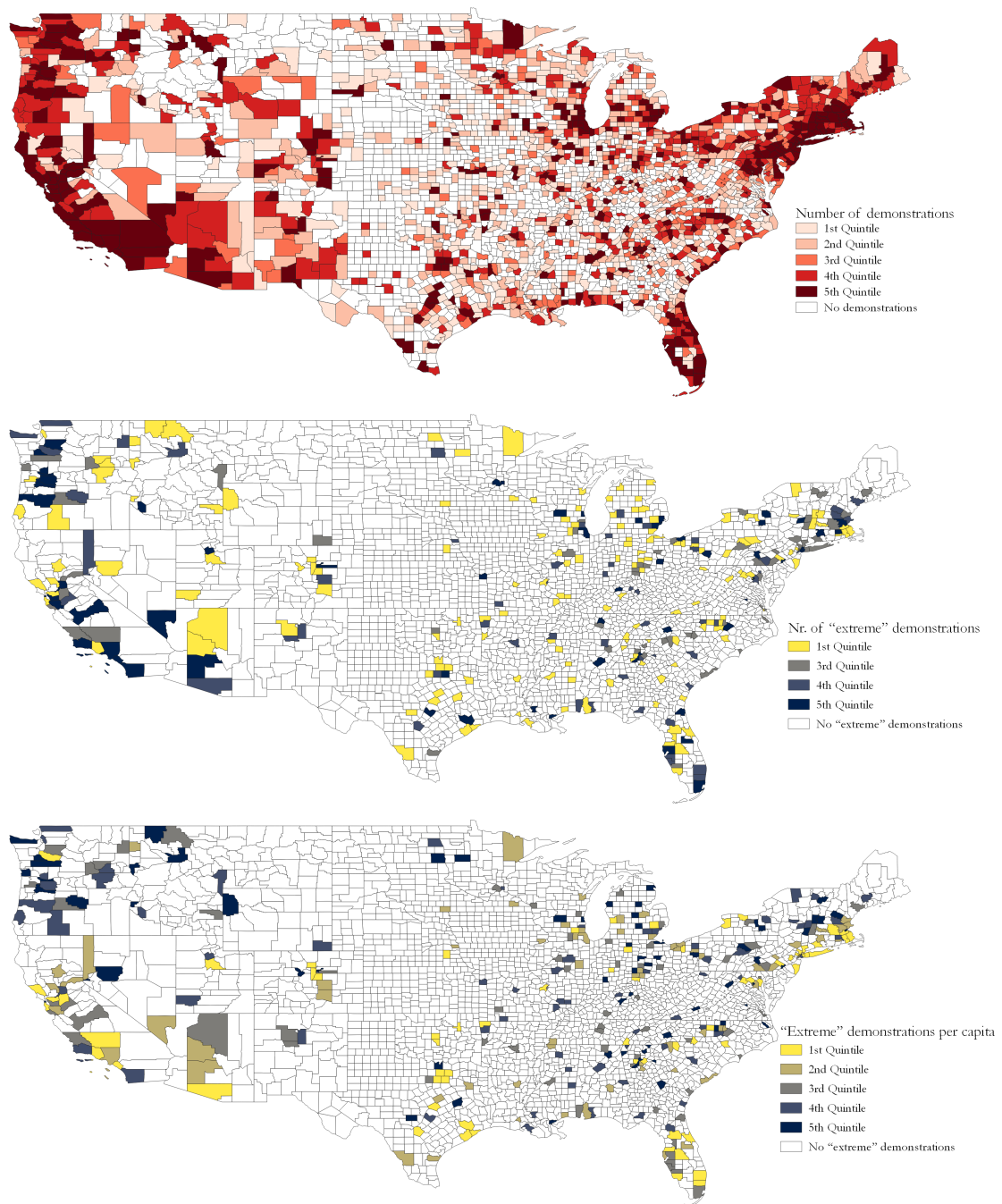


Figure A1: Spatial distribution of protest events and extreme protest events

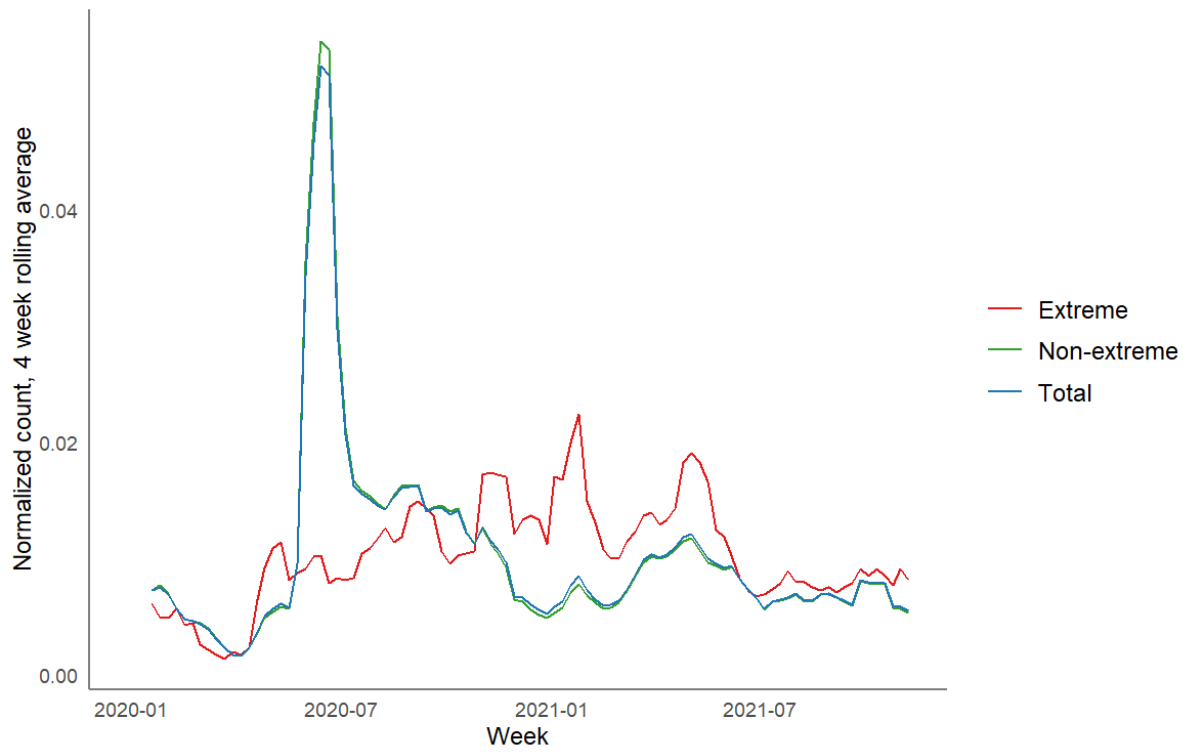
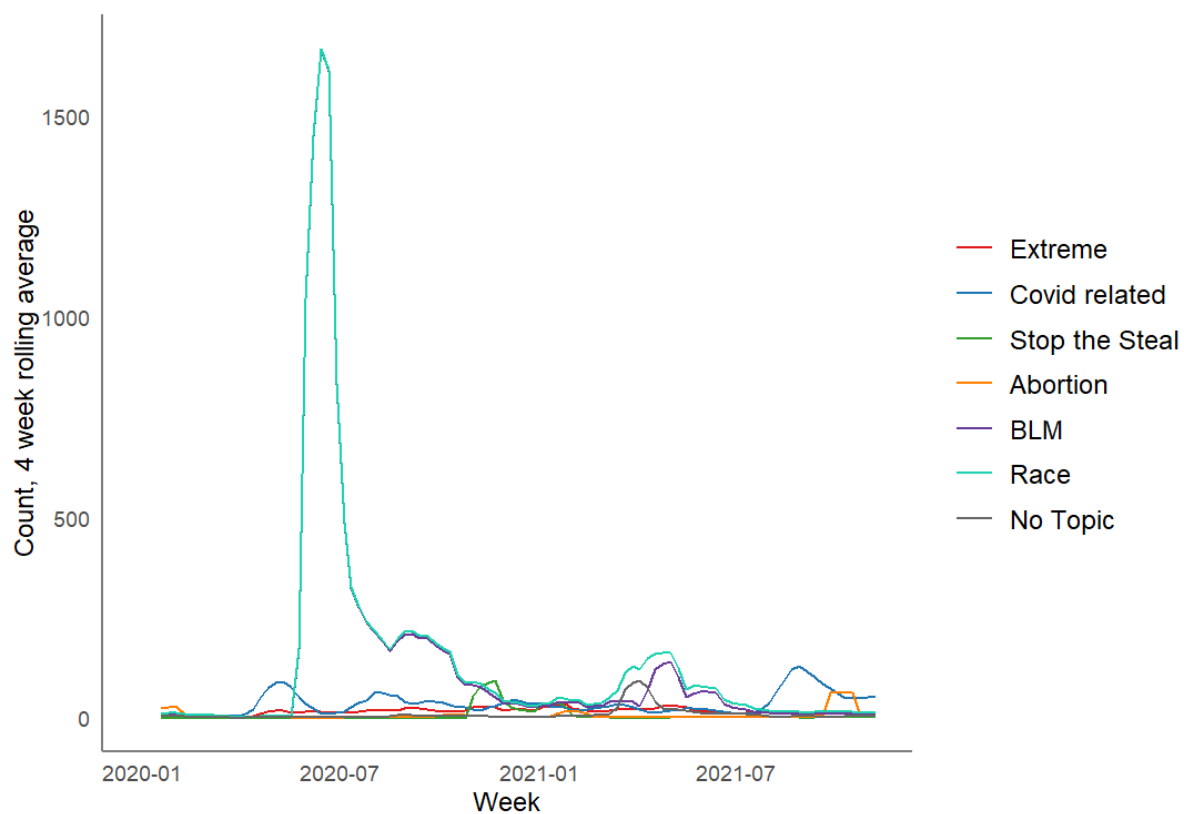
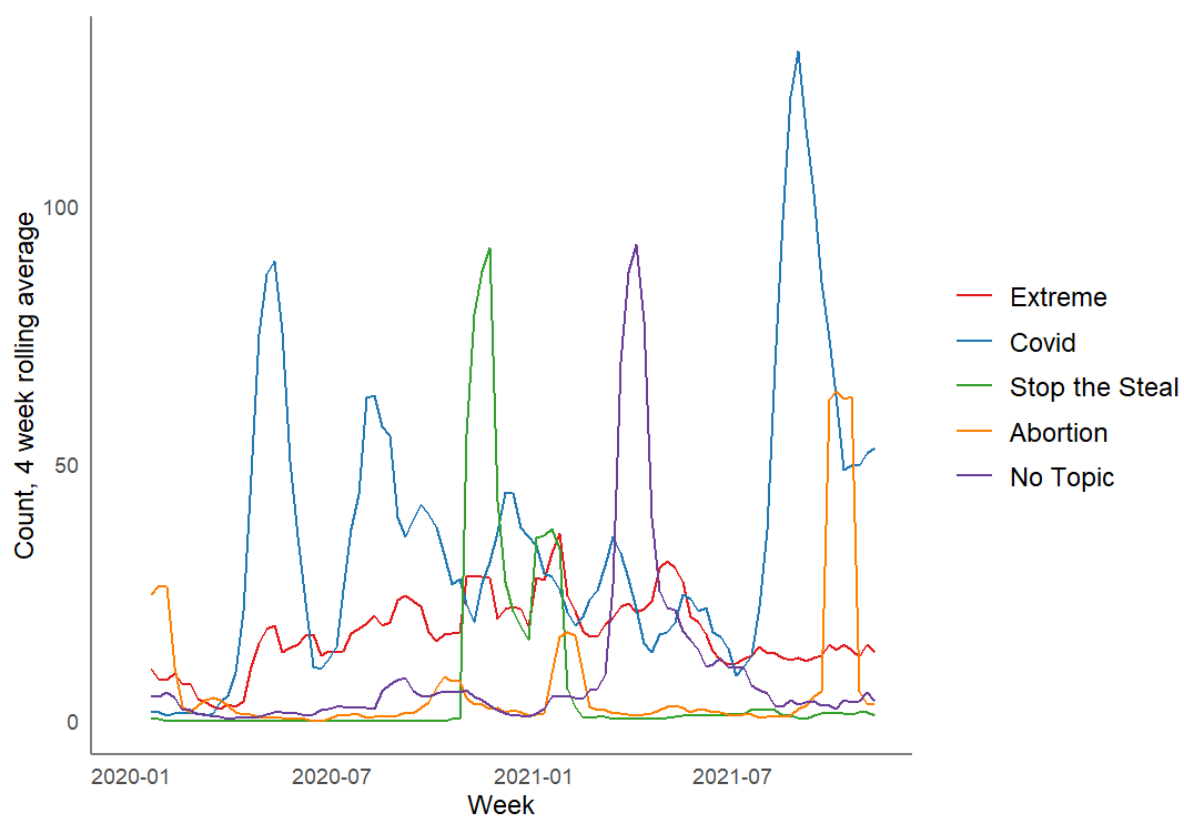


Figure A2: Normalized weekly number of protests events by extreme group involvement

Note: The figure shows 4 week rolling averages. Weekly frequency of protests in a given category is normalized by the total number of protest in that category over the sample period.



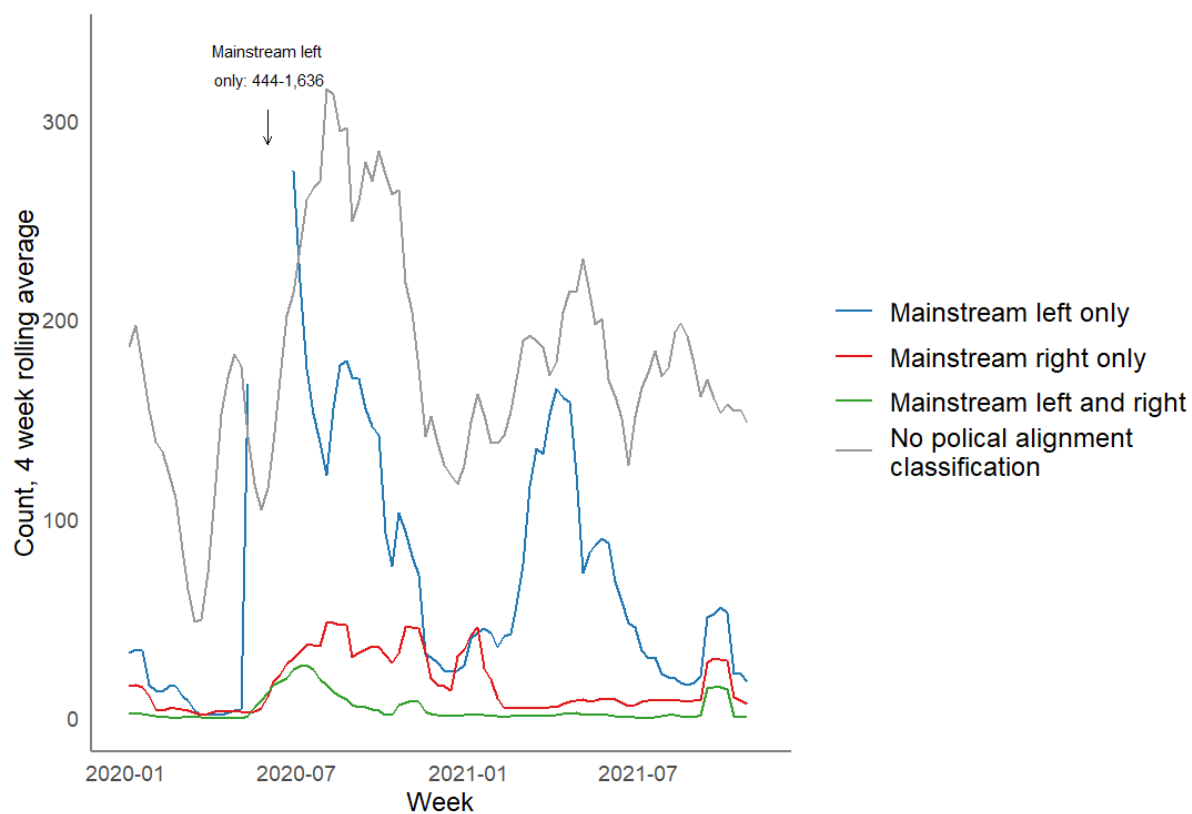
(a) With BLM and racial topics



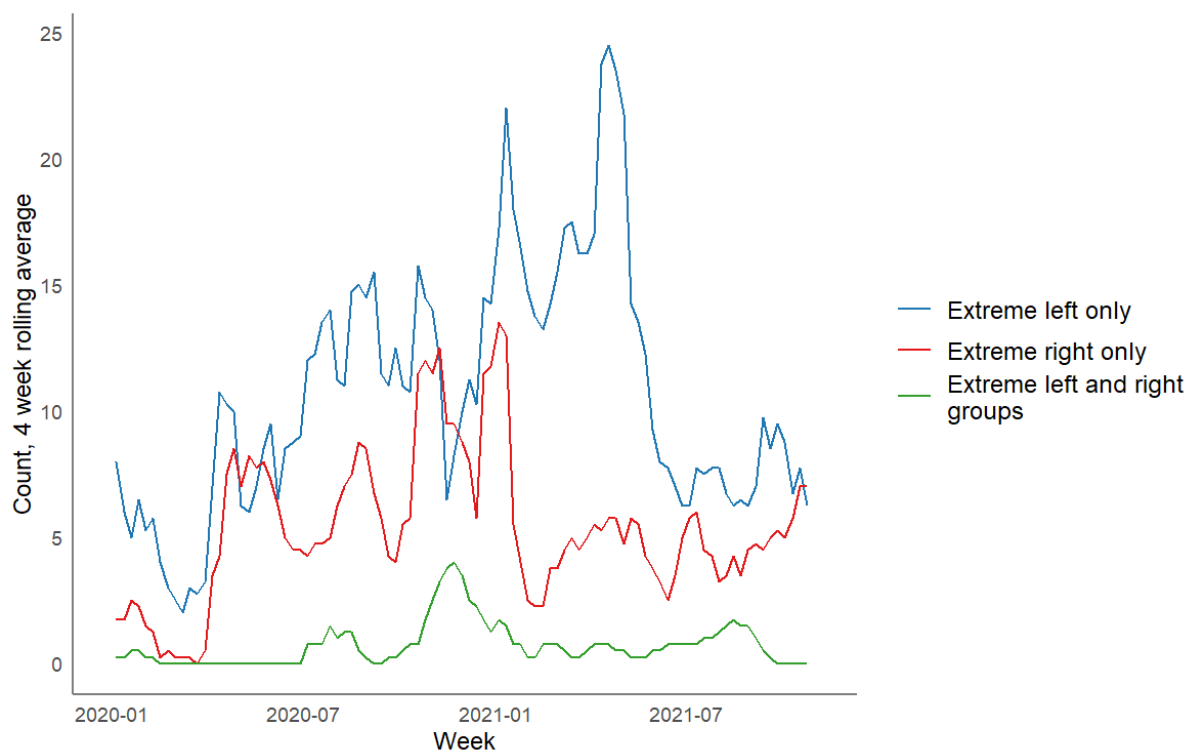
(b) Without BLM and racial topics

Figure A3: Weekly number of protests events by topic and extreme group involvement

Note: The figures show 4 week rolling averages.



(a) Mainstream protests by political leaning



(b) Extreme protests by political leaning

Figure A4: Weekly number of protests events by political leaning and extreme group involvement

Note: The figures show 4 week rolling averages.

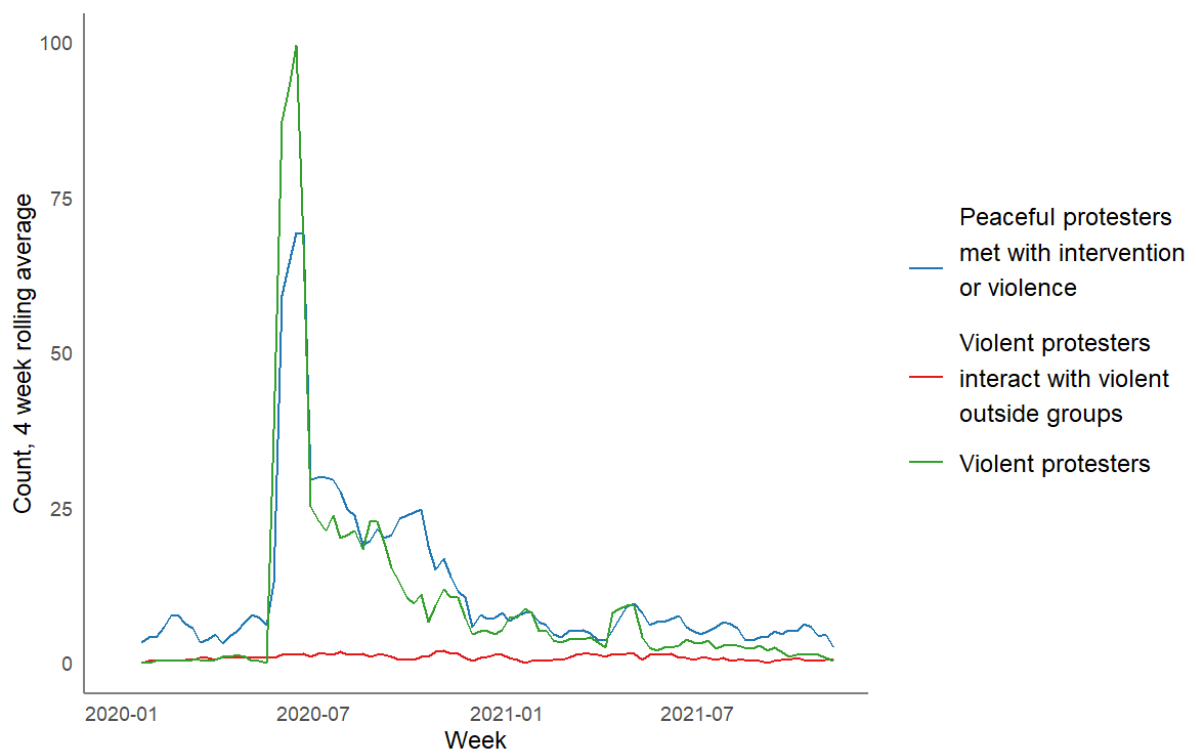


Figure A5: Weekly number of protests events by nature of violence

Note: The figure shows 4 week rolling averages.

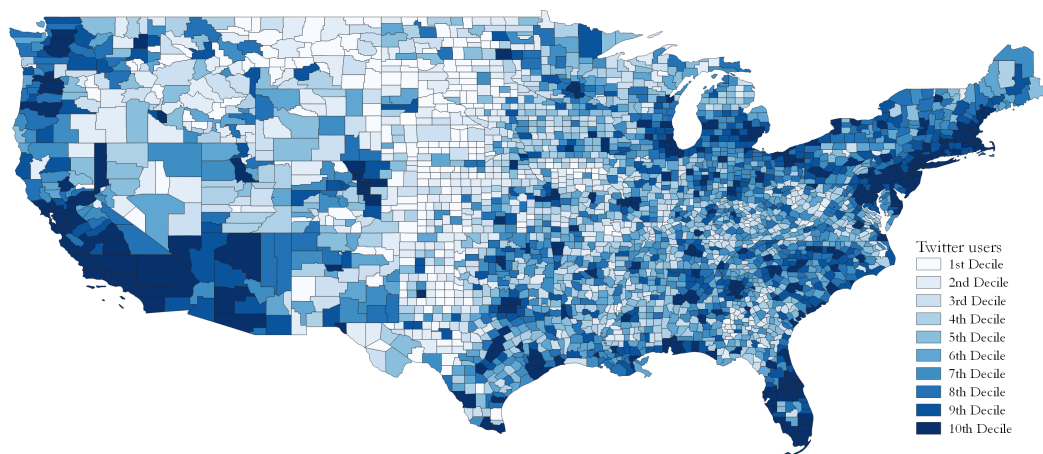


Figure A6: Spatial distribution of Twitter users

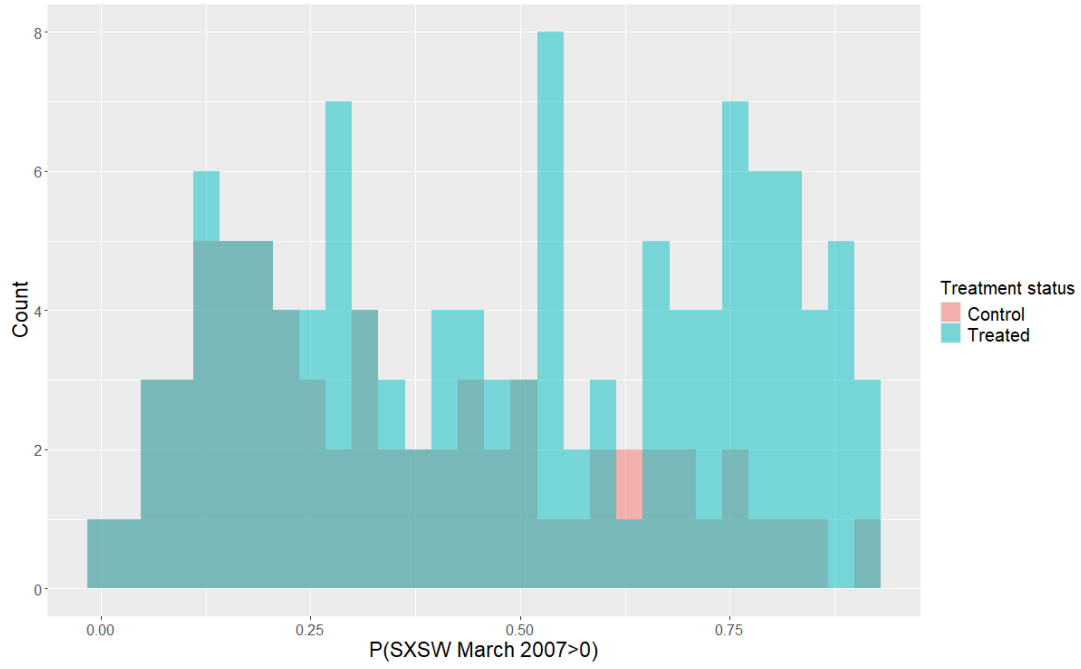


Figure A7: Common Support for matched sample

Common support: [0.002-0.93]

Observations: 120 Treated & 67 Control

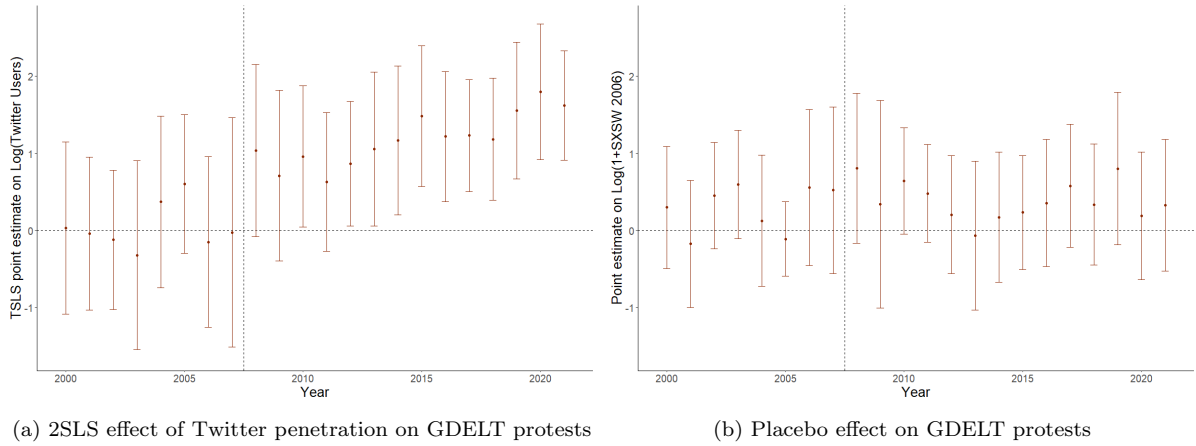


Figure A8: Twitter and GDELT protest, not de-duplicated – 2SLS

Note: This figure presents point estimates and 95% confidence intervals for λ_1 and λ_2 from the 2SLS model represented in equation (4), estimated on the matched sample ($N=187$) where each control observation is weighted by the number of treated observations it is matched to. The outcome variables are the log number of yearly protests in the GDELT database, without a de-duplication strategy, from 2000-2021. An increase from 0 to 1 is weighted as a 100% increase. Panel (a) plots the point estimate and standard error for λ_1 , measuring the effect of Twitter penetration (measured in 2014-2015) on protest frequency. Panel (b) plots the point estimate and standard error for λ_2 from the same regressions, measuring the correlation between the log number of (one plus) SXSW followers who created their account in 2006, and the outcome variable. All regressions control for state fixed effects, population deciles, demographic, geographic, socio-economic, media, crime, and pre-2010 election controls (see description of controls in Table A9 in the Appendix). Red dots represent point estimates and whiskers represent 95% confidence intervals based on robust standard errors clustered by state. For ease of interpretation, a horizontal dashed line is placed at 0 and a vertical dashed line is placed between the years 2007 and 2008.

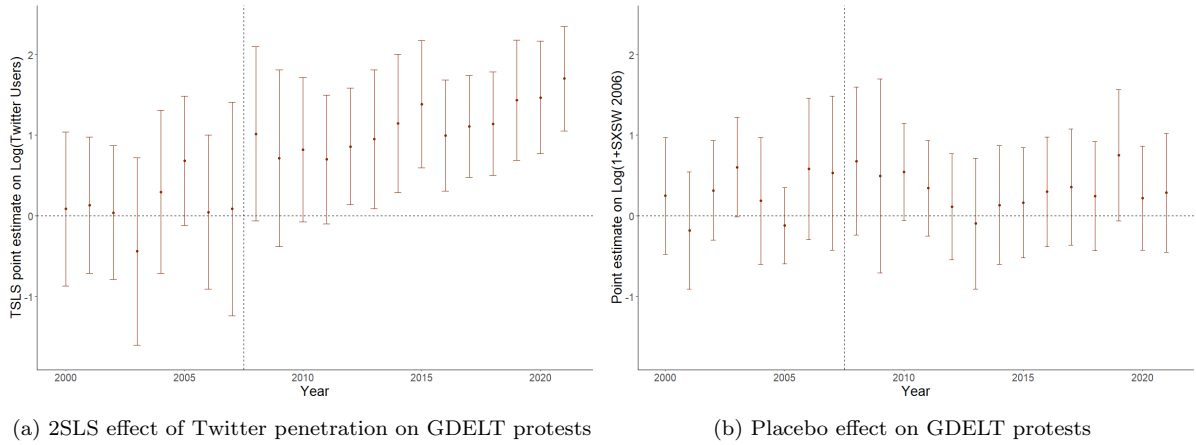


Figure A9: Twitter and GDELT protest, de-duplicated at the coordinates X event type X date level – 2SLS

Note: This figure presents point estimates and 95% confidence intervals for λ_1 and λ_2 from the 2SLS model represented in equation (4), estimated on the matched sample ($N=187$) where each control observation is weighted by the number of treated observations it is matched to. The outcome variables are the log number of yearly protests in the GDELT database, where all events occurring in the same county and at the same date are counted as one, from 2000-2021. An increase from 0 to 1 is weighted as a 100% increase. Panel (a) plots the point estimate and standard error for λ_1 , measuring the effect of Twitter penetration (measured in 2014-2015) on protest frequency. Panel (b) plots the point estimate and standard error for λ_2 from the same regressions, measuring the correlation between the log number of (one plus) SXSW followers who created their account in 2006, and the outcome variable. All regressions control for state fixed effects, population deciles, demographic, geographic, socio-economic, media, crime, and pre-2010 election controls (see description of controls in Table A9 in the Appendix). Red dots represent point estimates and whiskers represent 95% confidence intervals based on robust standard errors clustered by state. For ease of interpretation, a horizontal dashed line is placed at 0 and a vertical dashed line is placed between the years 2007 and 2008.

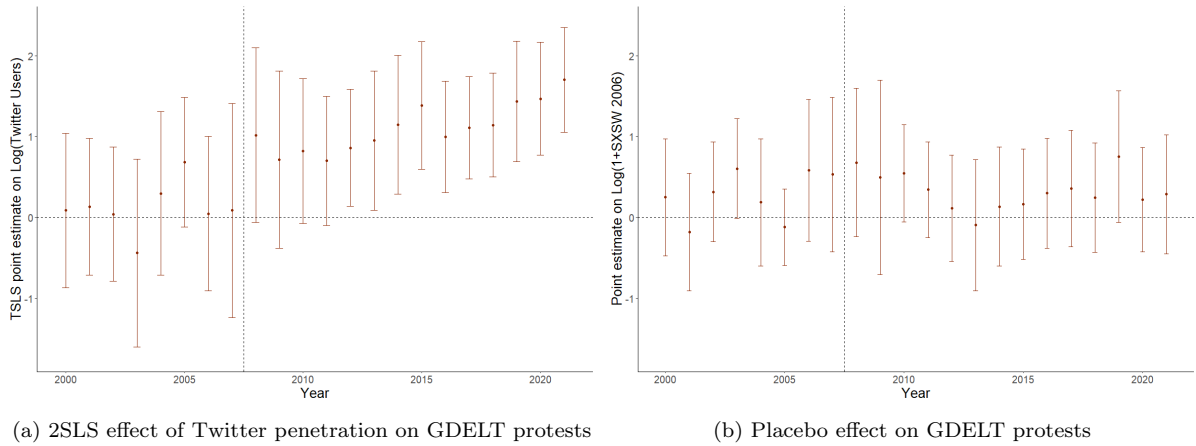


Figure A10: Twitter and GDELT protest, de-duplicated at the county X date level – 2SLS

Note: This figure presents point estimates and 95% confidence intervals for λ_1 and λ_2 from the 2SLS model represented in equation (4), estimated on the matched sample ($N=187$) where each control observation is weighted by the number of treated observations it is matched to. The outcome variables are the log number of yearly protests in the GDELT database, where all events occurring in the same county and at the same date are counted as one, from 2000-2021. An increase from 0 to 1 is weighted as a 100% increase. Panel (a) plots the point estimate and standard error for λ_1 , measuring the effect of Twitter penetration (measured in 2014-2015) on protest frequency. Panel (b) plots the point estimate and standard error for λ_2 from the same regressions, measuring the correlation between the log number of (one plus) SXSW followers who created their account in 2006, and the outcome variable. All regressions control for state fixed effects, population deciles, demographic, geographic, socio-economic, media, crime, and pre-2010 election controls (see description of controls in Table A9 in the Appendix). Red dots represent point estimates and whiskers represent 95% confidence intervals based on robust standard errors clustered by state. For ease of interpretation, a horizontal dashed line is placed at 0 and a vertical dashed line is placed between the years 2007 and 2008.

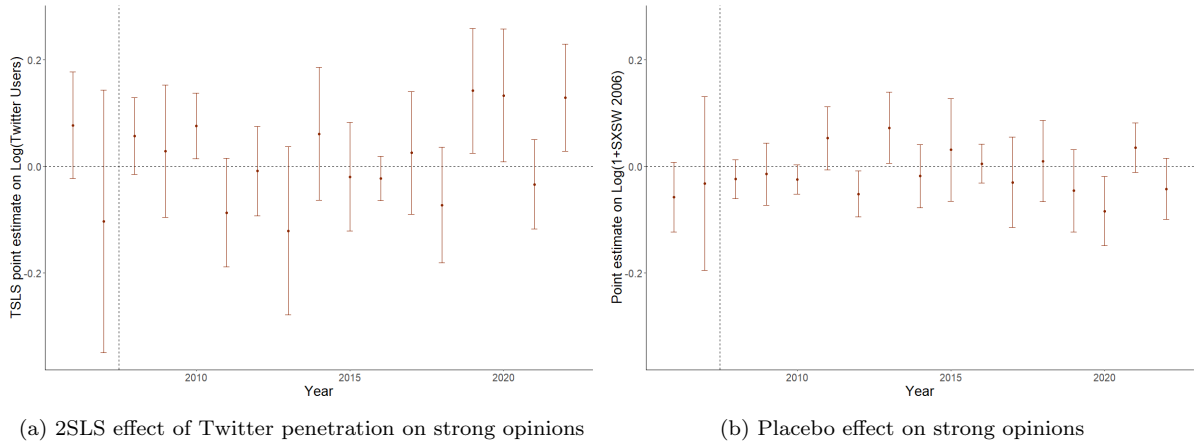


Figure A11: Twitter and the first principal component of strong opinions – 2SLS

Note: This figure presents point estimates and 90% confidence intervals for π_1 and π_2 from the 2SLS model represented in equation (6), estimated on a sample of all respondents from the Cooperative election study with non-missing answers, from the full sample of counties ($N=5,873-63,584$, depending on year). The outcome variables are the first principal component of strong opinions to questions related to race issues, gender/sexuality/environmental issues, or partisanship. Panel (a) plots the point estimate and standard error for π_1 , measuring the effect of Twitter penetration in one's local environment (measured in 2014-2015) on holding strong opinions. Panel (b) plots the point estimate and standard error for π_2 from the same regressions, measuring the correlation between the log number of (one plus) SXSW followers who created their account in 2006, and the outcome variable. All regressions control for state fixed effects and the full set of county-characteristics (see description of controls in Table A9 in the Appendix), and the following individual-level controls: Age, gender, race, ethnicity, religion, education, marital status, employment status, family income, home ownership, union membership, military service, and number of children. Red dots represent point estimates and whiskers represent 90% confidence intervals based on robust standard errors clustered by county. For ease of interpretation, a horizontal dashed line is placed at 0 and a vertical dashed line is placed between the years 2007 and 2008.

Table A1: Description of filtering processes for topic categorization: Extreme groups

Category	Filter by
Extreme left	
a) Radical racial justice:	"Antifa" or "RBPP: Revolutionary Black Panther Party" or "NFAC: Not Fucking Around Coalition" or "TPR: The People's Revolution" or "Take 'Em Down" or "NBPP: New Black Panther Party" or "Black Panthers" or "Black Man Army" or "NABPP: New Afrikan Black Panther Party" or "Detroit Will Breathe" or "No Sleep Till Justice" or "ADOS: American Descendants of Slavery" or "NAIP: New Afrikan Independence Party"
b) Other:	"Cancel the Rents Movement" or "PSL: Party for Socialism and Liberation" or "Refuse Fascism" or "DSA: Democratic Socialists of America" or "Extinction Rebellion" or "APSP: African People's Socialist Party" or "SAP: Socialist Alternative Party" or "CPUSA: Communist Party of the United States of America" or "RCP: Revolutionary Communist Party" or "SWP: Socialist Workers Party"
Extreme right	
a) White supremacists:	"White Nationalists" or "Boogaloo boys" or "Proud Boys" or "Groypers" or "White Defense Force" or "Aryan Nations" or "ABC: Aryan Cowboys Brotherhood" or "QAnon" or "Super Happy Fun America" or "GDL: Goyim Defense League" or "NSC: Nationalist Social Club" or "Boogaloo Boys" or "KKK: Ku Klux Klan" or "Minnesota Patriot Alliance" or "NSM: National Socialist Movement" or "Patriot Front" or "UADF: United American Defence Force" or "Oath Keepers" or ("Washington State III%ers" or "Patriot Prayer" and not "coronavirus" or "gun regulations" in notes)

Table A2: Description of filtering processes for topic categorization: Coronavirus and Presidential elections

Topic	Filter by
Coronavirus public health measures	Notes include: "vaccine" or "vaccinated" or "vaccination" or "mask mandate" or "mask-mandate" or "must wear masks" or "coronavirus mandate" or "coronavirus prevention measures" or "COVID-related restrictions" or "COVID restrictions" or "Covid restrictions" or "public health protocols" or "coronavirus protocols" or "health protection" or "slow the spread" or "wearing masks" or "mask wearing" or "anti-mask" or "pro-mask" or "universal masking" or "coronavirus related safety" or "mask policy" or ("mask" and "coronavirus") or "coronavirus pandemic restrictions" or "restrictions related to the coronavirus" or "contact tracing" or "health and safety measures" or "coronavirus protection" or "coronavirus-related policies" or "public health measures" or "coronavirus test" or "coronavirus precautions" or "quarantine" or "Quarantine" or ("safety protocols" and "corona" or "coronavirus-related mandates" or "stricter measures" or ((("mandates" or "measures") and ("corona" or "virus" or "covid"))) or "coronavirus restrictions" or "policies on coronavirus" or "protections against the spread" or ("restrictions" and ("corona" or "pandemic" or "covid" or "COVID"))) or "in-person learning" or "remote learning" or "Reopen Our Cities" or "reopen businesses" or "reopening businesses" or "business closure" or "school closure" or ("lockdown" and "corona") or "reopening schools" or "reopen schools" or "in-person school" or "reopening of schools" or "in-person classes" or "reopen bars" or "reopening of bars" or "closing of bars" or "restaurant closure" or ("stay-at-home order" and "corona")
Presidential elections (Left leaning)	Notes include: count every vote" tag
Presidential elections (Right leaning)	Notes include: "stop the steal" tag

Table A3: Description of filtering processes for topic categorization: Abortion related

Category	Filter by
Abortion rights, Pro-Choice (Left-leaning)	Notes include: ("reproductive rights" <i>or</i> "support of abortion") <i>or</i> Associated actors include: ("PP: Planned Parenthood")
Abortion rights, Pro-Life (Right-leaning)	Notes include: ("pro-life" <i>or</i> "right to live" <i>or</i> "anti abortion" <i>or</i> "Anti abortion" <i>or</i> "anti-abortion" <i>or</i> "against abortion" <i>or</i> "protesting abortion" <i>or</i> ("abortion" <i>and</i> "murder")) <i>or</i> Associated actors include: ("RTL: Right To Life" <i>or</i> "Family Policy Alliance") <i>or</i> (associated actors include "Christian Group" <i>and</i> notes include "abortion")

Table A4: Description of filtering processes for topic categorization: BLM related

Category	Filter by
BLM or related (Left-leaning)	Associated actors include: "BLM: Black Lives Matter" <i>or</i> "Antifa" <i>or</i> "BU: Black Unity" <i>or</i> "RBPP: Revolutionary Black Panther Party" <i>or</i> "NFAC: Not Fucking Around Coalition" <i>or</i> "Expect US" <i>or</i> "NOI: Nation of Islam" <i>or</i> "TPR: The People's Revolution"
Pro-Police/Counter to BLM (Right-leaning)	Associated actors include: ("Back the Blue" <i>or</i> "Blue Lives Matter" <i>or</i> "Gastonia Watchmen" <i>or</i> "West Virginia 34 Mountain Milita" <i>or</i> "Pro-Police Group") <i>or</i> Notes include: "back the blue"

Table A5: Description of filtering processes for topic categorization: Racial issues

Category	Filter by
Anti Racism (Left leaning)	The same associated actors as "BLM or related," and in addition: "Stop Asian Hate" <i>or</i> "Take 'Em Down" <i>or</i> "BVM: Black Voters Matter" <i>or</i> "NAACP: National Association for the Advancement of Colored People" <i>or</i> "NBPP: New Black Panther Party" <i>or</i> "Black Panthers"
Pro-Police/Counter to BLM/Racist (Right leaning)	The same associated actors as "Pro-Police/Counter to BLM," and in addition: "White Nationalists" <i>or</i> "Boogaloo boys" <i>or</i> "Proud Boys" <i>or</i> "Groypers" <i>or</i> "White Defence Force" <i>or</i> "Aryan Nations" <i>or</i> "ABC: Aryan Cowboys Brotherhood" <i>or</i> "QAnon" <i>or</i> "Super Happy Fun America" <i>or</i> "GDL: Goyim Defense League" <i>or</i> "NSC: Nationalist Social Club" <i>or</i> "KKK: Ku Klux Klan" <i>or</i> "Minnesota Patriot Alliance" <i>or</i> "NSM: National Socialist Movement" <i>or</i> "Patriot Front" <i>or</i> "UADF: United American Defence Force" <i>or</i> ("Washington State III%ers" <i>or</i> "Patriot Prayer" <i>and</i> not "coronavirus" <i>or</i> "gun regulations" in notes)

Table A6: Description of filtering processes for topic categorization

Category	Filter by
Left leaning	Protest is "Count Every Vote" related <i>or</i> Pro-Choice related <i>or</i> BLM or Anti Racism related <i>or</i> "Extreme left." In addition: Associated actors include: "LGBT" <i>or</i> "NLM: Native Lives Matter" <i>or</i> "FFF: Fridays For Future" <i>or</i> "Cancel the Rents Movement" <i>or</i> "Abolish ICE" <i>or</i> "DEM: Democratic Party"
Right leaning	Protest is "Stop The Steal" related <i>or</i> Pro-Life related <i>or</i> Pro-Police/Counter to BLM/Racist <i>or</i> "Extreme right." In addition: Associated actors include: "Occupy Movement" <i>or</i> "Save Our Children" <i>or</i> "GOA: Gun Owners Of America" <i>or</i> "Everytown for Gun Safety" <i>or</i> "Sons of Liberty" <i>or</i> "Iron City Citizen Response Unit" <i>or</i> "GOP: Republican Party" <i>or</i> Notes include: "right to bear arms" <i>or</i> "2nd amendment" <i>or</i> "second amendment" <i>or</i> "against gun regulation" <i>or</i> "gun-rights" <i>or</i> "pro-gun"

Table A7: Distribution of SXSW March 2007 followers

		(1)	(2)
		Full sample	At least one protest
SXSW March 2007 followers	$\log(1+\text{SXSW March 2007})$	Frequency	
0	0	2951	1606
1	0.693	90	85
2	1.099	25	25
3	1.386	11	11
4	1.609	3	3
5	1.792	2	2
6	1.946	2	2
7	2.079	3	3
8	2.197	1	1
9	2.303	2	2
11	2.485	1	1
12	2.565	2	2
13	2.639	1	1
14	2.708	1	1
15	2.773	1	1
16	2.833	1	1
19	2.996	1	1
20	3.045	1	1
24	3.219	3	1
64	4.174	1	1
87	4.477	1	1
102	4.635	1	1
144	4.977	1	1
N		3,106	1,758
Percent non-zero:		4.99%	8.65%

Note: This table shows the distribution of SXSW March 2007 followers in the full sample and in the sample of counties where at least one protest occurred.

Each row represents a unique value for SXSW March 2007 followers and columns (1) and (2) show the frequency of that value in each sample.

Table A8: Description of survey questions

Question	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Race issues																	
Some people think that if a company has a history of discriminating against blacks when making hiring decisions, then they should be required to have an affirmative action program that gives blacks preference in hiring. What do you think? Should companies that have discriminated against blacks have to have an affirmative action program?	x	x	x	x													
Affirmative action programs give preference to racial minorities in employment and college admissions in order to correct for past discrimination. Do you support or oppose affirmative action ?					x	x	x	x	x								
The Irish, Italians, Jews and many other minorities overcame prejudice and worked their way up. Blacks should do the same without any special favors								x	x				x		x		x
Generations of slavery and discrimination have created conditions that make it difficult for Blacks to work their way out of the lower class								x	x				x		x		x
Do the police make you feel? (Mostly safe to mostly unsafe)											x				x		x
I am angry that racism exists												x					
I often find myself fearful of people of other races												x					
Over the past few years, blacks have gotten less than they deserve												x					
It's really a matter of some people not trying hard enough, if blacks would only try harder they could be just as well off as whites.												x					
White people in the US have certain advantages because of the color of their skin													x	x	x	x	x
Racial problems in the U.S. are rare, isolated situations													x	x	x	x	x
Gender/sexuality issues																	
President Bush recently spoke out in favor of a Constitutional Amendment defining marriage as strictly between a man and a woman. Do you support or oppose a Constitutional amendment banning gay marriage?	x	x															
There has been some discussion about abortion during recent years. Which one of the opinions on this page best agrees with your view on this issue? (From abortion should never be permitted to always permitted)	x	x	x	x	x	x	x	x		x	x	x	x	x		x	
When women lose to men in a fair competition, they typically complain about being discriminated against													x	x			
Feminists are making entirely reasonable demands of men													x	x			
Women seek to gain power by getting control over men															x		x
Women are too easily offended															x		x
Environmental issues																	
From what you know about global climate change or global warming, which one of the following statements comes closest to your opinion? (From "Immediate and drastic action necessary" to "this is not a serious problem" or "climate change is not occurring")	x	x		x	x	x	x	x									
Some people think it is important to protect the environment even if it costs some jobs or otherwise reduces our standard of living. Other people think that protecting the environment is not as important as maintaining jobs and our standard of living. Which is closer to the way you feel, or haven't you thought much about this? (Environment much more important to jobs much more important, 5 point scale)			x		x		x	x									
Partisanship/Ideology																	
Thinking about politics these days, how would you describe your own political viewpoint? (Very liberal to very conservative)	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Party affiliation, from strong democrat to strong republican	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
President approval: Do you approve or disapprove of the way XXXX is handling his job as president?	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

Table A9: Description of control variables

Variables	Description	Source
Demographic controls	Include total population, the share of people in age buckets 20-24, 25-29, 30-34, 35-39, 40-44, 45-49 and 50+, the share of Women, the share of Black or African American, American Indian and Alaska Native, Asian, Native Hawaiian or other Pacific Islander and the share of Hispanic among the total population.	2010 US Census
Geographic controls	Log(Area) in square kilometers, Log(Distance from Austin, Texas), population density 2016	US Census Tigerline File
Socio-economic controls 1	Poverty rates and median household income	2009 US Census Bureau's SAIGE program
Socio-economic controls 2	Share of adult population with at least high school diploma, share of adult population with at least Bachelor degree	2010 American Community Survey
Socio-economic controls 3	Unemployment rate and Employment-to-Population rate in 2010	Bureau of Labor Statistics
Media controls	Ratio of prime time TV viewership to population, cable spending to population, and the share of Fox News viewership	SimplyAnalytics
Crime controls	The number of violent crime per capita, the number of property crimes per capita	FBI UCR Data
Pre-2010 election controls	Share of votes towards the Republican candidate in 2000 (George W. Bush), 2004 (George W. Bush), and 2008 (John McCain)	MIT Election Lab
Pre-Twitter protest controls	Number of yearly protests from 2000 to 2007	GDELT

Table A10: IV balance

Variable:	(1) Both	(2) March 2007 <i>only</i>	(3) 2006 <i>only</i>	(4) T-test (2)-(3) P-value	(5) T-test (2)-(3) Q-value
log(Population in 2019)	13.38	12.72	12.34	(0.238)	(1.000)
Share aged 20-24	0.080	0.081	0.079	(0.831)	(1.000)
Share aged 25-29	0.083	0.069	0.073	(0.495)	(1.000)
Share aged 30-34	0.075	0.063	0.065	(0.654)	(1.000)
Share aged 35-39	0.071	0.063	0.063	(1.000)	(1.000)
Share aged 40-44	0.069	0.066	0.065	(0.981)	(1.000)
Share aged 45-49	0.072	0.072	0.073	(0.697)	(1.000)
Share 50 or older	0.293	0.320	0.322	(0.872)	(1.000)
Share women	0.512	0.509	0.505	(0.189)	(1.000)
Share white	0.683	0.804	0.815	(0.718)	(1.000)
Share black	0.191	0.125	0.089	(0.060)*	(1.000)
Share native american	0.010	0.009	0.018	(0.280)	(1.000)
Share Asian	0.088	0.039	0.044	(0.675)	(1.000)
Share Hispanic	0.189	0.146	0.135	(0.714)	(1.000)
Population growth 2000-2019	0.203	0.205	0.177	(0.578)	(1.000)
Log(Area)	7.26	7.59	7.49	(0.755)	(1.000)
Log(Distance from Austin, Texas)	7.22	7.31	7.18	(0.545)	(1.000)
Population Density	1,551.7	360.5	632.3	(0.354)	(1.000)
Share Poor	0.144	0.145	0.132	(0.284)	(1.000)
Log median hh income	10.92	10.83	10.89	(0.280)	(1.000)
Share adults with highschool degree	0.858	0.870	0.882	(0.297)	(1.000)
Share adults with bachelor degree	0.373	0.316	0.332	(0.628)	(1.000)
Unemployment rate	0.088	0.087	0.075	(0.028)**	(1.000)
Employment to population ratio	0.474	0.464	0.478	(0.325)	(1.000)
Primetime TV viewership to population	0.42	0.43	0.43	(0.910)	(1.000)
Cable spending per capita	423.22	746.65	750.27	(0.577)	(1.000)
Fox TV viewership share	0.251	0.256	0.256	(0.900)	(1.000)
Fox online viewership share	0.127	0.123	0.126	(0.420)	(1.000)
Violent crime per capita	0.006	0.004	0.004	(0.802)	(1.000)
Property crime per capita	0.034	0.030	0.024	(0.035)**	(1.000)
Republican vote share 2000	0.347	0.479	0.489	(0.777)	(1.000)
Republican vote share 2004	0.383	0.501	0.511	(0.790)	(1.000)
Republican vote share 2008	0.327	0.444	0.470	(0.478)	(1.000)
log(Pre2007 protest activity)	2.90	1.313	1.001	(0.202)	(1.000)
Observations	47	108	20		

Note: This table presents mean values of the control variables for three types of counties: (1) counties with new SXSW followers in March 2007 *and* in 2006; (2) counties with new SXSW followers in March 2007 but no new followers in 2006; (3) counties with new SXSW followers in 2006 but no new followers in March 2007. P-value represents the p-value from a simple t-test for the equality of means between the counties with the key identification strategy and the q-value is a FDR adjusted p-value from the same test in the style of [Anderson \(2008\)](#). *p<0.1; **p<0.05; ***p<0.01

Table A11: First stage analysis, inverse hyperbolic sine of SXSW follower

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent variable:	Log(Twitter Users)							
Inverse hyperbolic sine of SXSW March2007	0.496*** (0.046)	0.373*** (0.041)	0.356*** (0.042)	0.345*** (0.041)	0.341*** (0.042)	0.335*** (0.041)	0.330*** (0.041)	0.308*** (0.041)
Inverse hyperbolic sine of SXSW 2006	0.196*** (0.055)	0.104* (0.059)	0.081 (0.060)	0.106* (0.061)	0.102* (0.060)	0.092 (0.057)	0.088 (0.057)	0.077 (0.057)
Inverse hyperbolic sine of Protests 2000-2007								0.044*** (0.012)
R ²	0.925	0.939	0.940	0.945	0.945	0.946	0.947	0.947
Cluster robust F-statistic	91.99	57.14	52.72	42.46	40.98	41.67	40.47	37.85
p-value: March 2007 = 2006	0.01**	0.02**	0.01**	0.04**	0.04**	0.03**	0.04**	0.04**
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Population deciles	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Demographic controls		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Geographic controls			Yes	Yes	Yes	Yes	Yes	Yes
Socio-economic controls				Yes	Yes	Yes	Yes	Yes
Media controls					Yes	Yes	Yes	Yes
Crime controls						Yes	Yes	Yes
Pre 2010 election controls							Yes	Yes
Observations	3,108	3,108	3,108	3,108	3,107	3,107	3,106	3,106

Note: This table presents county-level regressions where the natural logarithm of our Twitter users from 2014-2015 is the dependent variable. SXSW March2007 is the number of Twitter users who joined Twitter in March 2007 and follow South by Southwest (@SXSW). SXSW 2006 is the number of Twitter users who joined in 2006 and follow SXSW. Regressions include the indicated control variables (see Table A9 in the Appendix for their descriptions). Robust standard errors in parentheses are clustered by state. *p<0.1; **p<0.05; ***p<0.01

Table A12: Effect of Twitter penetration on the frequency of overall protests

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent variable:	Log(Protests)							
Panel A: 2SLS Second Stage								
Log(Twitter Users)	1.029*** (0.126)	0.904*** (0.119)	1.004*** (0.136)	0.954*** (0.144)	0.931*** (0.143)	0.930*** (0.143)	0.916*** (0.146)	0.843*** (0.162)
Log(1+SXSW2006)	−0.123 (0.111)	−0.121 (0.094)	−0.083 (0.082)	−0.046 (0.085)	−0.053 (0.084)	−0.056 (0.083)	−0.054 (0.083)	−0.063 (0.085)
Log(1+Protests 2000-2007)								0.054* (0.030)
Cluster robust F-statistic	84.46	53.33	49.18	39.48	38.11	38.60	37.50	34.99
Panel B: Reduced-form								
Log(1+SXSW March2007)	0.636*** (0.099)	0.421*** (0.066)	0.445*** (0.065)	0.410*** (0.075)	0.396*** (0.075)	0.388*** (0.072)	0.377*** (0.074)	0.321*** (0.073)
Log(1+SXSW 2006)	0.096 (0.116)	−0.031 (0.115)	−0.010 (0.110)	0.054 (0.115)	0.040 (0.113)	0.025 (0.112)	0.021 (0.114)	−0.006 (0.114)
Log(1+Protests 2000-2007)								0.100*** (0.026)
R ²	0.769	0.801	0.802	0.808	0.809	0.810	0.811	0.812
Panel C: OLS								
Log(Twitter Users)	0.576*** (0.055)	0.392*** (0.043)	0.395*** (0.044)	0.340*** (0.044)	0.330*** (0.043)	0.313*** (0.042)	0.311*** (0.043)	0.295*** (0.042)
Log(1+SXSW 2006)	0.303*** (0.075)	0.187* (0.097)	0.236** (0.099)	0.287*** (0.099)	0.266*** (0.097)	0.259*** (0.097)	0.248** (0.100)	0.177* (0.134)
Log(1+Protests 2000-2007)								0.093*** (0.026)
R ²	0.788	0.808	0.808	0.812	0.813	0.814	0.814	0.815
Observations	3,108	3,108	3,108	3,108	3,107	3,107	3,106	3,106
Mean number of protests	11.00	11.00	11.00	11.00	11.00	11.00	11.00	11.00
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Population deciles	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Demographic controls		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Geographic controls			Yes	Yes	Yes	Yes	Yes	Yes
Socio-economic controls				Yes	Yes	Yes	Yes	Yes
Media controls					Yes	Yes	Yes	Yes
Crime controls						Yes	Yes	Yes
Pre 2010 election controls							Yes	Yes

Note: This table presents county-level regressions for an OLS and a 2SLS model of the effect of Twitter penetration on protest frequency between January 2020 and November 2021. An extensive margin increase in protests from 0 to 1 is valued as a 150% increase. Log(Twitter Users) is the natural logarithm of our Twitter users from 2014-2015. SXSW March2007 is the number of Twitter users who joined Twitter in March 2007 and follow South by Southwest (@SXSW). SXSW 2006 is the number of Twitter users who joined in 2006 and follow SXSW. Regressions include the indicated control variables (see Table A9 in the Appendix for their descriptions). Robust standard errors in parentheses are clustered by state. *p<0.1; **p<0.05; ***p<0.01

Table A13: Reduced form effect on protests by extremity

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A								
Dependent variable:	Log(Number of extreme protests)							
Log(1+SXSW March2007)	0.878*** (0.151)	0.781*** (0.140)	0.773*** (0.141)	0.765*** (0.142)	0.760*** (0.142)	0.761*** (0.141)	0.761*** (0.140)	0.696*** (0.138)
Log(1+SXSW 2006)	0.207 (0.182)	0.134 (0.185)	0.118 (0.189)	0.130 (0.189)	0.124 (0.190)	0.110 (0.191)	0.110 (0.192)	0.079 (0.190)
Log(1+Protests 2000-2007)								0.114*** (0.026)
R ²	0.562	0.582	0.583	0.585	0.586	0.587	0.587	0.595
Mean number of protests	0.524	0.524	0.524	0.524	0.524	0.524	0.524	0.524
Panel B								
Dependent variable:	Log(Number of non-extreme protests)							
Log(1+SXSW March2007)	0.618*** (0.096)	0.406*** (0.064)	0.431*** (0.063)	0.396*** (0.073)	0.382*** (0.073)	0.374*** (0.070)	0.362*** (0.072)	0.306*** (0.071)
Log(1+SXSW 2006)	0.103 (0.114)	-0.017 (0.112)	0.004 (0.108)	0.068 (0.113)	0.053 (0.110)	0.039 (0.109)	0.034 (0.110)	0.007 (0.111)
Log(1+Protests 2000-2007)								0.099*** (0.026)
R ²	0.767	0.799	0.800	0.806	0.807	0.809	0.809	0.810
Mean number of protests	10.48	10.48	10.48	10.48	10.48	10.48	10.48	10.48
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Population deciles	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Demographic controls		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Geographic controls			Yes	Yes	Yes	Yes	Yes	Yes
Socio-economic controls				Yes	Yes	Yes	Yes	Yes
Media controls					Yes	Yes	Yes	Yes
Crime controls						Yes	Yes	Yes
Pre 2010 election controls							Yes	Yes
Observations	3,108	3,108	3,108	3,108	3,107	3,107	3,106	3,106
p-value: $\beta_{Extreme} = \beta_{Non-extreme}$	0.027**	0.000***	0.001***	0.000***	0.000***	0.000***	0.000***	0.000***

Note: This table presents county-level regressions for the reduced form effect of South by Southwest followers on protest frequency from January 2020 to November 2021, depending on extreme group involvement. An extensive margin increase in protests from 0 to 1 is valued as a 150% increase. SXSW March2007 is the number of Twitter users who joined Twitter in March 2007 and follow South by Southwest (@SXSW). SXSW 2006 is the number of Twitter users who joined in 2006 and follow SXSW. Extreme refers to demonstrations involving a political group defined as extreme. The groups classified as extreme are listed in Table A1. Regressions include the indicated control variables (see Table A9 in the Appendix for their descriptions). Robust standard errors in parentheses are clustered by state. *p<0.1; **p<0.05; ***p<0.01

Table A14: Reduced form effect on protests by extremity, inverse hyperbolic sine of SXSW followers

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A								
Dependent variable:	Log(Number of extreme protests)							
Inverse hyperbolic sine of SXSW March2007	0.698*** (0.119)	0.621*** (0.111)	0.613*** (0.111)	0.607*** (0.112)	0.603*** (0.112)	0.603*** (0.111)	0.603*** (0.110)	0.545*** (0.109)
Inverse hyperbolic sine of SXSW 2006	0.210 (0.143)	0.152 (0.147)	0.136 (0.151)	0.146 (0.151)	0.140 (0.152)	0.130 (0.153)	0.130 (0.153)	0.106 (0.152)
Inverse hyperbolic sine of Protests 2000-2007								0.094*** (0.022)
R ²	0.563	0.582	0.584	0.585	0.586	0.587	0.588	0.595
Mean number of protests	0.524	0.524	0.524	0.524	0.524	0.524	0.524	0.524
Panel B								
Dependent variable:	Log(Number of non-extreme protests)							
Inverse hyperbolic sine of SXSW March2007	0.500*** (0.075)	0.329*** (0.050)	0.347*** (0.050)	0.318*** (0.057)	0.306*** (0.057)	0.300*** (0.055)	0.291*** (0.057)	0.248*** (0.056)
Inverse hyperbolic sine of SXSW 2006	0.106 (0.086)	0.007 (0.089)	0.025 (0.086)	0.074 (0.090)	0.062 (0.088)	0.050 (0.087)	0.046 (0.088)	0.026 (0.089)
Inverse hyperbolic sine of Protests 2000-2007								0.084*** (0.022)
R ²	0.768	0.800	0.800	0.806	0.807	0.809	0.809	0.810
Mean number of protests	10.48	10.48	10.48	10.48	10.48	10.48	10.48	10.48
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Population deciles	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Demographic controls		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Geographic controls			Yes	Yes	Yes	Yes	Yes	Yes
Socio-economic controls				Yes	Yes	Yes	Yes	Yes
Media controls					Yes	Yes	Yes	Yes
Crime controls						Yes	Yes	Yes
Pre 2010 election controls							Yes	Yes
Observations	3,108	3,108	3,108	3,108	3,107	3,107	3,106	3,106
p-value: $\beta_{Extreme} = \beta_{Non-extreme}$	0.033**	0.001***	0.001***	0.001***	0.000***	0.000***	0.000***	0.000***

Note: This table presents county-level regressions for the reduced form effect of South by Southwest followers on protest frequency from January 2020 to November 2021, depending on extreme group involvement. An extensive margin increase in protests from 0 to 1 is valued as a 150% increase. SXSW March2007 is the number of Twitter users who joined Twitter in March 2007 and follow South by Southwest (@SXSW). SXSW 2006 is the number of Twitter users who joined in 2006 and follow SXSW. Extreme refers to demonstrations involving a political group defined as extreme. The groups classified as extreme are listed in Table A1. Regressions include the indicated control variables (see Table A9 in the Appendix for their descriptions). Robust standard errors in parentheses are clustered by state.

*p<0.1; **p<0.05; ***p<0.01

Table A15: 2SLS effect on protests by extremity, alternative outcome transformation

	(1)	(2)	(3)	(4)
Functional form of dependent variable:	D=1 if X>0	Log(X), 0 ->1 valued as a 0% increase	Log(X), 0 ->1 valued as a 100% increase	Log(X), 0 ->1 valued as a 200% increase
Panel A				
Dependent variable:	Extreme protests			
Log(Twitter Users)	0.232*** (0.080)	1.480*** (0.280)	1.712*** (0.317)	1.944*** (0.368)
Log(1+SXSX 2006)	-0.029 (0.056)	-0.001 (0.164)	-0.030 (0.164)	-0.059 (0.182)
Log(1+Protests 2000-2007)	0.018* (0.011)	-0.013 (0.029)	0.005 (0.032)	0.024 (0.039)
Mean dependent variable	0.11	0.09	-0.80	-1.69
Panel B				
Dependent variable:	Non-extreme protests			
Log(Twitter Users)	-0.188*** (0.048)	1.086*** (0.157)	0.898*** (0.155)	0.710*** (0.167)
Log(1+SXSX 2006)	0.054 (0.041)	-0.128 (0.084)	-0.074 (0.078)	-0.020 (0.092)
Log(1+Protests 2000-2007)	0.013 (0.009)	0.036 (0.025)	0.049* (0.027)	0.062* (0.032)
Mean dependent variable	0.56	0.92	0.48	0.05
Cluster robust F-statistic	34.99	34.99	34.99	34.99
Observations	3,106	3,106	3,106	3,106
p-value: $\beta_{Extreme}^{RF} = \beta_{Non-extreme}^{RF}$	0.000***	0.002***	0.000***	0.000***

Note: This table presents county-level regressions from the second stage of an 2SLS model of the effect of Twitter penetration on protest frequency from January 2020 to November 2021, depending on extreme group involvement. In column (1), the outcome is an indicator variable equal to 1 if a protest occurred, and in columns (2)-(4) the outcome is the log number of protests, with an extensive margin increase in protests from 0 to 1 valued as a 0%, 100%, and a 200% increase respectively. Log(Twitter Users) is the natural logarithm of unique Twitter users from a sample of Tweets from 2014-2015. They are instrumented for via the log number of (one plus) Twitter users who joined Twitter in March 2007 and follow South by Southwest (@SXSX). SXSX 2006 is the number of Twitter users who joined in 2006 and follow SXSX. Extreme refers to demonstrations involving a political group defined as extreme. The groups classified as extreme are listed in Table A1. All regressions control for State fixed effects and the full set of controls (as in column (8) of Table 2), see Table A9 in the Appendix for their descriptions). Robust standard errors in parentheses are clustered by state. *p<0.1; **p<0.05; ***p<0.01

Table A16: First stage, sample of counties with at least one protest

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent variable:	Log(Twitter Users)							
Log(1+SXS March 2007)	0.642*** (0.064)	0.483*** (0.053)	0.469*** (0.050)	0.456*** (0.052)	0.455*** (0.051)	0.441*** (0.050)	0.442*** (0.049)	0.405*** (0.049)
Log(1+SXS 2006)	0.190** (0.074)	0.084 (0.072)	0.068 (0.076)	0.103 (0.077)	0.103 (0.077)	0.094 (0.073)	0.095 (0.073)	0.076 (0.074)
Log(1+Protests 2000-2007)								0.068*** (0.016)
R ²	0.905	0.928	0.930	0.937	0.937	0.939	0.940	0.941
Cluster robust F-statistic	88.40	57.60	60.64	50.13	50.04	50.16	51.11	47.88
p-value: March 2007 = 2006	0.004***	0.006***	0.004***	0.014**	0.014**	0.014**	0.013**	0.014**
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Population deciles	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Demographic controls		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Geographic controls			Yes	Yes	Yes	Yes	Yes	Yes
Socio-economic controls				Yes	Yes	Yes	Yes	Yes
Media controls					Yes	Yes	Yes	Yes
Crime controls						Yes	Yes	Yes
Pre 2010 election controls							Yes	Yes
Observations	1,758	1,758	1,758	1,758	1,757	1,757	1,756	1,756

Note: This table presents county-level regressions where the natural logarithm of our Twitter users from 2014-2015 is the dependent variable. The sample only includes counties where at least one protest occurred. SXS March 2007 is the number of Twitter users who joined Twitter in March 2007 and follow South by Southwest (@SXS). SXS 2006 is the number of Twitter users who joined in 2006 and follow SXS. Regressions include the indicated control variables (see Table A9 in the Appendix for their descriptions). Robust standard errors in parentheses are clustered by state. *p<0.1; **p<0.05; ***p<0.01

Table A17: 2SLS: Effect of Twitter penetration on the share of extreme protests, inverse hyperbolic sine of SXS followers

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent variable:	Share of extreme protests							
Log(Twitter Users)	0.027*** (0.007)	0.030*** (0.009)	0.029*** (0.011)	0.031*** (0.011)	0.032*** (0.011)	0.034*** (0.011)	0.036*** (0.011)	0.037*** (0.013)
Inverse hyperbolic sine of SXS 2006	-0.010 (0.007)	-0.014** (0.007)	-0.014** (0.007)	-0.015** (0.007)	-0.015** (0.007)	-0.015** (0.007)	-0.015** (0.007)	-0.014** (0.007)
Inverse hyperbolic sine of Protests 2000-2007								-0.001 (0.002)
Mean dependent variable	0.022	0.022	0.022	0.022	0.022	0.022	0.022	0.022
Cluster robust F-statistic	96.58	61.83	63.72	52.76	52.6	52.93	53.03	50.55
Observations	1,758	1,758	1,758	1,758	1,757	1,757	1,756	1,756
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Population deciles	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Demographic controls		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Geographic controls			Yes	Yes	Yes	Yes	Yes	Yes
Socio-economic controls				Yes	Yes	Yes	Yes	Yes
Media controls					Yes	Yes	Yes	Yes
Crime controls						Yes	Yes	Yes
Pre 2010 election controls							Yes	Yes

Note: This table presents county-level second stage regression from a 2SLS model of the effect of Twitter penetration on the share of protests from January 2020 to November 2021 with extreme group involvement. The groups classified as extreme are listed in Table A1. The sample only includes counties where at least one protest occurred. Log(Twitter Users) is the natural logarithm of our Twitter users from 2014-2015. They are instrumented for via the inverse hyperbolic sine of Twitter users who joined Twitter in March 2007 and follow South by Southwest (@SXS). SXS 2006 is the number of Twitter users who joined in 2006 and follow SXS. Regressions include the indicated control variables (see Table A9 in the Appendix for their descriptions). Robust standard errors in parentheses are clustered by state. *p<0.1; **p<0.05; ***p<0.01

Table A18: 2SLS Effects on the share of protests with extreme group involvement, alternative controls for pre-Twitter protests

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Functional form of pre-Twitter protest controls:	Linear		Log(1+X)		Inverse hyperbolic sine		
Dependent variable:	Share of protests with extreme group involvement						
Log(Twitter Users)	0.037*** (0.011)	0.037*** (0.012)	0.039*** (0.011)	0.038*** (0.013)	0.038*** (0.013)	0.038*** (0.013)	0.037*** (0.013)
Log(1+SXSW 2006)	-0.019** (0.009)	-0.019** (0.009)	-0.016** (0.008)	-0.019** (0.009)	-0.018** (0.008)	-0.019** (0.009)	-0.018** (0.008)
Protests 2000-2007 ^a		-0.0000 (0.0000)		-0.001 (0.003)		-0.001 (0.002)	
Protests 2000 ^a			0.0005 (0.001)		0.013* (0.007)		0.010* (0.005)
Protests 2001 ^a			-0.003** (0.001)		-0.011* (0.006)		-0.008* (0.005)
Protests 2002 ^a			-0.003 (0.002)		-0.004 (0.007)		-0.003 (0.005)
Protests 2003 ^a			0.002 (0.001)		-0.001 (0.005)		-0.001 (0.004)
Protests 2004 ^a			0.006*** (0.002)		0.010 (0.007)		0.007 (0.006)
Protests 2005 ^a			0.001 (0.002)		0.003 (0.005)		0.003 (0.004)
Protests 2006 ^a			-0.002* (0.001)		-0.005 (0.008)		-0.004 (0.006)
Protests 2007 ^a			0.0002 (0.0004)		-0.002 (0.004)		-0.002 (0.003)
Cluster robust F-statistic	51.11	50.37	50.02	47.88	47.79	48.04	47.85
Mean dependent variable	0.022	0.022	0.022	0.022	0.022	0.022	0.022
Observations	1,756	1,756	1,756	1,756	1,756	1,756	1,756

Note: This table presents county-level regressions, on a subsample of counties where at least one protest occurred. The outcome variable is the share of protests with extreme group involvement from January 2020 to November 2021. The groups classified as extreme are listed in Table A1. SXSX March2007 is the number of Twitter users who joined Twitter in March 2007 and follow South by Southwest (@SXSX). SXSX 2006 is the number of Twitter users who joined in 2006 and follow SXSX. All regressions control for population via decile dummies, and the full set of controls (as in column (8) in Table 2), see Table A9 in the Appendix for their descriptions. Robust standard errors in parentheses are clustered by state. *p<0.1; **p<0.05; ***p<0.01

^a In columns (2)-(3), this variable is included as the linear number of protests. In columns (4)-(5), this variable is included as log(1+number of protests). In columns (6)-(7), this variable is included as the inverse hyperbolic sine of the number of protests.

Table A19: 2SLS on the share of protests with extreme group involvement, alternative de-duplication strategies for pre-Twitter protests

	(1)	(2)	(3)	(4)
GDELT de-duplication	None	Same date + event type + coordinates	Same date + coordinates	Same date + county
Dependent variable:	Share of protests with extreme group involvement			
Log(Twitter Users)	0.038*** (0.013)	0.038*** (0.013)	0.038*** (0.013)	0.038*** (0.013)
Log(1+SXSW 2006)	-0.019** (0.009)	-0.019** (0.009)	-0.019** (0.009)	-0.019** (0.009)
Log(1+Protests 2000-2007)	-0.001 (0.002)	-0.001 (0.002)	-0.001 (0.003)	-0.001 (0.003)
Cluster robust F-statistic	47.88	47.88	47.88	47.88
Mean dependent variable	0.022	0.022	0.022	0.022
Mean number GDELT protests	13.62	7.27	6.24	6.00
Observations	1,756	1,756	1,756	1,756

Note: This table presents county-level regressions, on a subsample of counties where at least one protest occurred. The outcome variable is the share among all protests with extreme group involvement from January 2020 to November 2021. The groups classified as extreme are listed in Table A1. Log(Twitter Users) is the natural logarithm of our Twitter users from 2014-2015. They are instrumented for via the log number of (one plus) Twitter users who joined Twitter in March 2007 and follow South by Southwest (@SXSW). SXSW 2006 is the number of Twitter users who joined in 2006 and follow SXSW. All regressions control for population via decile dummies, and the full set of controls (as in column (8) in Table 2), see Table A9 in the Appendix for their descriptions. Each column controls for protests in 2000-2007 from the GDELT database using different deduplication strategies. In column (1) the GDELT data is not de-duplicated, column (2) considers all events of the same type (types include for example “Demonstrate or rally for leadership change”, “demonstrate or rally for policy change”, “obstruct passage to demand rights”) occurring on the same date and at the same coordinates as the same protest, column (3) considers all events occurring on the same date and at the same coordinates as the same protest, and column (4) considers all protests occurring in the same county on the same date as one protest. Robust standard errors in parentheses are clustered by state. *p<0.1; **p<0.05; ***p<0.01

Table A20: Twitter penetration & IV correlation with Pre-Twitter protest activity

	(1)	(2)
Dependent variable:	log(Protests 2000-2007)	
Log(Twitter Users)	0.223*** (0.058)	
Log(1+ SXSW March 2007)		0.706*** (0.122)
Log(1+SXSW 2006)	0.831*** (0.156)	0.230 (0.221)
Observations	3,106	3,106
R ²	0.412	0.417
State FE	Yes	Yes
Population deciles	Yes	Yes
Demographic controls	Yes	Yes
Geographic controls	Yes	Yes
Socio-economic controls	Yes	Yes
Media controls	Yes	Yes
Crime controls	Yes	Yes
Pre 2010 election controls	Yes	Yes

Note: This table presents county-level OLS regressions. An extensive margin increase in protests from 0 to 1 is valued as a 150% increase. Log(Twitter Users) is the natural logarithm of our Twitter users from 2014-2015. All regressions control for population via decile dummies, and the full set of controls (as in column (8) in Table 2), see Table A9 in the Appendix for their descriptions. Robust standard errors in parentheses are clustered by state. *p<0.1; **p<0.05; ***p<0.01

Table A21: Balance table for matched sample

	(1)	(2)	(3)	(4)
	SXSW 2007 followers	No SXSW 2007 followers	Weighted difference	Weighted t-test adjusted-q-value
log(1+SXSW 2006)	0.147	0.169	-0.022	(1.000)
log(Population in 2016)	12.98	12.67	0.283	(0.832)
Share aged 20-24	0.082	0.081	0.001	(1.000)
Share aged 25-29	0.070	0.075	-0.005	(1.000)
Share aged 30-34	0.064	0.066	-0.002	(1.000)
Share aged 35-39	0.064	0.065	-0.001	(1.000)
Share aged 40-44	0.066	0.067	-0.001	(1.000)
Share aged 45-49	0.072	0.072	-0.000	(1.000)
Share 50 or older	0.314	0.313	0.001	(1.000)
Share women	0.510	0.509	0.001	(1.000)
Share white	0.791	0.807	-0.016	(1.000)
Share black	0.132	0.110	0.022	(1.000)
Share native american	0.009	0.010	-0.001	(1.000)
Share Asian	0.044	0.049	-0.005	(1.000)
Share Hispanic	0.153	0.152	0.001	(1.000)
Population growth 2000-2016	0.220	0.200	0.020	(1.000)
Log(Area)	7.59	7.43	0.16	(1.000)
Log(Distance from Austin, Texas)	7.33	7.52	-0.18	(0.832)
Population Density	4.914	5.624	0.710	(1.000)
Share Poor	14.44	13.33	1.10	(1.000)
Log median hh income	10.84	10.92	-0.08	(0.832)
Share adults with highschool degree	86.91	87.67	-0.46	(1.000)
Share adults with bachelor degree	32.19	35.57	-3.39	(0.832)
Unemployment rate	0.087	0.086	0.000	(1.000)
Employment to population ratio	46.47	48.11	-1.64	(0.832)
Primetime TV viewership to population	0.427	0.426	0.001	(1.000)
Cable spending per capita	745.42	749.60	-4.18	(1.000)
Fox TV viewership share	25.50	25.50	0.000	(1.000)
Fox online viewership share	12.60	12.70	-0.10	(1.000)
Violent crime per capita	0.004	0.003	0.001	(0.832)
Property crime per capita	0.031	0.028	0.002	(1.000)
Republican vote share 2000	0.470	0.449	0.021	(1.000)
Republican vote share 2004	0.491	0.466	0.024	(1.000)
Republican vote share 2008	0.432	0.411	0.021	(1.000)
log(1+Pre2007 protest activity)	1.508	1.725	-0.217	(1.000)
Observations	120	67	187	

Note: This table plots the weighted mean values of the control variables for counties with new SXSW followers in March 2007 (treated counties) and those without such followers. The counties without followers are weighted by the number of treatment counties they are matched to. The t-statistic q-value is a FDR-adjusted p-value in the style of [Anderson \(2008\)](#). *p<0.1; **p<0.05; ***p<0.00

Table A22: 2SLS protests by topic, alternative outcome transformation

	(1)	(2)	(3)	(4)	(5)
Topic of protest:	Coronavirus	Presidential elections	Abortion rights	BLM related	Racism related
Panel A					
Dependent variable:	Log(Number of protests) [Where 0->1 is valued as a 0% increase]				
Log(Twitter Users)	0.839*** (0.172)	0.639*** (0.174)	0.507*** (0.116)	1.235*** (0.167)	1.232*** (0.169)
Log(1+SXSW 2006)	0.031 (0.108)	0.084 (0.085)	0.054 (0.100)	-0.182* (0.104)	-0.187* (0.099)
Log(1+Protests 2000-2007)	0.062*** (0.024)	0.014 (0.017)	-0.003 (0.012)	-0.007 (0.022)	-0.001 (0.023)
p-value: $\beta^{RF} = \beta_{Extreme}^{RF}$	0.000***	0.000***	0.000***	0.028**	0.026**
Panel B					
Dependent variable:	Log(Number of protests) [Where 0->1 is valued as a 100% increase]				
Log(Twitter Users)	0.884*** (0.219)	0.964*** (0.266)	0.976*** (0.177)	1.033*** (0.179)	1.037*** (0.180)
Log(1+SXSW 2006)	-0.057 (0.146)	0.052 (0.120)	-0.040 (0.131)	-0.115 (0.094)	-0.119 (0.090)
Log(1+SXSW 2006)	0.072** (0.030)	0.048* (0.029)	-0.012 (0.023)	0.0003 (0.025)	0.005 (0.026)
p-value: $\beta^{RF} = \beta_{Extreme}^{RF}$	0.000***	0.000***	0.000***	0.000***	0.000***
Panel C					
Dependent variable:	Log(Number of protests) [Where 0->1 is valued as a 200% increase]				
Log(Twitter Users)	0.928*** (0.274)	1.289*** (0.376)	1.445*** (0.255)	0.830*** (0.209)	0.842*** (0.208)
Log(1+SXSW 2006)	-0.146 (0.188)	0.020 (0.165)	-0.133 (0.179)	-0.048 (0.107)	-0.051 (0.103)
Log(1+Protests 2000-2007)	0.082** (0.039)	0.081* (0.042)	-0.021 (0.036)	0.008 (0.031)	0.012 (0.032)
p-value: $\beta^{RF} = \beta_{Extreme}^{RF}$	0.002***	0.017**	0.054*	0.001***	0.001***
Observations	3,106	3,106	3,106	3,106	3,106
Cluster robust F-statistic	34.99	34.99	34.99	34.99	34.99
Mean number of protests	1.10	0.20	0.19	4.03	4.35

Note: This table presents county-level 2SLS regressions where the outcome is the log number of protests fitting the given column category, from January 2020 to November 2021. In Panel A, an extensive margin change from 0 to 1 is valued as a 0% increase. In Panel B and C, an extensive margin change from 0 to 1 is valued as a 100% and 200% increase respectively. Log(Twitter Users) is the natural logarithm of our Twitter users from 2014-2015. They are instrumented for via the log number of (one plus) Twitter users who joined Twitter in March 2007 and follow South by Southwest (@SXSW). SXSW 2006 is the number of Twitter users who joined Twitter in 2006 and follow South by Southwest (@SXSW). Coronavirus refers to protests related to public health measures due to the Covid pandemic. Presidential elections refers to protests related to the 2020 post-presidential election unrest (including those calling for results to be respected). Abortion rights refers to protests related to reproductive rights. BLM related refers to protests concerning police brutality against minorities, and counter movements. Race related refers to protests concerning racial tensions. A detailed description of topic categorization is presented in Tables A2 to A6 in the Appendix. All regressions control for population via decile dummies, and the full set of controls (as in column (8) in Table 2), see Table A9 in the Appendix for their descriptions. Robust standard errors in parentheses are clustered by state. *p<0.1; **p<0.05; ***p<0.01

Table A23: 2SLS by political leaning and extremity, inverse hyperbolic sine of SXSW followers

	(1)	(2)	(3)	(4)
Political leaning:	Left		Right	
Extremity:	Mainstream	Extreme	Mainstream	Extreme
Panel A				
Dependent variable:	Log(Number of protests)			
Log(Twitter Users)	1.008*** (0.182)	1.859*** (0.354)	0.869*** (0.233)	1.225*** (0.288)
Inverse hyperbolic sine of SXSW 2006	-0.061 (0.071)	-0.050 (0.154)	-0.076 (0.109)	0.065 (0.127)
Inverse hyperbolic sine of Protests 2000-2007	-0.004 (0.026)	-0.002 (0.032)	0.055* (0.029)	0.013 (0.023)
Mean number protests	4.40	0.36	0.68	0.19
Cluster robust F-statistic	38.09	38.09	38.09	38.09
Observations	3,106	3,106	3,106	3,106
p-value: $\beta_{Extreme}^{RF} = \beta_{Mainstream}^{RF}$		0.000***		0.087*
Panel B				
Dependent variable:	Share among all protests			
Log(Twitter Users)	-0.013 (0.040)	0.024*** (0.009)	0.001 (0.020)	0.015** (0.008)
Inverse hyperbolic sine of SXSW 2006	0.024 (0.019)	-0.007 (0.006)	-0.0003 (0.007)	-0.008** (0.003)
Inverse hyperbolic sine of Protests 2000-2007	-0.022** (0.009)	0.0001 (0.001)	0.001 (0.004)	-0.001 (0.002)
Mean dependent variable	0.495	0.009	0.078	0.013
Cluster robust F-statistic	50.55	50.55	50.55	50.55
Observations	1,756	1,756	1,756	1,756

Note: This table presents county-level 2SLS regressions. Panel A presents results where the log number of protests from January 2020 to November 2021, fitting the given column category is the dependent variable, weighting a change from 0 to 1 as a 150% increase. Panel B presents results on a subsample of counties where at least one protest event occurred, where the dependent variable is the share of protests that fit the column-category. Log(Twitter Users) is the natural logarithm of our Twitter users from 2014-2015. They are instrumented for via the inverse hyperbolic sine of Twitter users who joined Twitter in March 2007 and follow South by Southwest (@SXSW). SXSW 2006 is the number of Twitter users who joined Twitter in 2006 and follow South by Southwest (@SXSW). Mainstream left refers to protests which are left-leaning and no extreme groups were involved, mainstream right refers to right-leaning protests where no extreme groups were involved, Extreme left and right refer to protests where extreme left or extreme right groups are involved, respectively. A detailed description of protest categorization can be found in Tables A2 to A6 in the Appendix. All regressions control for population via decile dummies, and the full set of controls (as in column (8) in Table 2), see Table A9 in the Appendix for their descriptions. Robust standard errors in parentheses are clustered by state. *p<0.1; **p<0.05; ***p<0.01

Table A24: 2SLS by political leaning and extremity, alternative outcome transformation

	(1)	(2)	(3)	(4)
Political leaning:	Left		Right	
Extremity:	Mainstream	Extreme	Mainstream	Extreme
Panel A	Log(Number of protests)			
Dependent variable:	[0->1 valued as a 0% increase]			
Log(Twitter Users)	1.253*** (0.170)	1.340*** (0.261)	0.641*** (0.176)	0.931*** (0.201)
Log(1+SXSW 2006)	-0.162 (0.101)	0.002 (0.189)	0.032 (0.106)	-0.084 (0.108)
Log(1+Protests 2000-2007)	-0.010 (0.026)	-0.028 (0.029)	0.034* (0.021)	-0.014 (0.020)
p-value: $\beta_{Extreme}^{RF} = \beta_{Mainstream}^{RF}$	0.401		0.000***	
Panel B	Log(Number of protests)			
Dependent variable:	[0->1 valued as a 100% increase]			
Log(Twitter Users)	1.089*** (0.176)	1.704*** (0.325)	0.797*** (0.214)	1.158*** (0.261)
Log(1+SXSW 2006)	-0.103 (0.086)	-0.042 (0.188)	-0.053 (0.124)	0.017 (0.138)
Log(1+Protests 2000-2007)	-0.009 (0.029)	-0.014 (0.036)	0.054* (0.030)	0.003 (0.025)
p-value: $\beta_{Extreme}^{RF} = \beta_{Mainstream}^{RF}$	0.000***		0.014**	
Panel C	Log(Number of protests)			
Dependent variable:	[0->1 valued as a 200% increase]			
Log(Twitter Users)	0.925*** (0.198)	2.068*** (0.398)	0.954*** (0.262)	1.385*** (0.337)
Log(1+SXSW 2006)	-0.044 (0.096)	-0.086 (0.197)	-0.137 (0.149)	0.119 (0.186)
Log(1+Protests 2000-2007)	-0.008 (0.035)	-0.0004 (0.044)	0.073* (0.041)	0.021 (0.034)
p-value: $\beta_{Extreme}^{RF} = \beta_{Mainstream}^{RF}$	0.000***		0.136	
Mean number protests	4.40	0.36	0.68	0.19
Cluster robust F-statistic	34.99	34.99	34.99	34.99
Observations	3,106	3,106	3,106	3,106

Note: This table presents county-level 2SLS regressions where the outcome is the log number of protests fitting the given column category, from January 2020 to November 2021. In Panel A, an extensive margin change from 0 to 1 is valued as a 0% increase. In Panel B and C, an extensive margin change from 0 to 1 is valued as a 100% and 200% increase respectively. Log(Twitter Users) is the natural logarithm of our Twitter users from 2014-2015. They are instrumented for via log number of (one plus) Twitter users who joined Twitter in March 2007 and follow South by Southwest (@SXSW). SXSW 2006 is the number of Twitter users who joined Twitter in 2006 and follow South by Southwest (@SXSW). Mainstream left refers to protests which are left-leaning and no extreme groups were involved, mainstream right refers to right-leaning protests where no extreme groups were involved, Extreme left and right refer to protests where extreme left or extreme right groups are involved, respectively. A detailed description of protest categorization can be found in Tables A2 to A6 in the Appendix. All regressions control for population via decile dummies, and the full set of controls (as in column (8) in Table 2), see Table A9 in the Appendix for their descriptions. Robust standard errors in parentheses are clustered by state. *p<0.1; **p<0.05; ***p<0.01

Table A25: 2SLS by topic and political leaning

	(1)	(2)	(3)	(4)
Dependent variable:	Share among all protests			
Topic:	Presidential elections	Abortion rights	BLM related	Race related
Panel A				
Political leaning:	Left			
Log(Twitter Users)	−0.0005 (0.003)	−0.002 (0.004)	−0.013 (0.046)	−0.017 (0.046)
Log(1+SXSW 2006)	−0.003 (0.002)	0.002 (0.003)	0.027 (0.027)	0.029 (0.027)
Log(1+Protests 2000-2007)	0.001 (0.001)	−0.001* (0.001)	−0.027** (0.011)	−0.026** (0.011)
Mean dependent variable	0.002	0.006	0.458	0.047
Panel B				
Political leaning:	Right			
Log(Twitter Users)	0.0001 (0.010)	0.002 (0.008)	0.006 (0.017)	0.019 (0.018)
Log(1+SXSW 2006)	−0.004 (0.005)	0.0001 (0.004)	−0.007 (0.009)	−0.014 (0.009)
Log(1+Protests 2000-2007)	0.004** (0.001)	−0.001 (0.001)	0.001 (0.004)	−0.001 (0.005)
Mean dependent variable	0.011	0.012	0.041	0.047
Cluster robust F-statistic	47.88	47.88	47.88	47.88
Observations	1,756	1,756	1,756	1,756

Note: This table presents county-level 2SLS regressions, on a subsample of counties where at least one protest occurred from January 2020 to November 2021. The outcome variables are the share of protests fitting the given category, from January 2020 to November 2021. Panel A looks at left leaning protests and Panel B looks at right leaning protests. Presidential elections refers to protests related to the 2020 post-presidential election unrest (those calling for results to be respected are left leaning). Abortion rights refers to protests related to reproductive rights. BLM related refers to protests concerning police brutality against minorities, and counter movements (which are considered right leaning). Race related refers to protests concerning racial tensions. A detailed description of topic categorization is presented in Tables A2 to A6 in the Appendix. All regressions control for population via decile dummies, and the full set of controls (as in column (8) in Table 2), see Table A9 in the Appendix for their descriptions. Robust standard errors in parentheses are clustered by state. *p<0.1; **p<0.05; ***p<0.01

Table A26: 2SLS Extreme left groups

	(1)	(2)
Political leaning:	“Radical racial justice”	Extreme left other
Panel A		
Dependent variable:	Log(Number of protests)	
Log(Twitter Users)	1.077*** (0.278)	1.832*** (0.319)
Log(1+SXSW 2006)	−0.109 (0.159)	−0.033 (0.196)
Log(1+Protests 2000-2007)	−0.002 (0.025)	−0.020 (0.037)
Mean number of protests	0.076	0.285
Cluster robust F-statistic	34.99	34.99
Observations	3,106	3,106
p-value: $\beta^{RF} = \beta_{MainstreamLeft}^{RF}$	0.727	0.000***
Panel B		
Dependent variable:	Share among all protests	
Log(Twitter Users)	0.008** (0.004)	0.017** (0.007)
Log(1+SXSW 2006)	−0.002 (0.002)	−0.007 (0.007)
Log(1+Protests 2000-2007)	−0.0004 (0.001)	0.0005 (0.002)
Mean dependent variable	0.003	0.006
Cluster robust F-statistic	47.88	47.88
Observations	1,756	1,756

Note: This table presents county-level 2SLS regressions. Panel A presents 2SLS results where the log number of protests from January 2020 to November 2021, fitting the given column category is the dependent variable, weighting a change from 0 to 1 as a 150% increase. Panel B presents 2SLS results on a subsample of counties where at least one protest event occurred between January 2020 and November 2021, where the dependent variable is the share of protest that fit the column-category. A detailed description of the groups considered under “Radical racial justice” and “Extreme left other” is found in in Table A1 in the Appendix. All regressions control for population via decile dummies, and the full set of controls (as in column (8) in Table 2), see Table A9 in the Appendix for their descriptions. Robust standard errors in parentheses are clustered by state. *p<0.1; **p<0.05; ***p<0.01

Table A27: 2SLS protests by aggression, alternative outcomes

	(1)	(2)	(3)	(4)
Type of protest:	Peaceful	Peaceful Protesters met with intervention or violence	Violent Protesters	Violent protesters interact with violent outside group
Panel A	Log(Number of protests)			
Dependent variable:	[0->1 valued as a 0% increase]			
Log(Twitter Users)	1.099*** (0.161)	1.013*** (0.190)	1.173*** (0.233)	0.119 (0.075)
Log(1+SXSW 2006)	-0.134 (0.086)	0.141 (0.107)	-0.017 (0.139)	0.079 (0.064)
Log(1+ Protests 2000-2007)	0.029 (0.024)	-0.014 (0.020)	-0.033 (0.022)	-0.004 (0.005)
p-value: $\beta^{RF} = \beta_{Peaceful}^{RF}$		0.471	0.549	0.000***
Panel B	Log(Number of protests)			
Dependent variable:	[0->1 valued as a 100% increase]			
Log(Twitter Users)	0.914*** (0.160)	1.285*** (0.236)	1.379*** (0.259)	0.278** (0.131)
Log(1+SXSW 2006)	-0.080 (0.081)	0.089 (0.146)	0.006 (0.178)	0.195* (0.105)
Log(1+Protests 2000-2007)	0.039 (0.026)	0.014 (0.031)	-0.019 (0.026)	0.007 (0.010)
p-value: $\beta^{RF} = \beta_{Peaceful}^{RF}$		0.054*	0.016**	0.000***
Panel C	Log(Number of protests)			
Dependent variable:	[0->1 valued as a 200% increase]			
Log(Twitter Users)	0.729*** (0.172)	1.558*** (0.304)	1.585*** (0.326)	0.436** (0.198)
Log(1+SXSW 2006)	-0.027 (0.096)	0.038 (0.203)	0.028 (0.240)	0.312** (0.158)
Log(1+Protests 2000-2007)	0.049 (0.031)	0.042 (0.044)	-0.005 (0.034)	0.017 (0.017)
p-value: $\beta^{RF} = \beta_{Peaceful}^{RF}$		0.012**	0.009***	0.208
Mean number of protests	10.31	0.37	0.30	0.03
Cluster robust F-statistic	34.99	34.99	34.99	34.99
Observations	3,106	3,106	3,106	3,106

Note: This table presents county-level 2SLS regressions where the outcome is the log number of protests fitting the given column category, from January 2020 to November 2021. In Panel A, an extensive margin change from 0 to 1 is valued as a 0% increase. In Panel B and C, an extensive margin change from 0 to 1 is valued as a 100% and 200% increase respectively. Log(Twitter Users) is the natural logarithm of our Twitter users from 2014-2015. They are instrumented for via the log number of (one plus) Twitter users who joined Twitter in March 2007 and follow South by Southwest (@SXSW). SXSW 2006 is the number of Twitter users who joined Twitter in 2006 and follow South by Southwest (@SXSW). All regressions control for population via decile dummies, and the full set of controls (as in column (8) in Table 2), see Table A9 in the Appendix for their descriptions. Robust standard errors in parentheses are clustered by state. *p<0.1; **p<0.05; ***p<0.01

Table A28: 2SLS Effects on strong opinions on individual statements on race or gender issues, Cooperative Election Study 2020

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Statement:	Police safe	White privilege	Racial problems rare	Black people don't deserve favors	Lingering effects of slavery	Women too easily offended	Women seek power
Dependent variable:	D=1 if strong opinion						
Log(Twitter Users)	-0.010 (0.014)	0.039* (0.022)	0.023 (0.016)	0.040** (0.018)	0.027* (0.015)	0.020 (0.017)	0.029** (0.014)
Log(1+SXSW 2006)	0.007 (0.009)	-0.026** (0.012)	-0.024*** (0.008)	-0.011 (0.010)	-0.023** (0.010)	-0.010 (0.009)	-0.013* (0.008)
Log(1+Protests 2000-2007)	0.002 (0.005)	-0.001 (0.005)	-0.001 (0.005)	-0.005 (0.005)	0.004 (0.004)	-0.002 (0.004)	-0.003 (0.004)
Mean dependent variable	0.52	0.52	0.48	0.50	0.52	0.38	0.41
Observations	60,403	51,099	49,863	51,093	51,088	50,857	50,866

Note: This table presents individual-level regressions. Log(Twitter Users) is the natural logarithm of unique Twitter users from a large sample of Tweets from 2014-2015. They are instrumented for via the log number of (one plus) Twitter users who joined Twitter in March 2007 and follow South by Southwest (@SXSW). SXSW 2006 is the number of Twitter users who joined Twitter in 2006 and follow South by Southwest (@SXSW). All columns present second stages. All regressions include the following individual-level controls: Age, gender, education, race, ethnicity, employment status, home-ownership, family income, number of children and internet connection, and the full set of county-characteristics (as in column (8) in Table 2, see Table A9 in the Appendix for their descriptions. The statements on race issues are the following: “Do the police make you feel...?” where “Mostly safe” is considered the strongly conservative option and “Mostly unsafe” is the strongly liberal option. “White people in the U.S. have certain advantages because of the color of their skin,” where “Strongly disagree” is considered the strongly conservative opinion and “Strongly agree” is the strongly liberal option. “Racial problems in the U.S. are rare, isolated situations,” “Irish, Italians, Jewish and many other minorities overcame prejudice and worked their way up. Blacks should do the same without any special favors,” “Generations of slavery and discrimination have created conditions that make it difficult for blacks to work their way out of the lower class,” where “Strongly agree” is the strongly liberal option and “Strongly disagree” is the strongly conservative option. The statements on gender issues are the following: Women are too easily offended, Women seek to gain power by getting control over men, where “Strongly agree” is the strongly conservative option and “Strongly disagree” is the strongly liberal option. Observations are weighted to represent adult Americans and robust standard errors are clustered by state. *p<0.1; **p<0.05; ***p<0.01

Table A29: 2SLS Effects on strong liberal or conservative opinions on individual statements on race or gender issues, Cooperative Election Study 2020

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Statement:	Police safe	White privilege	Racial problems rare	Black people don't deserve favors	Lingering effects of slavery	Women too easily offended	Women seek power
Dependent variable:	D=1 if strongly conservative opinion						
Log(Twitter Users)	-0.018 (0.012)	0.006 (0.016)	0.001 (0.016)	0.024 (0.015)	0.006 (0.014)	0.010 (0.013)	0.024*** (0.008)
Log(1+SXSW 2006)	-0.0001 (0.008)	-0.005 (0.007)	-0.008 (0.008)	-0.006 (0.007)	-0.011 (0.008)	0.00005 (0.008)	-0.008* (0.005)
Log(1+Protests 2000-2007)	0.004 (0.004)	0.001 (0.005)	-0.0002 (0.004)	-0.003 (0.003)	0.003 (0.004)	-0.002 (0.003)	-0.004* (0.002)
Mean dependent variable	0.45	0.18	0.39	0.27	0.23	0.10	0.06
Dependent variable:	D=1 if strongly liberal opinion						
Log(Twitter Users)	0.008 (0.010)	0.032** (0.015)	0.022* (0.011)	0.016 (0.010)	0.021* (0.013)	0.010 (0.011)	0.005 (0.012)
Log(1+SXSW 2006)	0.007 (0.007)	-0.021* (0.012)	-0.015** (0.006)	-0.005 (0.007)	-0.012 (0.009)	-0.010 (0.010)	-0.004 (0.007)
Log(1+Protests 2000-2007)	-0.002 (0.003)	-0.001 (0.004)	-0.001 (0.003)	-0.002 (0.003)	0.002 (0.003)	0.0002 (0.003)	0.001 (0.004)
Mean dependent variable	0.08	0.35	0.09	0.24	0.30	0.28	0.36
Observations	60,403	51,099	49,863	51,093	51,088	50,857	50,866

Note: This table presents individual-level regressions. Log(Twitter Users) is the natural logarithm of unique Twitter users from a large sample of Tweets from 2014-2015. They are instrumented for via the log number of (one plus) Twitter users who joined Twitter in March 2007 and follow South by Southwest (@SXSW). SXSW 2006 is the number of Twitter users who joined Twitter in 2006 and follow South by Southwest (@SXSW). All columns present second stages. All regressions include the following individual-level controls: Age, gender, education, race, ethnicity, employment status, home-ownership, family income, number of children and internet connection, and the full set of county-characteristics (as in column (8) in Table 2, see Table A9 in the Appendix for their descriptions. The statements on race issues are the following: “Do the police make you feel...?” where “Mostly safe” is considered the strongly conservative option option and “Mostly unsafe” is the strongly liberal option. “White people in the U.S. have certain advantages because of the color of their skin,” where “Strongly disagree” is considered the strongly conservative opinion and “Strongly agree” is the strongly liberal option. “Racial problems in the U.S. are rare, isolated situations,” “Irish, Italians, Jewish and many other minorities overcame prejudice and worked their way up. Blacks should do the same without any special favors,” “Generations of slavery and discrimination have created conditions that make it difficult for blacks to work their way out of the lower class,” where “Strongly agree” is the strongly liberal option and “Strongly disagree” is the strongly conservative option. The statements on gender issues are the following: Women are too easily offended, Women seek to gain power by getting control over men, where “Strongly agree” is the strongly conservative option and “Strongly disagree” is the strongly liberal option. Observations are weighted to represent adult Americans and robust standard errors are clustered by state. *p<0.1; **p<0.05; ***p<0.01

Table A30: 2SLS Effects on political knowledge, Cooperative Election Study 2020

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Knows party of:	Governor	Senator, senior	Senator, junior	Congressman/woman	House of representatives	US senate		
Knows majority party in:								
Dependent variable:	D=1 if correct party						Sum correct answers	1st principal component
Log(Twitter Users)	0.001 (0.016)	0.012 (0.021)	-0.010 (0.012)	0.001 (0.019)	0.020 (0.024)	0.003 (0.022)	0.034 (0.082)	0.030 (0.075)
Log(1+SXSW 2006)	-0.011 (0.011)	-0.018* (0.011)	0.009 (0.007)	-0.010 (0.010)	-0.018 (0.011)	-0.009 (0.013)	-0.056 (0.046)	-0.050 (0.043)
Log(1+Protests 2000-2007)	0.008* (0.005)	-0.003 (0.006)	0.006 (0.004)	-0.002 (0.004)	0.004 (0.005)	0.006 (0.005)	0.017 (0.021)	0.016 (0.020)
Mean dependent variable	0.82	0.73	0.75	0.67	0.64	0.70	4.33	
Observations	60,246	60,235	60,238	59,864	60,470	60,456	59,590	

Note: This table presents individual-level regressions. Log(Twitter Users) is the natural logarithm of unique Twitter users from a large sample of Tweets from 2014-2015. They are instrumented for via the log number of (one plus) Twitter users who joined Twitter in March 2007 and follow South by Southwest (@SXSW). SXSW 2006 is the number of Twitter users who joined Twitter in 2006 and follow South by Southwest (@SXSW). All columns present second stages. All regressions include the following individual-level controls: Age, gender, education, race, ethnicity, employment status, home-ownership, family income, number of children and internet connection, and the full set of county-characteristics (as in column (8) in Table 2, see Table A9 in the Appendix for their descriptions. Observations are weighted to represent adult Americans and robust standard errors are clustered by state. *p<0.1; **p<0.05; ***p<0.01